

MOTION IN A STRAIGHT LINE

Grade 9 Study Guide - Source-Complete Integration

Concept -> application -> visual model -> first move -> guided practice -> transfer.

WORDS -> PICTURE -> PHYSICS FACTS -> TARGET -> RELATION

The equation is the compression of the model, not the starting point.

Source bank Q1-Q68 is mapped before publication; handwritten scan markings are not authority.

NAV-1

Every numerical problem: first 20 seconds

ROUTINE

1. Draw a tiny line/axis or functional picture.
2. Choose + direction if signs matter.
3. Write the known physical quantities.
4. Translate hidden words: starts $\rightarrow u=0$; stops $\rightarrow v=0$; highest point $\rightarrow v=0$.
5. Circle the target.
6. Decide the model.
7. Choose the relation that avoids unnecessary unknowns.
8. Check units, sign, magnitude and model validity.

Physics hint ladder

H1 - NOTICE

What clue in the words or picture identifies the structure?

H2 - MODEL

What diagram, timeline, graph, UVATS strip or relative frame represents it?

H3 - START

What is the first executable relation? Stop the hint as soon as you can continue alone.

I'M STUCK - WHICH RELATION?

Model gate before formula selection

NAV-2

1

MODEL

constant acceleration? average? cycle? delayed start? relative motion? graph?

2

REPRESENT

line / journey strip / UVATS / timeline / vertical axis / dual track / graph

3

TRANSLATE

rest $\rightarrow u=0$; stop/top $\rightarrow v=0$; delay $\rightarrow$ local elapsed time

4

TARGET

what exactly is asked?

5

EXCLUDE

which variable does not need to appear?

6

CHOOSE

pick the relation or graph operation that avoids it

7

CHECK

units, sign, magnitude, assumptions

TOC

Clean titles | Start = page in this integrated 296-page edition

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NAV-3

SOURCE BANK

All supplied source questions Q1-Q68 are linked to at least one learner concept. Source defects remain flagged rather than silently corrected.

SCOPE

Core topics are those supported by the supplied Motion source. Extra JEE surface forms may be used only when they transfer the same source concept; unrelated calculus / two-dimensional / mechanics extensions are not silently added.

EXAMSIDE

The question book is the current audited in-scope primary set, not yet a line-by-line classification of every one of the 123 items visible on the ExamSIDE chapter index.

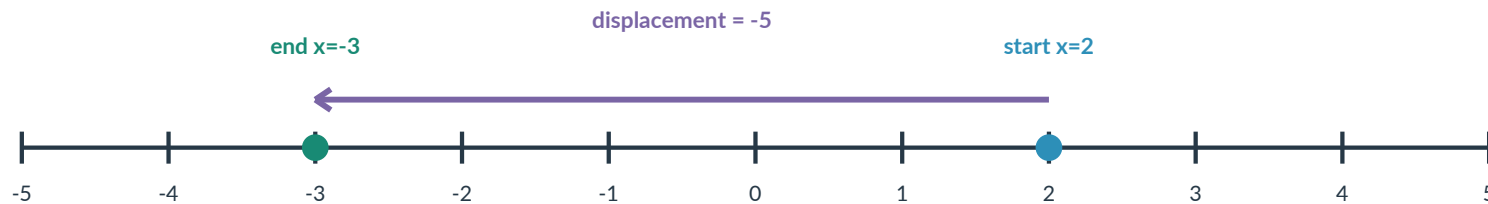
OWNERSHIP

Each ExamSIDE question has one primary concept home. Cross-links do not create duplicate-primary counts.

MOTION IN A STRAIGHT LINE

M-ST01A - Reference Frame, Distance and Displacement

FOUNDATION | See the motion first, then compress it into mathematics.



MEMORY ANCHOR

Distance follows the path. Displacement compares only start and end coordinates.

SOURCE FOCUS

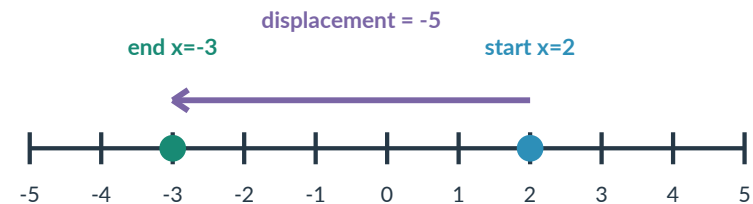
Supplied-bank anchors: Q1, Q2, Q4.

APPLY

When do I see this?

- 1 start/end position is given
- 2 question asks distance vs displacement
- 3 motion reverses direction
- 4 closed trip / returns to start

Represent it first



FIRST MOVE

Choose an origin and + direction. Mark start, end, and actual path separately.

HELP

Helper

CONCEPT HELPER

Distance = total path length. Displacement = final coordinate - initial coordinate. Always $|\text{displacement}| \leq \text{distance}$.

Worked contrast

EXAMPLE

Walk 4 m right, then 7 m left. Distance=11 m. With right positive, displacement=-3 m.

COMMON WRONG IDEA

“One-dimensional motion means distance always equals displacement.”

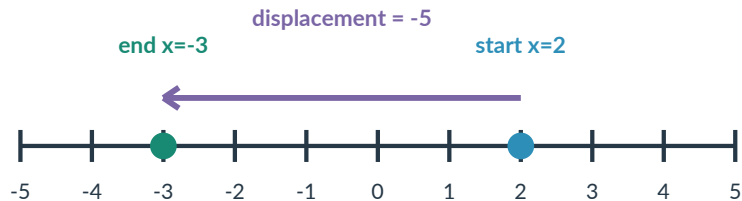
WHEN NOT TO USE A SHORTCUT

Do not set $\text{distance} = |\text{displacement}|$ if the path reverses or curves.

GUIDED

Guided 1 - full support

A runner completes one lap and returns to the start. Which must be zero: distance or displacement?



Guided 2 - faded

A student walks 6 m right, 2 m left, then 5 m left. Find distance and displacement.

Representation: _____

Key relation / idea: _____

Check: _____

H1 -> H2 -> H3

H1: compare start/end. H2: closed path -> endpoint change zero. H3: distance is path length.

H1 -> H2 -> H3

Notice the structure -> build the physical model -> write only the first relation you need.

CHECK

Independent transfer

A delivery rider ends 2 km east of the start after travelling 9 km. Which quantity tells how much road was covered, and which tells the net change of position?

NO HINT INITIALLY

Draw the representation yourself before writing a relation.

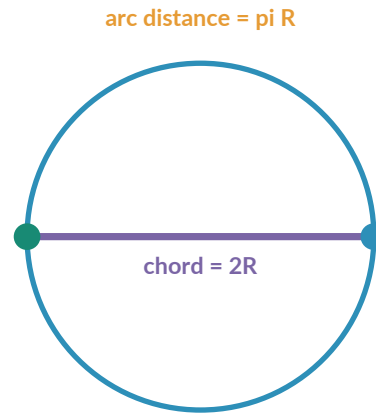
Self-check

- 1 I choose an origin and + direction.
- 2 I separate path from endpoint change.
- 3 I use $|\Delta x| \leq \text{distance}$.
- 4 I recognise closed-path displacement.

MOTION IN A STRAIGHT LINE

M-ST01B - Circular Paths: Arc Distance vs Chord Displacement

FOUNDATION | See the motion first, then compress it into mathematics.



MEMORY ANCHOR

On a curved path, distance follows the arc while displacement is the straight chord joining endpoints.

SOURCE FOCUS

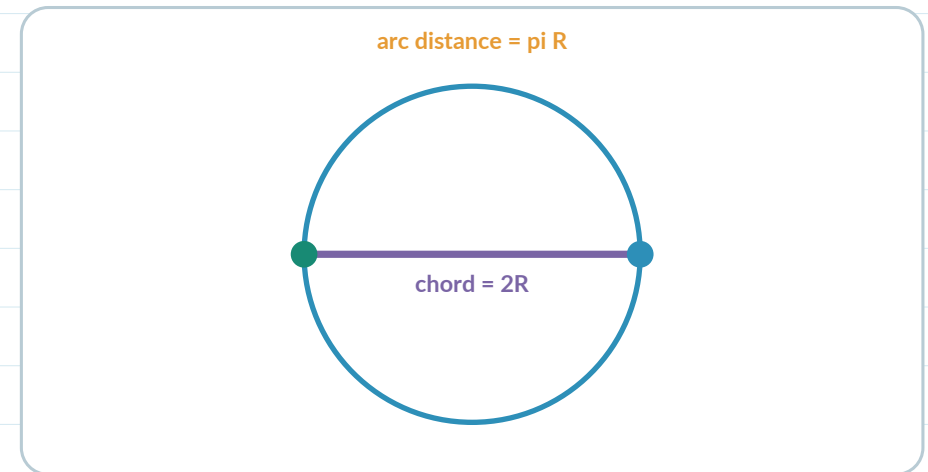
Supplied-bank anchors: Q3, Q5.

APPLY

When do I see this?

- 1 circle/semicircle/revolution language
- 2 path is curved but displacement asks endpoints
- 3 complete turns plus a final fraction

Represent it first



FIRST MOVE

Draw the circle, mark endpoints, trace the arc actually travelled, then draw the straight chord.

HELP

Helper

CONCEPT HELPER

For a half-circle: distance= πR , displacement magnitude= $2R$. Complete revolutions add path length but contribute zero net displacement.

Worked contrast

EXAMPLE

Travel a semicircle radius R . The journey length is πR , but endpoint separation is $2R$.

COMMON WRONG IDEA

"The object travelled along the displacement chord."

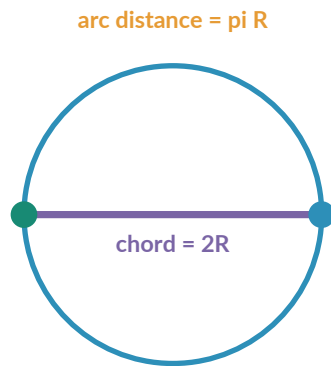
WHEN NOT TO USE A SHORTCUT

Do not use circumference/arc length when the question asks displacement.

GUIDED

Guided 1 - full support

A body moves along a semicircle of radius R . Compare distance with displacement magnitude.



Guided 2 - faded

A body completes one full revolution and then a quarter revolution on a circle. What should you compute for distance and for displacement?

Representation: _____

Key relation / idea: _____

Check: _____

H1 -> H2 -> H3

H1: path is an arc. H2: endpoints form a diameter. H3: write πR and $2R$.

H1 -> H2 -> H3

Notice the structure -> build the physical model -> write only the first relation you need.

CHECK

Independent transfer

After 2.5 revolutions of a circle, identify which complete turns affect distance and which affect final displacement.

NO HINT INITIALLY

Draw the representation yourself before writing a relation.

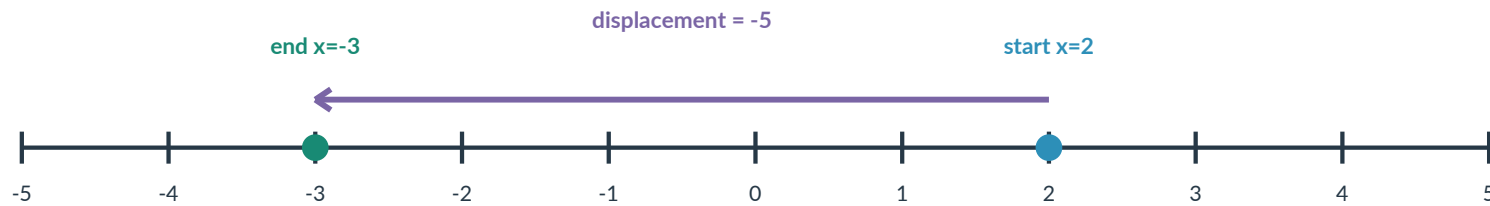
Self-check

- 1 I distinguish arc from chord.
- 2 I handle complete revolutions.
- 3 I use endpoints for displacement.
- 4 I avoid adding chord lengths as path.

MOTION IN A STRAIGHT LINE

M-ST02A - Speed vs Velocity: Different Numerators

FOUNDATION | See the motion first, then compress it into mathematics.



MEMORY ANCHOR

Average speed uses total distance; average velocity uses displacement. A round trip can have nonzero average speed and zero average velocity.

SOURCE FOCUS

Supplied-bank anchors: Q6, Q7.

APPLY

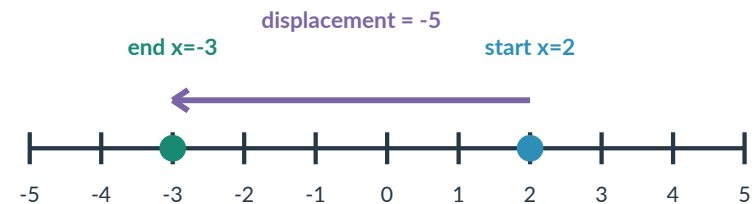
When do I see this?

1 average speed vs average velocity

2 round trip

3 distance and displacement differ

Represent it first



FIRST MOVE

Write two numerators explicitly: total distance and displacement. They are not interchangeable.

HELP

Helper

CONCEPT HELPER

average speed = total distance / total time; average velocity = displacement / total time.

Worked contrast

EXAMPLE

Go 30 m out and 30 m back in 10 s: average speed=6 m/s; average velocity=0.

COMMON WRONG IDEA

“Average speed and average velocity are the same formula.”

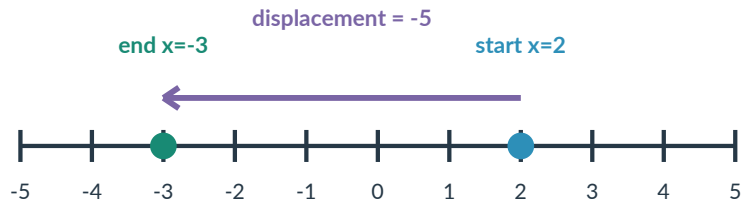
WHEN NOT TO USE A SHORTCUT

Do not replace signed displacement with total path.

GUIDED

Guided 1 - full support

A cyclist travels 100 m out and 100 m back in 40 s. Find average speed and average velocity.



Guided 2 - faded

A motion has positive total distance but zero displacement. What must be true about average velocity?

Representation: _____

Key relation / idea: _____

Check: _____

H1 -> H2 -> H3

H1: totals have different numerators. H2: displacement is zero. H3: divide each by 40 s.

H1 -> H2 -> H3

Notice the structure -> build the physical model -> write only the first relation you need.

CHECK

Independent transfer

Construct your own round-trip example in which average speed is nonzero but average velocity is zero.

NO HINT INITIALLY

Draw the representation yourself before writing a relation.

Self-check

- 1 I use distance for average speed.
- 2 I use displacement for average velocity.
- 3 I handle round trips.
- 4 I keep direction in velocity.

MOTION IN A STRAIGHT LINE

M-ST02B - Equal Time vs Equal Distance Averages

FOUNDATION | See the motion first, then compress it into mathematics.

EQUAL TIME

v1 for T

v2 for T

same time weight -> arithmetic mean

EQUAL DISTANCE

L at v1

L at v2

slower leg lasts longer

MEMORY ANCHOR

Equal times give equal time weights; equal distances give unequal times and a harmonic-type average.

SOURCE FOCUS

Supplied-bank anchors: Q8, Q9.

APPLY

When do I see this?

- 1 'equal times' or 'first T, next T'
- 2 'half the distance' / equal distances
- 3 same speeds but different weighting structure

Represent it first

EQUAL TIME

v1 for T

v2 for T

same time weight -> arithmetic mean

EQUAL DISTANCE

L at v1

L at v2

slower leg lasts longer

FIRST MOVE

Mark what is equal: time or distance. Build total distance / total time before using any shortcut.

HELP

Helper

CONCEPT HELPER

Equal times: $(v_1 + v_2)/2$. Equal distances: $2v_1v_2/(v_1 + v_2)$. If neither condition holds, return to total distance / total time.

Worked contrast

EXAMPLE

10 and 15 m/s: equal times $\rightarrow$ 12.5 m/s; equal distances $\rightarrow$ 12 m/s.

COMMON WRONG IDEA

“Average of two speeds is always the arithmetic mean.”

WHEN NOT TO USE A SHORTCUT

Do not use the equal-time or equal-distance shortcut unless that weighting condition is actually true.

GUIDED

Guided 1 - full support

A cyclist moves at 8 m/s for 3 min and then 12 m/s for the next 3 min. Find average speed.

EQUAL TIME

v1 for T

v2 for T

same time weight -> arithmetic mean

EQUAL DISTANCE

L at v1

L at v2

slower leg lasts longer

Guided 2 - faded

Half a journey is at 12 m/s and the other half at 18 m/s. Find average speed.

Representation: _____

Key relation / idea: _____

Check: _____

H1 -> H2 -> H3

H1: times are equal. H2: equal time weights. H3: arithmetic mean.

H1 -> H2 -> H3

Notice the structure -> build the physical model -> write only the first relation you need.

CHECK

Independent transfer

Two journeys use the same two speeds. In one the times are equal; in the other the distances are equal. Predict which average is smaller before calculating.

NO HINT INITIALLY

Draw the representation yourself before writing a relation.

Self-check

- 1 I ask what is equal first.
- 2 I derive equal-distance average from times.
- 3 I know slower legs get more time weight.
- 4 I return to total distance/total time when needed.

MOTION IN A STRAIGHT LINE

M-ST02C - Repeated Motion: Complete Cycles + Final Partial Cycle

FOUNDATION | See the motion first, then compress it into mathematics.

5 steps forward



3 steps backward



one cycle: +2 steps net

MEMORY ANCHOR

Analyse one complete cycle, count full cycles, then inspect the unfinished final cycle.

SOURCE FOCUS

Supplied-bank anchors: Q10.

APPLY

When do I see this?

- 1 repeated forward/back pattern
- 2 cycles or laps
- 3 target may be reached partway through final cycle

Represent it first

5 steps forward



3 steps backward



one cycle: +2 steps net

FIRST MOVE

Write one complete cycle: cycle time, cycle distance, net progress. Then handle the leftover motion.

HELP

Helper

CONCEPT HELPER

One cycle creates a repeatable bookkeeping unit. Do not replace the final incomplete cycle by an average.

Worked contrast

EXAMPLE

5 forward, 3 back: one 8-step cycle gives +2 net steps. A target may be reached during the next forward run.

COMMON WRONG IDEA

“Use net progress per cycle and ignore where the target is reached inside the final cycle.”

WHEN NOT TO USE A SHORTCUT

Do not assume the target is reached only at cycle boundaries.

GUIDED

Guided 1 - full support

A walker repeats 5 forward steps and 3 backward steps. After 3 complete cycles, what is the net progress?

5 steps forward



3 steps backward



one cycle: +2 steps net

Guided 2 - faded

The same walker wants to reach a point 9 steps ahead. Explain why counting complete cycles alone is not enough.

Representation: _____

Key relation / idea: _____

Check: _____

H1 -> H2 -> H3

H1: build one cycle. H2: net +2 per cycle. H3: multiply by 3.

H1 -> H2 -> H3

Notice the structure -> build the physical model -> write only the first relation you need.

CHECK

Independent transfer

Create a forward/back repeated pattern where a target is reached during the final incomplete cycle.

NO HINT INITIALLY

Draw the representation yourself before writing a relation.

Self-check

- 1 I model one cycle.
- 2 I count full cycles.
- 3 I inspect the partial final cycle.
- 4 I do not overuse cycle averages.

MOTION IN A STRAIGHT LINE

M-ST03A - Acceleration as Change in Velocity

FOUNDATION | See the motion first, then compress it into mathematics.

equal velocity change each second $\rightarrow$ constant acceleration

4 m/s

t=0s

6 m/s

t=1s

8 m/s

t=2s

10 m/s

t=3s

MEMORY ANCHOR

Acceleration measures change in velocity per time; direction changes matter even when speed does not.

SOURCE FOCUS

Supplied-bank anchors: Q12, Q14, Q28, Q30, Q31.

APPLY

When do I see this?

- 1 velocity changes with time
- 2 same speed but direction changes
- 3 average acceleration asks for $\Delta v / \Delta t$

Represent it first

equal velocity change each second -> constant acceleration



FIRST MOVE

Draw velocity arrows at start and finish before subtracting them.

HELP

Helper

CONCEPT HELPER

$a_{avg} = (v_{final} - v_{initial}) / \Delta t$. Velocity is signed/directional.

Worked contrast

EXAMPLE

Velocity +12 to -4 m/s in 4 s: $\Delta v = -16$ m/s, so average acceleration = -4 m/s².

COMMON WRONG IDEA

“Acceleration only measures change of speed.”

WHEN NOT TO USE A SHORTCUT

Do not subtract speed magnitudes when direction changes; subtract velocities.

GUIDED

Guided 1 - full support

A particle changes velocity from +12 to -4 m/s in 4 s. Find average acceleration.

equal velocity change each second -> constant acceleration



Guided 2 - faded

A body moves at constant speed but reverses its velocity direction. Can average acceleration be nonzero? Explain.

Representation: _____

Key relation / idea: _____

Check: _____

H1 -> H2 -> H3

H1: use signed velocities. H2: $\Delta v = -4 - (+12)$. H3: divide by 4.

H1 -> H2 -> H3

Notice the structure -> build the physical model -> write only the first relation you need.

CHECK

Independent transfer

Explain how a body can have constant speed but nonzero average acceleration over an interval.

NO HINT INITIALLY

Draw the representation yourself before writing a relation.

Self-check

- 1 I compute signed Δv .
- 2 I interpret m/s^2 physically.
- 3 I allow direction-only velocity change.
- 4 I distinguish speed from velocity.

MOTION IN A STRAIGHT LINE

M-ST03B - Sign Logic: Negative Acceleration Can Speed Up

FOUNDATION | See the motion first, then compress it into mathematics.

CASE A: v right, a left $\rightarrow$ slowing



CASE B: v left, a left $\rightarrow$ speeding



MEMORY ANCHOR

Signs describe direction relative to the chosen axis. Speed increases when velocity and acceleration point the same way.

SOURCE FOCUS

Supplied-bank anchors: Q29 + Unit 3.2.

APPLY

When do I see this?

1 negative velocity / negative acceleration

2 speeding or slowing decision

3 object reverses direction

Represent it first

CASE A: v right, a left -> slowing



CASE B: v left, a left -> speeding



FIRST MOVE

Draw separate arrows for v and a. Compare directions, not just plus/minus symbols.

HELP

Helper

CONCEPT HELPER

Away from $v=0$ in 1D: v and a same direction $\rightarrow$ speed increases; opposite directions $\rightarrow$ speed decreases.

Worked contrast

EXAMPLE

$v=-6$ and $a=-2$: both point left, so speed increases even though a is negative.

COMMON WRONG IDEA

“Negative acceleration always means slowing down.”

WHEN NOT TO USE A SHORTCUT

At $v=0$, the same-sign/opposite-sign speed rule needs care; inspect what happens immediately after.

GUIDED

Guided 1 - full support

A body moves left with $v = -8 \text{ m/s}$ and $a = -2 \text{ m/s}^2$. Is its speed increasing or decreasing?

CASE A: v right, a left -> slowing



CASE B: v left, a left -> speeding



Guided 2 - faded

A body has $v = +10 \text{ m/s}$ and $a = -3 \text{ m/s}^2$. What happens to speed before it reaches $v = 0$?

Representation: _____

Key relation / idea: _____

Check: _____

H1 -> H2 -> H3

H1: v and a have same sign. H2: arrows point same way. H3: speed increases.

H1 -> H2 -> H3

Notice the structure -> build the physical model -> write only the first relation you need.

CHECK

Independent transfer

Give one example with negative acceleration that speeds up, and one that slows down.

NO HINT INITIALLY

Draw the representation yourself before writing a relation.

Self-check

1 I compare v and a directions.

2 I know negative a can speed up.

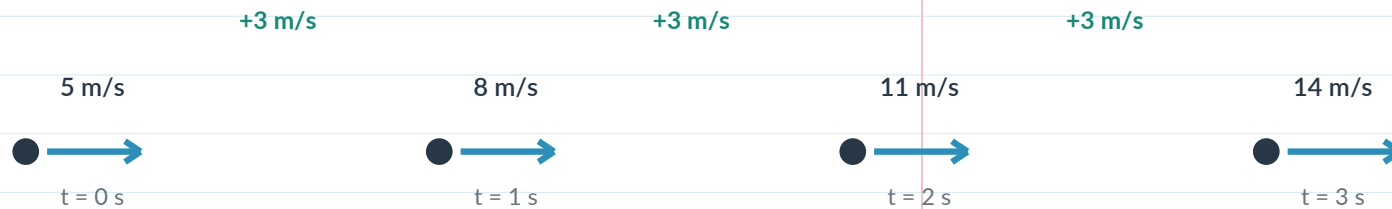
3 I treat $v=0$ as a boundary.

4 I use arrows before signs.

MOTION IN A STRAIGHT LINE

M-ST04A - Constant Acceleration and $v = u + at$

A visual, real-life route from changing velocity to equation choice and stopping time.



MEMORY ANCHOR

Equal change in velocity during equal time intervals is the visual signature of constant acceleration.

READING RHYTHM

Large explanations, large labels, and fewer words per box. The learner should not need to zoom the PDF to understand a diagram or helper.

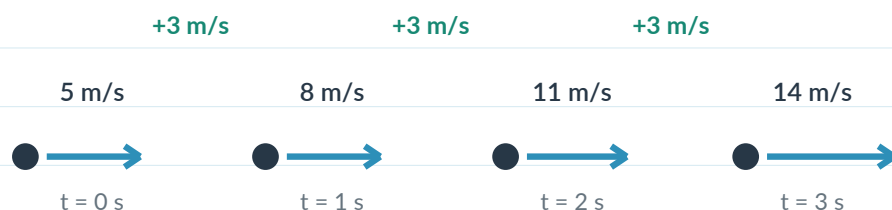
A SCOOTER LEAVING A TRAFFIC LIGHT

Real-life anchor before the equation

M4A-C1

Watch the speedometer once each second

A scooter is moving along a straight road. Imagine its velocity readings are 5, 8, 11 and 14 m/s at one-second intervals.



WHAT DO YOU NOTICE?

Each second adds the same 3 m/s. The scooter is not moving at constant velocity; the velocity is changing in a regular way.

Where the model appears in real life

A train pulling away from a station, a bicycle speeding up smoothly, or a car braking steadily can often be approximated by constant acceleration

PHYSICS MODEL, NOT A PROMISE

A real vehicle does not keep exactly the same acceleration forever. We use the model only over an interval where the change is sufficiently steady.

SAY IT IN ORDINARY LANGUAGE

Constant acceleration means velocity changes by the same amount in equal time intervals.

THE CONSTANT-ACCELERATION MEMORY HELPER

Same Δv / same Δt

M4A-C2

The idea in one picture

SAME Δv

during

SAME Δt

→ constant acceleration

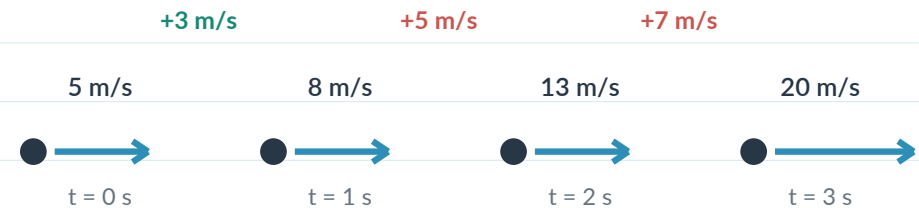
$$a = \Delta v / \Delta t$$

THINGS TO KNOW

Acceleration is built from velocity, not speed alone. Direction therefore matters when you calculate Δv .

Close contrast: acceleration is changing

Now compare velocity readings 5, 8, 13 and 20 m/s at one-second intervals.



WHY THIS IS DIFFERENT

The gains are +3, +5 and +7 m/s. Equal time intervals do not give equal velocity changes, so one constant-acceleration value cannot describe the whole interval.

M4A-C3

From the motion story to a number

If velocity changes from +4 m/s to +10 m/s in 3 s, the velocity change is +6 m/s.

$$a = (v - u) / t$$

$$a = (10 - 4) / 3 = +2 \text{ m/s}^2$$

READ THE UNIT

2 m/s² means: velocity changes by 2 m/s during each second. The sign tells the direction of acceleration relative to your chosen positive axis.

A sign trap worth seeing

Suppose velocity changes from +12 m/s to -4 m/s in 4 s.

DO NOT SUBTRACT SPEEDS

$\Delta v = \text{final} - \text{initial} = -4 - (+12) = -16 \text{ m/s}$, not 8 m/s.

$$a = -16 / 4 = -4 \text{ m/s}^2$$

CONCEPT CHECK

The negative sign does not yet tell you whether the object is speeding up or slowing down. For that, compare the direction of v and a .

THE SAME MOTION IN FOUR LANGUAGES

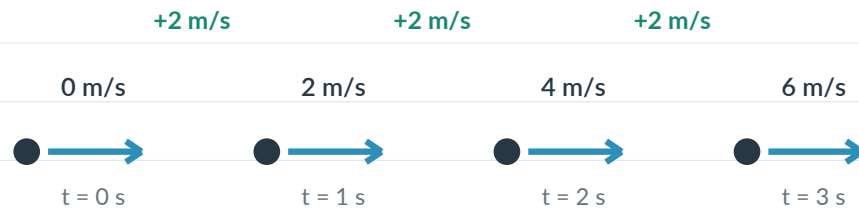
Words ↔ snapshots ↔ graphs

M4A-C4

Start with the story

WORDS

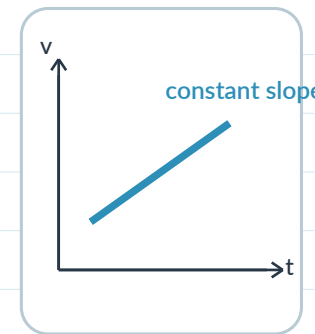
Starts from rest and speeds up uniformly in the positive direction.



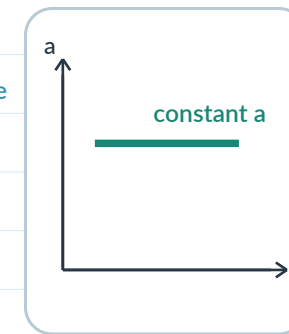
MEMORY LINK

Uniform acceleration $\rightarrow$ equal Δv in equal $\Delta t \rightarrow$ straight-line v - t graph.

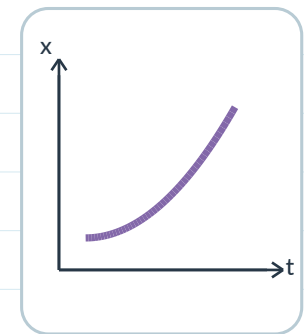
Translate into graphs



v - t : straight



a - t : horizontal



x - t : curves up

WHY x - t CURVES

As the object gets faster, it covers more displacement during each later second. So the position-time graph becomes progressively steeper.

WHY $v = u + at$

Build the relation instead of memorising it

M4A-C5

Say it in ordinary language

THE STORY

Begin with velocity u . Acceleration a is the velocity change per second. If it acts for t seconds, the accumulated velocity change is at .

**final velocity = starting velocity +
velocity gained**

$$\mathbf{v = u + at}$$

QUICK CHECKS

If $a = 0$, then $v = u$. If $t = 0$, then $v = u$. Units: $(\text{m/s}^2)(\text{s}) = \text{m/s}$, so at really is a velocity change.

Rebuild it from the definition

1. Start with acceleration

$$\mathbf{a = (v - u) / t}$$

2. Multiply by t

$$\mathbf{at = v - u}$$

3. Add u

$$\mathbf{v = u + at}$$

WHY THIS MATTERS

If the equation is forgotten, you can rebuild it. That is stronger than recalling a label such as “first equation of motion.”

WORKED EXAMPLE: A CYCLIST SPEEDS UP

Start + Change = Finish

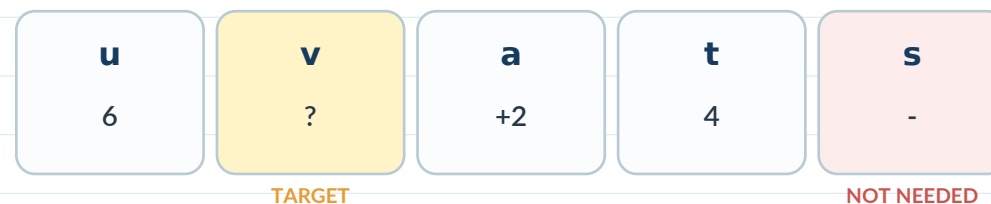
M4A-C5E

See the physical situation

A cyclist is already moving at 6 m/s. For the next 4 s, the cyclist accelerates uniformly at +2 m/s². Find the final velocity.



Compress the same reasoning



$$v = u + at$$

$$v = 6 + (2)(4) = 14 \text{ m/s}$$

REASONING STORY

Start: 6 m/s.

Change: 2 m/s each second $\times$ 4 s = 8 m/s.

Finish: 6 + 8 = 14 m/s.

PHYSICAL CHECK

Velocity and acceleration point the same way, so the speed should increase. 14 m/s is consistent with the motion story.

M4A-C6

The tempting wrong idea

WRONG IDEA

“Acceleration is negative, so the object must be slowing down.”

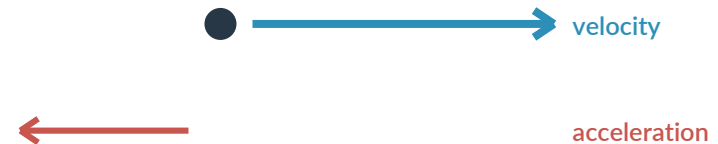
Many classroom braking examples choose the direction of motion as positive. In those examples braking happens to produce a negative acceleration, so the learner can accidentally attach “negative” to “slowing”.

WHAT THE SIGN ACTUALLY MEANS

A negative acceleration only means the acceleration arrow points in the negative direction of the chosen axis.

Case A: the familiar braking case

v points right; a points left



RESULT

The arrows oppose each other, so the speed decreases. This is slowing down - but the reason is the opposite directions, not the negative sign by itself.

M4A-C6

Counterexample: negative a can speed you up

v points left; a also points left

velocity



acceleration



WHY THIS SPEEDS UP

Both arrows point in the same direction. The magnitude of the leftward velocity grows, so the speed increases even though both v and a are negative.

Correct model + retry

CORRECT MODEL

Away from the instant $v = 0$: same sign for v and a → speed increases; opposite signs → speed decreases.

RETRY

A cart has $v = -6 \text{ m/s}$ and $a = -2 \text{ m/s}^2$. Predict the speed change before doing any calculation.

DIAGNOSTIC LINK

If the learner still says “slows”, the failure is at P2 FRAME / sign interpretation, not at algebra.

APPLICATION: HOW LONG UNTIL IT STOPS?

Translate the story before choosing a relation

M4A-C7

A bicycle brakes steadily

A bicycle is moving at 20 m/s. The rider brakes so that the acceleration is a constant 5 m/s^2 opposite the motion. How long until the bicycle stops?



HIDDEN WORD

Stops $\rightarrow v = 0$. Choosing motion to the right as positive makes the braking acceleration negative.

Use the UVATS helper

u	v	a	t	s
20	0	-5	?	-
			TARGET	NOT NEEDED

WHY THIS RELATION?

Time is the target and displacement is not needed. $v = u + at$ connects exactly u , v , a and t .

$$0 = 20 - 5t \rightarrow t = 4 \text{ s}$$

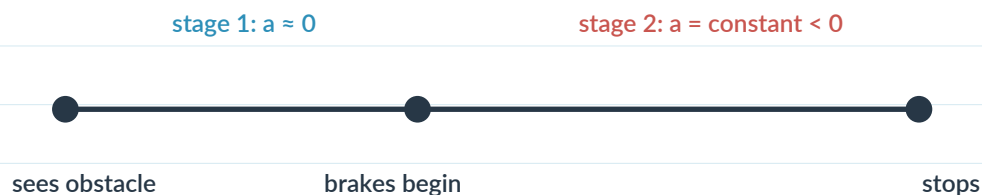
WHEN NOT TO USE ONE CONSTANT- a EQUATION

Split a real journey when the model changes

M4A-C8

A driver reacts, then brakes

A driver sees an obstacle, continues at nearly constant speed for 0.8 s while reacting, and only then brakes with approximately constant acceleration.



WHAT THIS IS NOT

One constant-acceleration interval does not describe the entire reaction + braking journey.

Model gate before formula choice

ASK FIRST

Am I modelling one interval with constant acceleration?

IF NO

Split the journey at the stage boundary. Then apply the correct model to each stage separately.

WHY THIS PREVENTS ERRORS

It stops the learner from treating SUVAT as a universal “moving object” formula.

A REPEATABLE FIRST-STEP ROUTINE

Make the equation obvious

M4A-C9

Five moves before arithmetic

1

SEE + FRAME

Describe the motion; choose a positive direction.

2

REPRESENT

Draw the smallest useful motion strip/arrows.

3

EXTRACT

Write u , v , a , t , s and translate hidden words.

4

MODEL

Check that acceleration is constant for this interval.

5

CHOOSE + CHECK

Circle the target; avoid unnecessary variables; solve and check.

Phrase decoder

starts from rest

$u = 0$

stops / comes to rest

$v = 0$

uniform acceleration

$a = \text{constant}$

slowing down

compare v and a directions

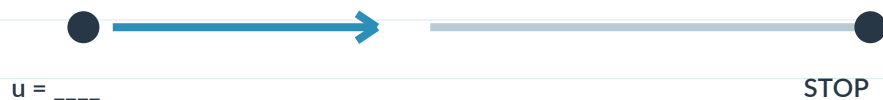
MEMORY QUESTION

Do not ask "Which formula do I remember?" Ask: What is happening?
What is known? What is asked? Which variable can I avoid?

M4A-G1

Question

A scooter moves at 18 m/s and slows uniformly at 3 m/s^2 . How long until it stops?



u ---	v ---	a ---	t ? TARGET	s ---
-----------------	-----------------	-----------------	-------------------------	-----------------

BEFORE HINTS

Choose the positive direction and fill as much of the strip as you can.

Hints - reveal one at a time

H1 - NOTICE

The word stops gives one hidden value.

H2 - MODEL

One constant-a interval: $u = 18, v = 0, a = -3, t = ?$.

H3 - START

Write $0 = 18 - 3t$. Finish the arithmetic yourself.

M4A-G2

Question

A cart is moving left at 12 m/s. Its constant acceleration is 4 m/s^2 to the right. After how long will its velocity become zero?

YOUR JOB

Draw the velocity and acceleration arrows. Choose right as positive. Then write signed u , v , a and the target.

positive direction: _____

$u = \text{_____}$ $v = \text{_____}$ $a = \text{_____}$ $t = \text{_____}$

first relation: _____

Hints - use only if stuck

H1 - NOTICE

Leftward velocity is negative if right is positive.

H2 - MODEL

$u = -12 \text{ m/s}$, $v = 0$, $a = +4 \text{ m/s}^2$.

H3 - START

Write $0 = -12 + 4t$.

NOW YOU TRY + CHECK YOUR UNDERSTANDING

Independent representation first

M4A-CHECK

Independent transfer

A trolley has velocity -5 m/s and constant acceleration -1.5 m/s^2 . Predict whether its speed initially increases or decreases. Then find its velocity

NO DIAGRAM PROVIDED

Draw your own velocity and acceleration arrows. Decide the speed change before calculating.

Quick retrieval

- 1 What does constant acceleration mean in words?
- 2 Why is Δv signed?
- 3 What does at mean physically?
- 4 What does “stops” translate to?
- 5 When must you split a journey before using constant- a equations?

Subtopic self-check

Real-life anchor	PASS
Concept + memory helper	PASS
Signed Δv explanation	PASS
Representation bridge	PASS
$v = u + at$ reconstruction	PASS
Worked example + physical check	PASS
Causal misconception repair	PASS
Model gate	PASS
Guided 1 / Guided 2 fade	PASS
Independent transfer	PASS

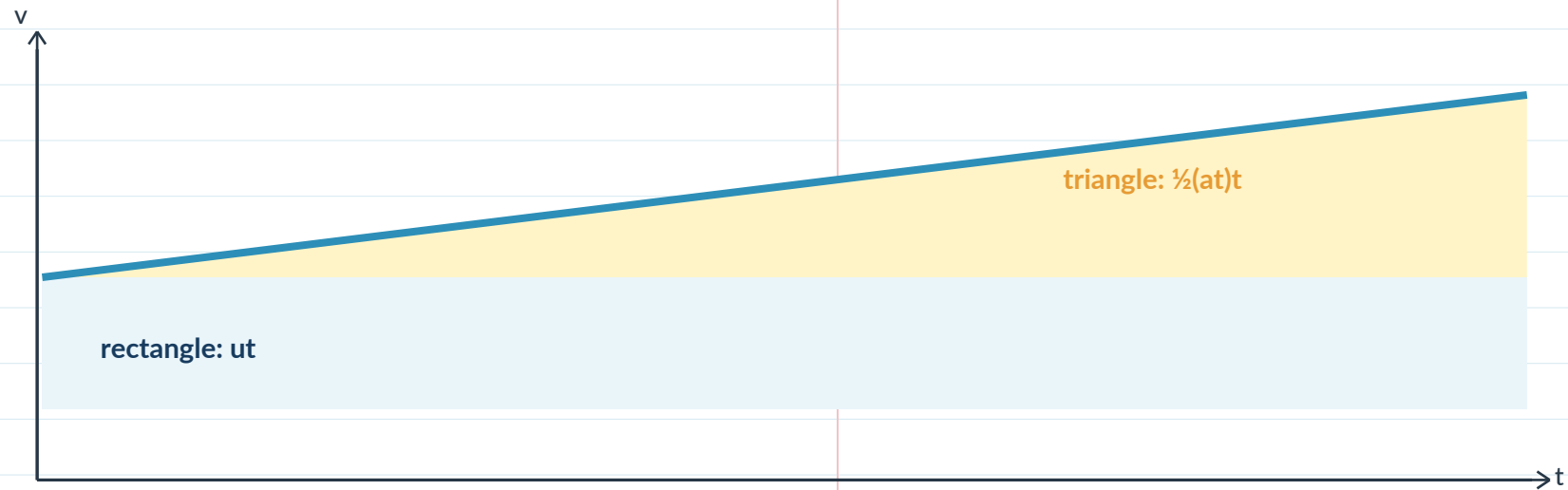
NEXT SUBTOPIC

M-ST04B: displacement under constant acceleration - v - t area and why $s = ut + \frac{1}{2}at^2$.

MOTION IN A STRAIGHT LINE

M-ST04B - Displacement under Constant Acceleration

See displacement as area under v - t , then build $s = ut + \frac{1}{2}at^2$.



MEMORY ANCHOR

Displacement = area under the velocity-time graph.

WHAT WE WILL EXPLAIN

Why the area splits into ut and $\frac{1}{2}at^2$, why $\frac{1}{2}$ appears, and why time is squared.

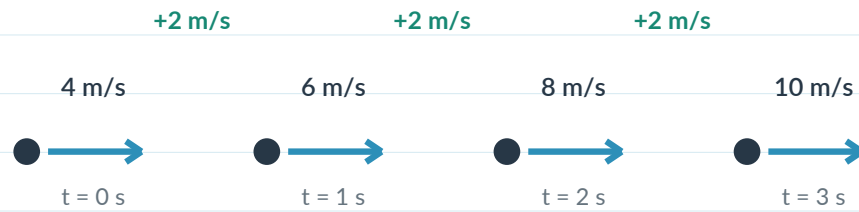
REAL-LIFE STORY: THE CYCLIST COVERS MORE EACH SECOND

Distance grows because velocity grows

M4B-C1

Watch equal one-second intervals

A cyclist starts at 4 m/s and accelerates uniformly at 2 m/s^2 . The velocity at successive seconds is 4, 6, 8, 10 m/s.



What displacement is accumulating?

If velocity were constant at 4 m/s for 3 s, displacement would be $4 \times 3 = 12 \text{ m}$. But acceleration adds extra velocity continuously, so there is additional displacement.

TWO CONTRIBUTIONS

1. Displacement from the starting velocity u .
2. Extra displacement created by acceleration.

PHYSICAL IDEA

Later one-second intervals begin at higher velocities, so the cyclist covers more displacement during later seconds.

MEMORY QUESTION

Can I see the displacement as starting-motion area + acceleration-added area?

DISPLACEMENT IS AREA UNDER v-t

The graph stores accumulation

M4B-C2

Why area has the right units

Velocity has units m/s and time has units s. Multiplying them gives metres, the unit of displacement.

$$(\text{m/s}) \times \text{s} = \text{m}$$

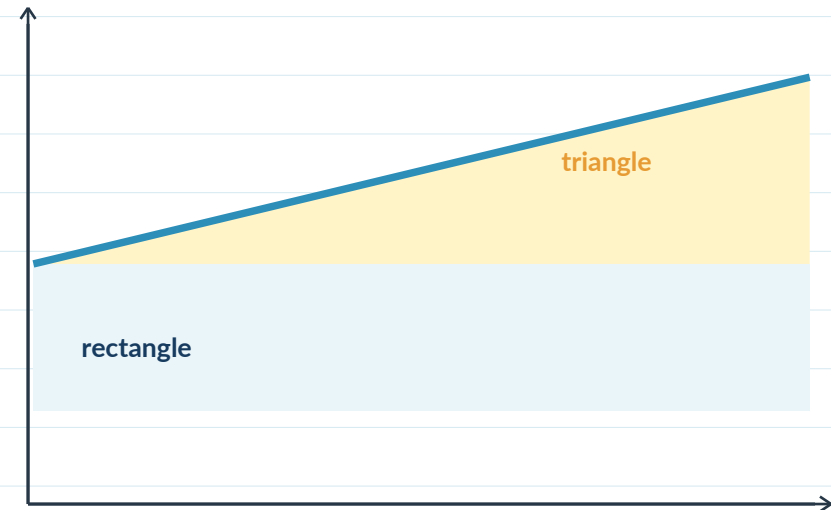
GRAPH HELPER

On a v-t graph, each thin vertical strip is approximately velocity $\times$ a tiny time interval. Adding all strips gives total displacement.

THINGS TO KNOW

Signed area gives displacement. If velocity goes below zero, that portion contributes negative displacement. Total distance is a later graph topic.

For constant acceleration, the graph is simple



VISUAL MEMORY ANCHOR

Constant acceleration turns displacement into rectangle + triangle.

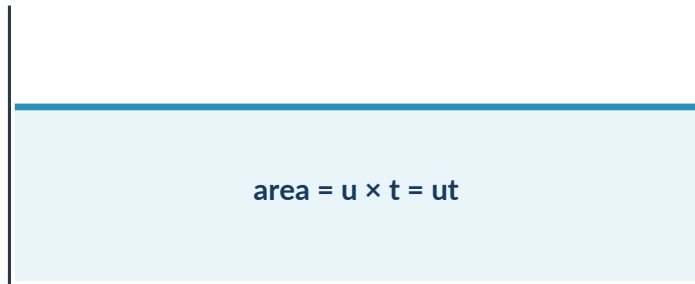
WHY THE TERM ut APPEARS

Starting velocity creates a rectangle

M4B-C3

Imagine acceleration temporarily removed

If the object simply kept its initial velocity u for time t , the v - t graph would be a rectangle of height u and width t .



Then acceleration adds something extra

Because velocity is increasing, the real v - t line rises above that rectangle. The extra region is a triangle.

NEXT QUESTION

What is the triangle's base? What is its height? Answer those two questions and the $\frac{1}{2}at^2$ term becomes inevitable.

PHYSICAL MEANING

ut is the displacement the object would have covered from its starting velocity alone.

HELPER

Baseline motion = rectangle.
Acceleration-added motion = triangle.

WHY $\frac{1}{2}$ AND t^2 APPEAR

The triangle explains both

M4B-C4

Read the triangle

Under constant acceleration, velocity rises from u to v . The triangle above the u -line has base t and height $v - u$.

$$\begin{aligned}\text{triangle area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times t \times (v - u)\end{aligned}$$

Now replace the height

From $v = u + at$, the velocity increase is $v - u = at$.

$$\begin{aligned}\text{extra displacement} &= \frac{1}{2} \times t \times at \\ &= \frac{1}{2}at^2\end{aligned}$$

WHERE $\frac{1}{2}$ COMES FROM

It is not a mysterious constant. The extra region is literally a triangle, whose area is half of base $\times$ height.

WHERE t^2 COMES FROM

One factor of t is the graph's width. The other appears because the velocity increase itself grows as at . Multiplying $t \times at$ gives at^2 .

BUILD THE DISPLACEMENT RELATION

Rectangle + triangle

M4B-C5

Put the two areas together

RECTANGLE

Starting-motion displacement = ut .

TRIANGLE

Acceleration-added displacement = $\frac{1}{2}at^2$.

$$s = ut + \frac{1}{2}at^2$$

Meaning of every term

s

Total displacement during the interval.

ut

Part produced by the initial velocity.

$\frac{1}{2}at^2$

Extra part produced because acceleration changes the velocity during the interval.

WORKED EXAMPLE: STARTING VELOCITY + ACCELERATION

Use the area story and the equation

M4B-C5E

Question

A scooter is moving at 4 m/s and accelerates uniformly at 2 m/s² for 3 s. Find its displacement.

u 4	v -	a +2	t 3	s ?
	NOT NEEDED			TARGET

PREDICT FIRST

The displacement must be more than $4 \times 3 = 12$ m because acceleration adds extra positive area.

Solve and interpret

$$s = ut + \frac{1}{2}at^2$$

$$s = (4)(3) + \frac{1}{2}(2)(3^2)$$

$$s = 12 + 9 = 21 \text{ m}$$

AREA STORY

12 m is the rectangle from the starting velocity. 9 m is the triangular extra displacement created by acceleration.

CHECK

21 m is greater than 12 m, matching our prediction that the object speeds up during the interval.

SPECIAL CASE: STARTS FROM REST

Why displacement scales with t^2

M4B-C6

Set $u = 0$

If the object starts from rest, the rectangle disappears. Only the acceleration-created triangle remains.

$$s = \frac{1}{2}at^2$$

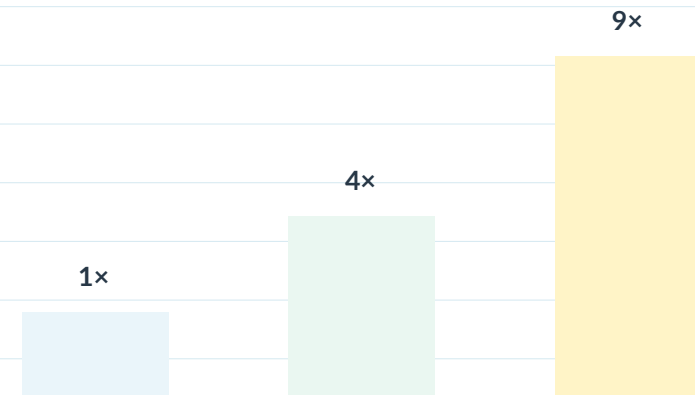
SCALING HELPER

For the same acceleration: double the time $\rightarrow$ four times the displacement.
Triple the time $\rightarrow$ nine times the displacement.

CUMULATIVE PATTERN

From rest, total displacement after 1, 2, 3 seconds scales as 1 : 4 : 9. The distances in individual seconds are a different pattern and belong to a later interval subtopic.

Visualise the growth



WHAT THIS IS NOT

Do not confuse cumulative 1:4:9 with the successive one-second distance pattern 1:3:5. That comes from interval subtraction.

M4B-C7

The tempting shortcut

WRONG IDEA

"The final velocity is 10 m/s for a 3 s journey, so displacement is $10 \times 3 = 30$ m."

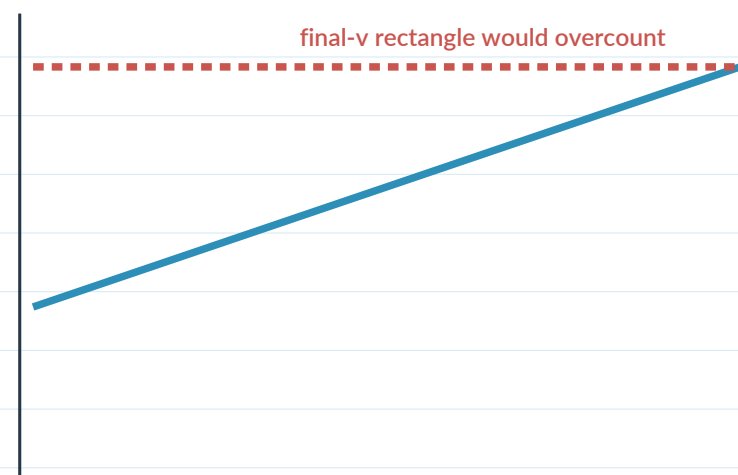
This treats the object as though it moved at 10 m/s for the entire 3 seconds. But under acceleration it spent earlier moments moving more slowly.

WHY IT FAILS

When velocity changes, displacement is not simply final velocity $\times$ time. Use the area under v-t or an appropriate constant-a relation.

Counterexample

Suppose $u = 4$ m/s, $a = 2$ m/s², $t = 3$ s. Then $v = 10$ m/s, but the true displacement is 21 m, not 30 m.



RETRY

If $u = 0$ and acceleration is constant, compare vt with $\frac{1}{2}at^2$. Why is using final $v \times t$ exactly twice the true displacement?

APPLICATION: UNIFORM BRAKING DISTANCE WHEN TIME IS KNOWN

Same equation, negative a

M4B-C8

A bus comes to rest

A bus travels at 20 m/s and comes to rest uniformly in 4 s. How far does it travel while braking?

u	v	a	t	s
20	0	?	4	?

TARGET

FIRST STEP

Find a from $v = u + at$: $0 = 20 + 4a$, so $a = -5 \text{ m/s}^2$.

Then find displacement

$$s = ut + \frac{1}{2}at^2$$

$$s = (20)(4) + \frac{1}{2}(-5)(4^2)$$

PHYSICAL CHECK

Because the bus slows from 20 to 0 m/s, its average speed must be less than 20 m/s. 40 m in 4 s corresponds to 10 m/s average speed - sensible.

CONCEPT LINK

The negative triangle subtracts from the starting-velocity rectangle.

WHEN NOT TO USE $s = ut + \frac{1}{2}at^2$ AS ONE STEP

Check the model first

M4B-C9

The formula assumes one constant-a interval

VALID

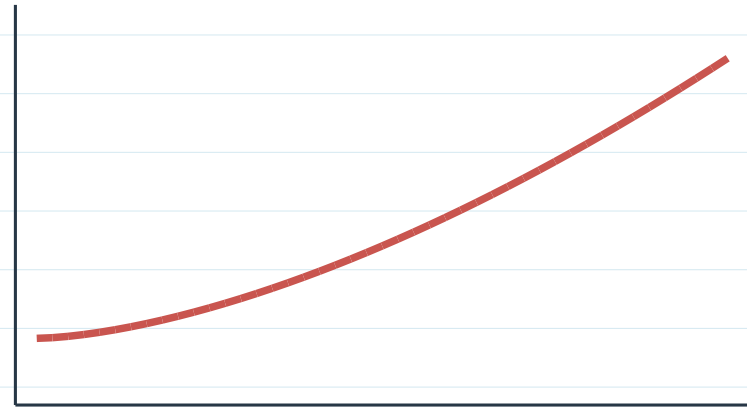
One interval with constant acceleration and a consistent sign convention.

NOT VALID AS ONE STEP

Acceleration changes during the interval, or the journey contains distinct stages such as reaction + braking.

Graph test

If the v-t graph is not a straight line over the interval, acceleration is not constant. Then rectangle + one triangle does not describe the whole area.



MODEL GATE

Straight v-t segment $\rightarrow$ constant a is plausible. Curved v-t $\rightarrow$ acceleration changes; use a different or piecewise model.

GUIDED 1 - FULL SCAFFOLD

Find displacement from a complete model

M4B-G1

Question

A trolley starts at 3 m/s and accelerates uniformly at 2 m/s² for 5 s. Find its displacement.

u 3	v -	a +2	t 5	s ?
	NOT NEEDED			TARGET

BEFORE HINTS

Predict: should the displacement be greater or less than $3 \times 5 = 15$ m?

Hints - reveal one at a time

H1 - NOTICE

Final velocity is not needed.

H2 - MODEL

One constant-a interval with u, a, t and target s.

H3 - START

Write $s = ut + \frac{1}{2}at^2$. Do the substitution yourself.

M4B-G2

Guided 2

A cart starts from rest and accelerates uniformly at 3 m/s^2 for 4 s. Find its displacement.

$u = \text{_____}$ $a = \text{_____}$ $t = \text{_____}$ $s = \text{_____}$

Which part of the v-t area disappears? _____

first relation: _____

Now you try - no helper

A car moves at 8 m/s and accelerates uniformly at 1.5 m/s^2 for 6 s. Without drawing a supplied template, find the displacement and explain which part came from the starting velocity.

TRANSFER EXPECTATION

Choose/draw your own representation, state the model gate, then solve. Explain the two terms physically.

HINT

Starts from rest $\rightarrow u = 0$. The rectangle disappears; the triangle remains.

SELF-CHECK

If you can explain why $\frac{1}{2}$ and t^2 appear, not just substitute numbers, the concept is transferring.

CHECK YOUR UNDERSTANDING

Before moving to the next equation family

M4B-CHECK

Retrieval

- 1
- Why does v-t area represent displacement?
- 2
- What physical contribution does ut represent?
- 3
- Why is there a factor $\frac{1}{2}$?
- 4
- Why does t^2 appear?
- 5
- From rest, how does displacement scale with time?
- 6
- Why is final velocity $\times$ time generally wrong during acceleration?

Subtopic self-check

- Real-life anchor
- PASS
- Area-under-v-t helper
- PASS
- Why ut
- PASS
- Why $\frac{1}{2}$
- PASS
- Why t^2
- PASS
- Equation reconstruction
- PASS
- Worked example
- PASS
- Scaling from rest
- PASS
- Misconception repair
- PASS
- Braking application
- PASS
- When-not-to-use gate
- PASS
- Guided fade + transfer
- PASS

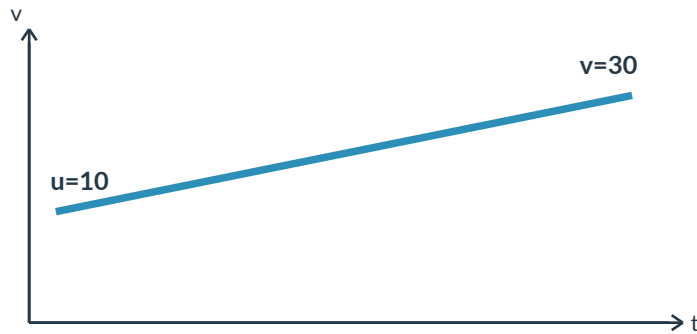
NEXT

Next natural split: average velocity under constant acceleration and/or the time-free relation $v^2 = u^2 + 2as$, depending on the source-question architecture.

MOTION IN A STRAIGHT LINE

M-ST04C - Average Velocity Under Constant Acceleration

Why a linear velocity ramp has a midpoint average - and when that shortcut fails.



LINEAR RAMP

average velocity = midpoint

$$v_{\text{avg}} = (u + v) / 2$$

MEMORY ANCHOR

LINEAR RAMP -> MIDPOINT AVERAGE. This is a constant-acceleration result, not the universal definition of average velocity.

SOURCE FOCUS

Source concept anchors: average velocity under constant acceleration (Q23); velocity-change + average-velocity route (Q33). Gravity examples are deferred to the gravity subtopic.

A CAR SPEEDS UP STEADILY

Real-life anchor before the shortcut

M4C-C1

The dashboard story

A car's velocity rises steadily from 10 m/s to 30 m/s over 5 s. Because the acceleration is constant, the velocity passes through the intermediate values at a uniform rate.



WHAT DO YOU EXPECT?

The car spends equal amounts of time below and above 20 m/s in a symmetric linear ramp. That makes 20 m/s a natural candidate for the average velocity.

But average velocity has a general definition

ALWAYS TRUE

average velocity = displacement / elapsed time. This remains the definition whether acceleration is constant, variable, or zero.

$$v_{\text{avg}} = \text{displacement} / \text{time}$$

SPECIAL SHORTCUT

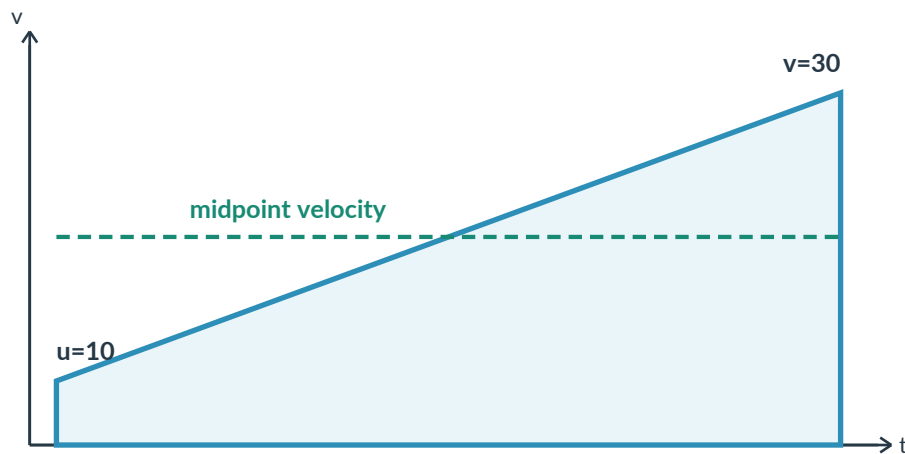
Only after we know the velocity changes linearly may we replace the general definition by the midpoint shortcut.

WHY THE MIDPOINT IS THE AVERAGE

See equal areas around the midpoint

M4C-C2

A linear velocity-time ramp



VISUAL IDEA

For a straight ramp, the part below the midpoint and the part above the midpoint balance. The mean height of the ramp is halfway between the endpoint heights.

Compress the picture

$$v_{\text{avg}} = (u + v) / 2$$

WHY CONSTANT a MATTERS

Constant acceleration makes v - t a straight line. If the graph bends, the midpoint of the endpoints need not equal the true time-average velocity.

UNIT CHECK

u and v are velocities, so their sum and half-sum are still velocities. No new unit appears.

FROM AVERAGE VELOCITY TO DISPLACEMENT

Why acceleration disappears from this relation

M4C-C3

Start with the definition

displacement = average velocity x time

SUBSTITUTE THE CONSTANT-*a* AVERAGE

For a linear velocity ramp, replace average velocity by the midpoint of u and v .

$$s = ((u + v) / 2) t$$

WHAT DISAPPEARED?

Acceleration does not appear explicitly because its effect is already encoded in how far apart u and v are over the interval.

When is this relation efficient?

u
known

v
known

t
known

s
?

USEFUL VARIABLE PATTERN

If u , v and t are known and acceleration is not needed, this is usually the cleanest constant-acceleration relation.

MODEL GATE

Check constant acceleration first. Do not use the endpoint average merely because the problem gives u and v .

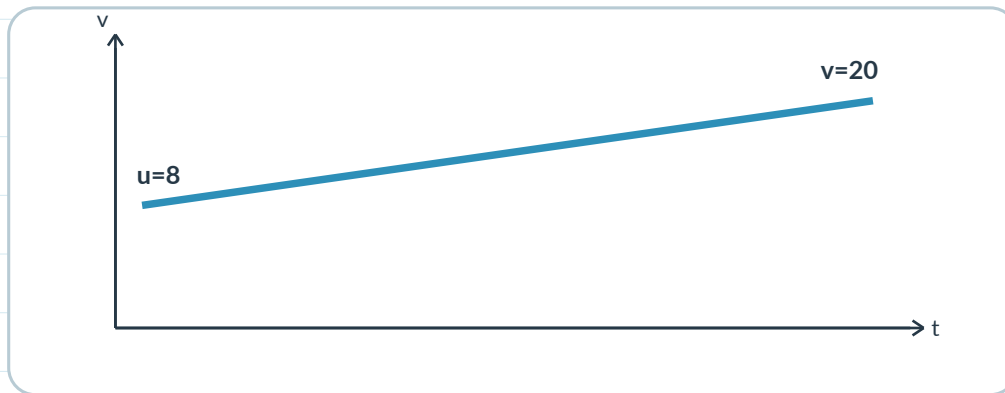
WORKED EXAMPLE: STEADY ACCELERATION

Reason first, compress second

M4C-E1

Question

A cyclist's velocity increases uniformly from 8 m/s to 20 m/s in 6 s. Find the displacement during this interval.



Solution

$$v_{\text{avg}} = (8 + 20) / 2 = 14 \text{ m/s}$$

$$s = 14 \times 6 = 84 \text{ m}$$

PHYSICAL CHECK

84 m in 6 s means an average of 14 m/s, which is sensibly between the initial and final velocities.

PREDICT

The average velocity must lie between 8 and 20 m/s. Because the ramp is linear, it should be exactly the midpoint.

CONCEPT HELPER: LINEAR RAMP -> MIDPOINT

A reusable first-step memory object

M4C-H1

Use this helper when you see...

uniform acceleration

v-t is a straight ramp

u and v both known

endpoint velocities are available

time known, displacement asked

midpoint average can avoid a

same constant-a interval

one model from start to finish

The three-line route

1. MODEL

constant a -> v changes linearly

2. AVERAGE

$$v_{\text{avg}} = (u + v) / 2$$

3. DISPLACEMENT

$$s = v_{\text{avg}} \times t$$

FIRST MOVE

Draw a tiny v-t ramp or write LINEAR RAMP. Then use the midpoint average only if the model gate passes.

WHY THIS HELPS

It keeps the model, the average, and the displacement step separate instead of turning the final relation into another isolated formula.

MISCONCEPTION CLINIC: ENDPOINT AVERAGE IS NOT UNIVERSAL

Same endpoints can hide different histories

M4C-C4

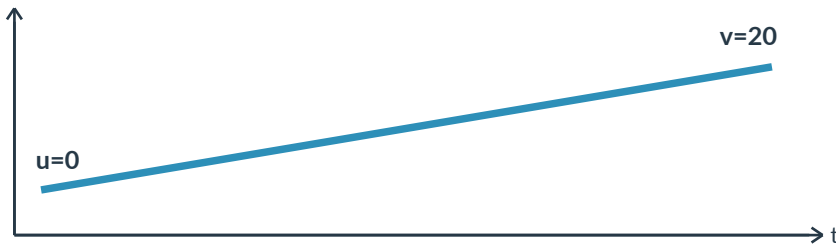
Wrong idea

WRONG IDEA

"If I know the initial and final velocity, the average is always $(u + v)/2$."

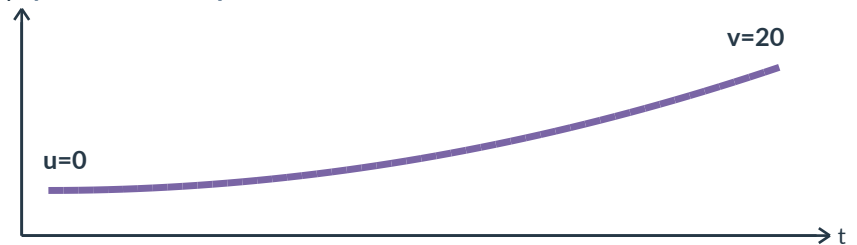
That works for a linear v-t graph. But the endpoints alone do not tell us how much time the object spent at intermediate velocities.

History A: linear ramp



Counterexample and repair

History B: curved ramp



WHY THE AVERAGES DIFFER

The curved graph spends more of the interval at lower velocity, so the area under it is smaller than the area under the straight ramp, even though both begin and end at the same velocities.

CORRECT MODEL

Use $(u+v)/2$ only when velocity varies linearly with time, which is the constant-acceleration condition for the interval.

DISPLACEMENT IS NOT ALWAYS DISTANCE

Signed velocity still matters

M4C-C5

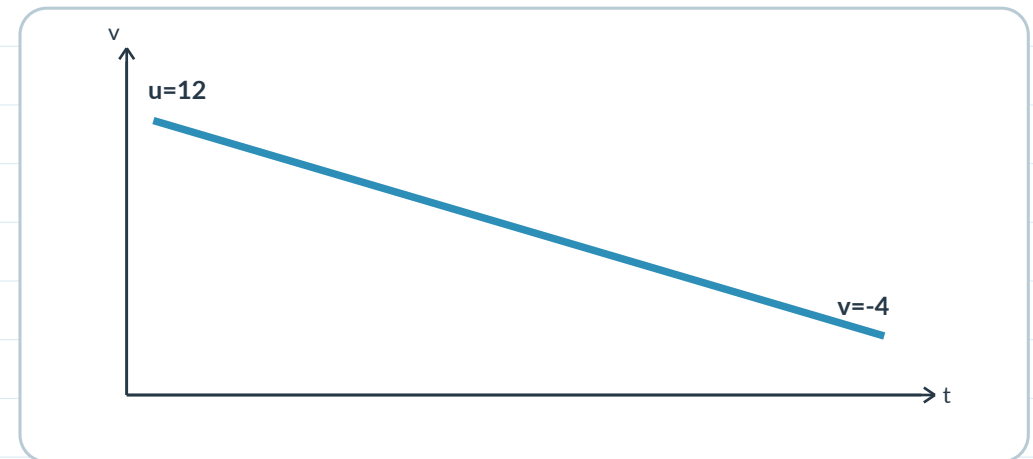
Constant acceleration can cross $v = 0$

Suppose velocity changes uniformly from +12 m/s to -4 m/s in 4 s. The midpoint-average relation is still valid for displacement.

$$v_{\text{avg}} = (12 + (-4)) / 2 = 4 \text{ m/s}$$

$$s = 4 \times 4 = 16 \text{ m}$$

See the sign change



WHAT DOES 16 m MEAN?

It is the signed net displacement over the interval. The object reverses direction, so the total distance travelled is larger than 16 m.

ANTI-TRIGGER

If a question asks for distance and the velocity changes sign, split the v-t area at the zero crossing and add absolute areas. Do not use signed displacement as distance.

APPLICATION: A TRAIN WITH TWO STAGES

Use midpoint average inside the acceleration stage

M4C-C6

The journey

A train starts from rest, accelerates uniformly to 80 km/h during time t , then travels at 80 km/h for another $3t$.

stage 1: 0 → 80 in t

stage 2: 80 for $3t$



STAGE 1 HELPER

Uniform acceleration from 0 to 80 means average speed during stage 1 is $(0+80)/2 = 40$ km/h.

Combine the stage distances

$$s_1 = 40t$$

$$s_2 = 80(3t) = 240t$$

$$v_{\text{avg, total}} = 280t / 4t = 70 \text{ km/h}$$

WHY THIS IS A VARIANT

The midpoint shortcut applies only to the first stage. The whole journey is multistage, so total average speed still requires total distance / total time.

M4C-C7

1. Acceleration is not constant

NO

A curved v-t graph means endpoint midpoint is not generally the time-average.

2. The trip has multiple stages

SPLIT FIRST

Use the shortcut separately only on stages that each have constant acceleration.

3. Distance is asked after reversal

SIGNED AREA WARNING

Average velocity gives displacement. If velocity changes sign, total distance requires absolute path/area accounting.

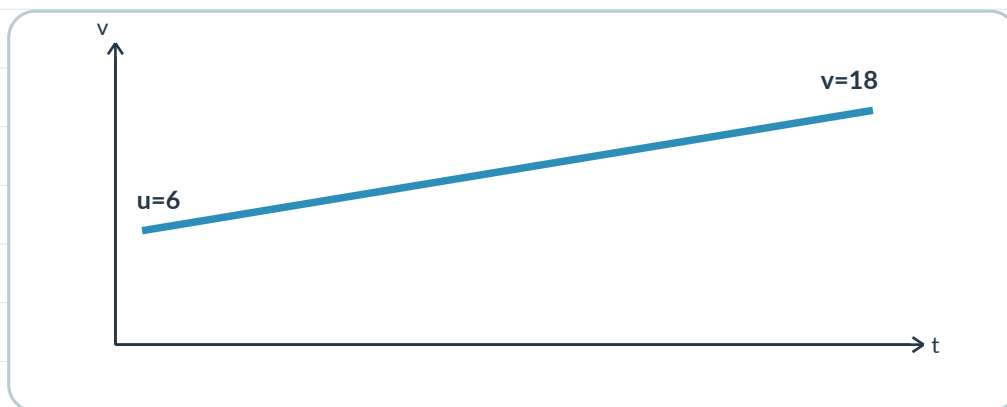
FAST MODEL CHECK

Ask: one interval? constant a? displacement or distance? Only then choose $s = ((u+v)/2)t$.

M4C-G1

Question

A car's velocity increases uniformly from 6 m/s to 18 m/s in 5 s. Find its displacement.



$u = 6$ $v = 18$ $t = 5$ $s = ?$

Hints - cover this column

H1 - NOTICE

"Uniformly" tells you the v-t graph is a straight ramp.

H2 - MODEL

Average velocity = midpoint = $(6+18)/2 = 12$ m/s.

H3 - START

Use $s = v_{\text{avg}} \times t$. Stop after writing $s = 12 \times 5$.

M4C-G2

Question

A cart slows uniformly from 20 m/s to rest in 4 s. Find the displacement during the braking interval.

YOUR JOB

Draw the v-t ramp yourself. Mark u, v and t. Decide whether the midpoint-average shortcut is valid.

model: _____

v_avg = _____

s = _____

Hints - use only if stuck

H1 - NOTICE

Rest means $v = 0$.

H2 - MODEL

Constant acceleration $\rightarrow v_{\text{avg}} = (20+0)/2$.

H3 - START

$s = 10 \times 4$.

NOW YOU TRY + CHECK YOUR UNDERSTANDING

Independent representation first

M4C-CHECK

Independent transfer

A trolley has velocity -6 m/s and constant acceleration. Four seconds later its velocity is $+10 \text{ m/s}$. Find the displacement. Then explain why your answer is not automatically the total distance travelled.

NO DIAGRAM PROVIDED

Decide the sign convention, sketch a v - t ramp, identify the zero crossing qualitatively, and choose the correct quantity: displacement or distance.

Check your understanding

- 1 State the general definition of average velocity.
- 2 Why does $(u+v)/2$ require constant acceleration?
- 3 What does $s=((u+v)/2)t$ calculate?
- 4 When does distance differ from |displacement|?
- 5 How should a multistage journey be handled?

Subtopic self-check

Real-life anchor	PASS
General definition preserved	PASS
Linear-ramp helper	PASS
Graph representation	PASS
Relation reconstruction	PASS
Worked example	PASS
Causal misconception repair	PASS
Distance/displacement anti-trigger	PASS
Multistage variant	PASS
Guided 1 / Guided 2 fade	PASS
Independent transfer	PASS

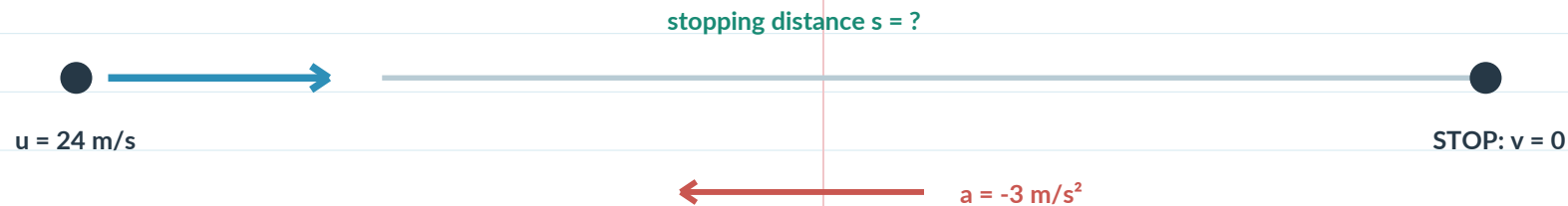
NEXT SUBTOPIC

M-ST04D: remove time - why the velocities become squared in $v^2 = u^2 + 2as$.

MOTION IN A STRAIGHT LINE

M-ST04D - Remove Time: Why Velocities Become Squared

The time-free constant-acceleration relation, stopping distance, and equal-distance v^2 thinking.



MEMORY ANCHOR

TIME ABSENT -> THINK VELOCITY-DISTANCE.
For constant acceleration, $v^2 = u^2 + 2as$ connects velocity directly to displacement.

SOURCE FOCUS

Source anchors: Q15, Q18, Q21 and Q25. The source explicitly derives the squares by eliminating time and then uses the relation for stopping and equal-distance retardation.

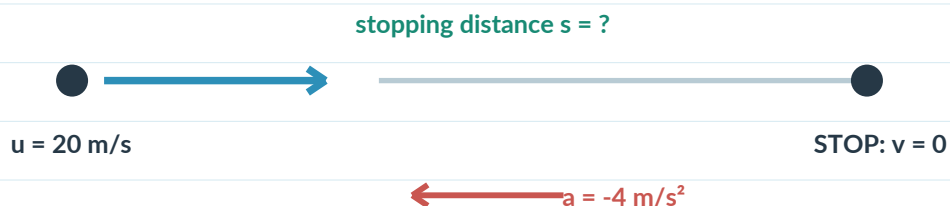
A DRIVER ASKS FOR DISTANCE, NOT TIME

Real-life reason to remove time

M4D-C1

The braking story

A car is moving at 20 m/s and brakes with constant acceleration -4 m/s^2 . The driver wants the stopping distance. The stopping time is not asked for.



WHAT IS KNOWN?

$u = 20$, $v = 0$, $a = -4$, $s = ?$. Time is the unnecessary variable.

The first-step helper

u	v	a	s	t
20	0	-4	?	-

WHY NOT START WITH $v = u + at$?

That relation introduces time. We could solve in two steps, but a time-free relation can connect the variables we actually care about in one step.

BIG QUESTION

Can we remove time from the constant-acceleration model without inventing a new law?

BUILD THE TIME-FREE RELATION

Use two relations you already understand

M4D-C2

Start from earlier subtopics

$$t = (v - u) / a$$

$$s = ((u + v) / 2) t$$

SUBSTITUTE FOR t

Put the time expression into the displacement relation. This is elimination, not a new assumption.

$$s = ((u + v)/2) ((v - u)/a)$$

Now multiply

$$(u + v)(v - u)$$

sum x difference

$$= v^2 - u^2$$

This is why squared velocities appear.

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as$$

WHY THIS MATTERS

You can rebuild the relation from earlier ideas, so the squares no longer need to be memorised as a mysterious pattern.

WHY DO THE VELOCITIES GET SQUARED?

The algebra has a physical job

M4D-C3

The mathematical reason

SUM x DIFFERENCE

Eliminating time multiplies $(u + v)$, which came from the average velocity, by $(v - u)$, which came from the velocity change.

$$(u + v)(v - u) = v^2 - u^2$$

NOT A MEMORY TRICK

The squared velocities appear because of the difference-of-squares identity. If you forget the formula, rebuild it from the two earlier relations.

The physical meaning

VELOCITY-DISTANCE RELATION

For constant acceleration, every displacement s changes v^2 by an amount $2as$. So velocity squared is the quantity that changes linearly with displacement.

$$\Delta(v^2) = 2a \Delta s$$

GRADE 9 CHECK

Can you explain in words why equal displacement intervals under the same acceleration give equal changes in v^2 ?

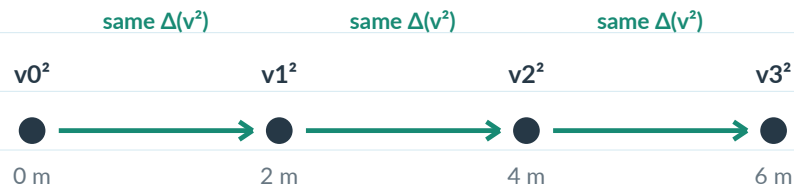
CONCEPT HELPER: EQUAL DISTANCE -> EQUAL CHANGE IN v^2

A powerful recognition pattern

M4D-H1

Imagine equal blocks of distance

A body moves under one constant acceleration through three equal 2 m intervals. Because $2a\Delta s$ is the same for each interval, v^2 changes by the same amount each time.



RECOGNITION CUE

If a question gives equal distances under the same constant acceleration or retardation, compare changes in v^2 before reaching for time.

Why this helps

AVOID EXTRA UNKNOWN TIME

Equal-distance questions often look awkward in t because the time spent in each segment is different. The v^2 relation speaks directly in distance.

SPECIAL RETARDATION VERSION

Under constant retardation, v^2 loses the same amount over equal distances. That is the hidden structure behind repeated penetration / resistance questions.

MEMORY OBJECT

distance steps -> v^2 steps

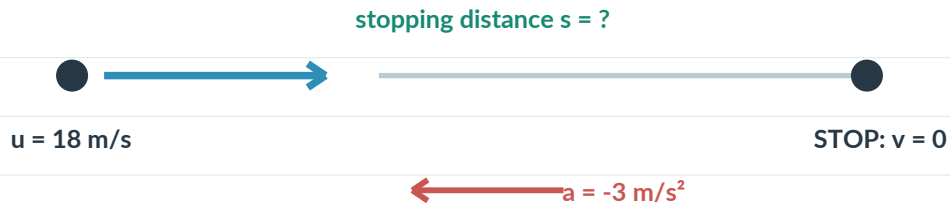
WORKED EXAMPLE: STOPPING DISTANCE

Time never needs to enter the calculation

M4D-E1

Question

A bicycle moves at 18 m/s and brakes uniformly with acceleration -3 m/s^2 . Find the stopping distance.



Solution

$$0^2 = 18^2 + 2(-3)s$$

$$0 = 324 - 6s$$

PHYSICAL CHECK

A moderate deceleration needs several seconds and tens of metres to remove an 18 m/s speed. 54 m is physically plausible.

PREDICT BEFORE CALCULATING

The answer must be positive because it is a forward stopping distance. The negative sign belongs to acceleration under our chosen direction.

STOPPING DISTANCE SCALES WITH SPEED SQUARED

Why twice the speed needs four times the distance

M4D-C4

Use the stop condition

$$0 = u^2 + 2as$$

$$s_{\text{stop}} = u^2 / (2|a|)$$

SAME BRAKING CONDITION

If $|a|$ is unchanged, stopping distance is proportional to u^2 . Double u -> four times the distance. Triple u -> nine times the distance.

Real-life contrast

Car A

speed = 40 km/h

stopping distance = 40 m

Car B

speed = 80 km/h

distance = $4 \times 40 = 160$ m

SAFETY INTERPRETATION

Higher speed is costly because stopping distance grows quadratically under the same braking acceleration.

M4D-C5

Wrong idea

WRONG IDEA

"If speed doubles, the stopping distance should also double."

That would be true for a quantity directly proportional to speed. But stopping distance under fixed braking acceleration depends on speed squared.

WHY IT FEELS PLAUSIBLE

Many everyday quantities scale linearly, so the learner may transfer that expectation without checking the equation.

Counterexample and repair

stopping distance $\propto u^2$

$$u \rightarrow 2u$$

$$u^2 \rightarrow 4u^2$$

$$s \rightarrow 4s$$

square the speed ratio

CORRECT MODEL

Compare the square of the speed ratio when braking conditions are the same.

RETRY

If speed becomes 1.5 times as large, by what factor does the stopping distance change?

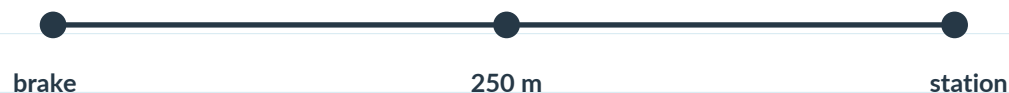
APPLICATION: A TRAIN BRAKES TOO LATE

Use a known stop to infer the new speed

M4D-C6

The story

A train at 20 m/s normally needs 500 m to stop with a fixed braking acceleration. If the brakes are applied only 250 m before the station, the train will still be moving at the station.



FIRST MOVE

Use the full 500 m stopping case to determine the braking acceleration, then apply the same v^2 relation over 250 m.

Why this is a good transfer

HIDDEN STRUCTURE

The acceleration is not given directly, but the first stopping case determines it. The second case reuses the same a with a shorter displacement.

CONCEPT LINK

This is not a new formula. It is the same velocity-distance relation used twice with one shared braking condition.

WHEN NOT TO USE ONE TIME-FREE RELATION

The model gate still comes first

M4D-C7

1. Acceleration changes with distance/time

NO SINGLE CONSTANT- a EQUATION

If resistance or acceleration changes substantially through the interval, $v^2 = u^2 + 2as$ with one constant a is not valid for the whole path.

2. The journey has separate stages

SPLIT FIRST

Reaction time + braking, different road surfaces, or different acceleration stages require separate models before recombination.

3. Time is actually the target

CHOOSE THE CLEAN ROUTE

The time-free relation may be true but inefficient if time is what you need. Use the relation that connects the target while avoiding unnecessary unknowns.

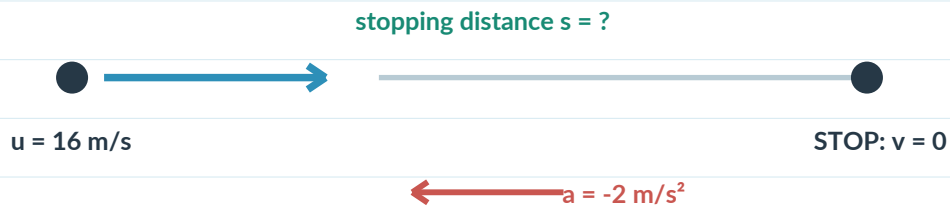
FAST CHECK

Ask: constant a ? one interval? distance/velocity target? time unnecessary?
If yes, the time-free relation is a strong candidate.

M4D-G1

Question

A scooter moving at 16 m/s brakes uniformly at 2 m/s^2 . Find the stopping distance.



u	v	a	s	t
16	0	-2	?	-

Hints - cover this column

H1 - NOTICE

Stops $\rightarrow v = 0$. Time is not asked for.

H2 - MODEL

One constant-acceleration braking interval.

H3 - START

Write $0 = 16^2 + 2(-2)s$.

M4D-G2

Question

A pellet enters a material with speed 30 m/s and experiences constant retardation. After travelling 5 cm its speed is 20 m/s. Without finding the time, determine the extra distance needed to reduce its speed from 20 m/s

YOUR JOB

Draw two distance segments. Use the first segment to determine how much v^2 is lost per centimetre, then apply the same constant-retardation pattern to the next speed change.

segment 1: $30^2 \rightarrow 20^2$ over 5 cm

segment 2: $20^2 \rightarrow 10^2$ over _____ cm

Hints - reveal progressively

H1 - NOTICE

Constant retardation means the same $2a$ factor applies throughout.

H2 - MODEL

Compare changes in v^2 , not changes in v .

H3 - START

First loss in $v^2 = 30^2 - 20^2 = 500$ over 5 cm.

M4D-CHECK

Independent transfer

A train of length L accelerates uniformly. Its engine passes a signal at speed u and its last carriage passes the same signal at speed v . Find the speed when the midpoint of the train passes the signal.

NO DIAGRAM PROVIDED

Recognise that the midpoint corresponds to half the train-length displacement. Use the velocity-distance relation twice; time never needs to appear.

Check your understanding

- 1 Why does eliminating time create squared velocities?
- 2 What changes linearly with displacement under constant a ?
- 3 Why does doubling speed quadruple stopping distance under the same brakes?
- 4 When do equal-distance segments suggest comparing v^2 ?
- 5 What model conditions must hold before using the time-free relation?

Subtopic self-check

Real-life anchor	PASS
Time-free first-step helper	PASS
Equation reconstruction	PASS
Why squares explanation	PASS
Equal-distance v^2 helper	PASS
Worked stopping example	PASS
Stopping scaling	PASS
Causal misconception repair	PASS
Inference variant	PASS
Model gates	PASS
Guided 1 / Guided 2 fade	PASS
Independent transfer	PASS

NEXT SUBTOPIC

M-ST05A: distance in the n th second - why interval displacement is not total displacement.

MOTION IN A STRAIGHT LINE

M-ST04E - Read a Motion Equation from Its Coefficients

SOURCE GAP REPAIR | Q24 and Q26: read x_0 , u and a before doing any algebra.

$$x(t) = x_0 + ut + \frac{1}{2}at^2$$

constant term $\rightarrow x_0$

coefficient of $t \rightarrow u$

coefficient of $t^2 \rightarrow a/2$

then use $v = u + at$ if velocity is asked

MEMORY ANCHOR

Match coefficients to the standard shape. $x(t) = x_0 + ut + \frac{1}{2}at^2$.

SOURCE FOCUS

Supplied source: Unit 4.4, Q24 and Q26.

M4E-C1

Standard shape

$$x(t) = x_0 + ut + \frac{1}{2} at^2$$

READ OFF

constant $\rightarrow x_0$; coefficient of $t \rightarrow u$; coefficient of $t^2 \rightarrow a/2$.

Example

$$x = 5t^2 - 4t + 5$$

TRANSLATE

$x_0 = 5$ m, $u = -4$ m/s, $a = 10$ m/s². The negative u tells the initial direction.

M4E-C2

WRONG IDEA

"Any $x(t)$ polynomial can be read using $x = x_0 + ut + \frac{1}{2}at^2$."

Functional helper

$$x(t) = x_0 + ut + \frac{1}{2}at^2$$

constant term $\rightarrow x_0$

coefficient of $t \rightarrow u$

coefficient of $t^2 \rightarrow a/2$

then use $v = u + at$ if velocity is asked

REPAIR

The coefficient shortcut applies when the position equation is exactly quadratic in time and represents constant acceleration. A t^3 term signals a different model.

CHECK

After reading a , verify that it is constant and that units match.

M4E-G

Guided 1

$x(t)=3+6t+2t^2$. Find initial position, initial velocity and acceleration.

$$x(t) = x_0 + ut + \frac{1}{2} at^2$$

constant term -> x_0

coefficient of t -> u

coefficient of t^2 -> $a/2$

then use $v = u + at$ if velocity is asked

Guided 2 - faded

$x(t)=7-5t+4t^2$. Find velocity at $t=3$ s without differentiating.

$u =$ _____

$a =$ _____

$v(3) =$ _____

H1 -> H2 -> H3

H1: compare standard shape. H2: $u=6$. H3: $\frac{1}{2}a=2 \rightarrow a=4$.

H1 -> H2 -> H3

Read $u=-5$ and $a=8$; then use $v=u+at$.

M4E-T

Given a quadratic position-time expression, first identify x_0 , u and a . Only then decide whether the question asks position, velocity, acceleration or a meeting condition.

ANTI-TRIGGER

If a t^3 or other non-quadratic term appears, do not silently force the constant-acceleration coefficient map.

Self-check

- 1 I match coefficients correctly.
- 2 I remember coefficient of t^2 is $a/2$.
- 3 I use signs as directions.
- 4 I verify the equation is quadratic in time.
- 5 I can use the read-off values in $v=u+at$.

MOTION IN A STRAIGHT LINE

M-ST05A - Distance During the nth Second

Why an interval is not a total - and how $S(n) - S(n-1)$ becomes the first step.



MEMORY ANCHOR

"during the nth second" = interval $(n-1) \rightarrow n$.
Start with cumulative displacement and subtract the two endpoints.

SOURCE FOCUS

The supplied Motion source maps Q16 directly to nth-second interval reasoning. Q20 and Q27 also use interval subtraction from cumulative displacement.

A STOPWATCH CAN HIDE TWO DIFFERENT QUESTIONS

Total time vs one-second interval

M5A-C1

These phrases are not the same

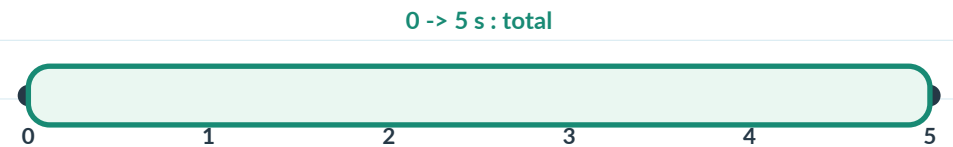
QUESTION A

"How far does the object travel in 5 seconds?"
This means the whole interval 0 -> 5 s.

QUESTION B

"How far does the object travel during the 5th second?"
This means only 4 -> 5 s.

See the difference on one timeline



FIRST RECOGNITION CUE

Words such as during the 7th second, in the nth second, or next t seconds are interval language.

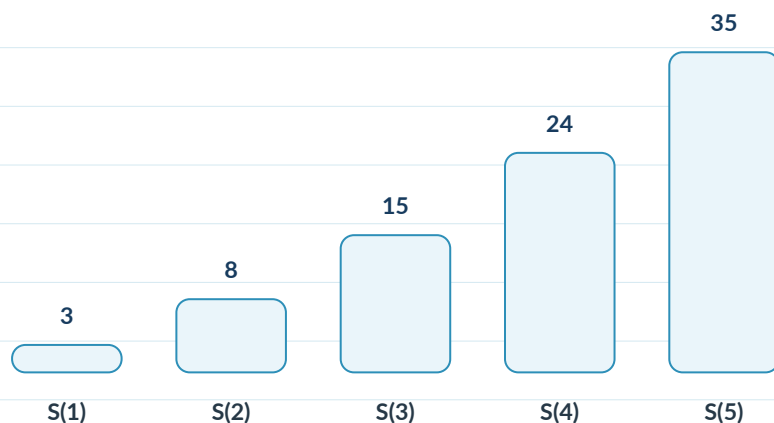
CUMULATIVE DISPLACEMENT IS A RUNNING TOTAL

Build the subtraction idea visually

M5A-C2

Think like a trip meter

Suppose the object's cumulative displacement from the start is $S(t)$. Then $S(5)$ includes everything from 0 to 5 s, while $S(4)$ includes everything from 0 to 4 s.



REMOVE WHAT CAME BEFORE

The displacement during the fifth second is the new total minus the old total.

The universal interval helper

$$\text{interval displacement} = S(\text{end}) - S(\text{start})$$

$$s_5 = S(5) - S(4)$$

WHY THIS IS BETTER THAN A SPECIAL FORMULA

The subtraction idea works for any interval. The n th-second formula is only a shortcut that comes from applying this idea to constant acceleration and a one-second interval.

DERIVE THE n th-SECOND RELATION

Start from $S(t)$, then subtract

M5A-C3

For constant acceleration

$$S(t) = ut + \frac{1}{2} at^2$$

THE n th SECOND

It begins at $t = n-1$ and ends at $t = n$. So take the cumulative displacement at those two times and subtract.

$$s_n = S(n) - S(n-1)$$

Simplify the subtraction

$$s_n = [un + \frac{1}{2} an^2] - [u(n-1) + \frac{1}{2} a(n-1)^2]$$

$$s_n = u + (a/2)(2n - 1)$$

WHAT THE FORMULA MEANS

Every later one-second interval starts with a different velocity. Under constant acceleration, the interval displacement therefore changes in a regular arithmetic pattern.

WORKED EXAMPLE: THE 4th SECOND

Use the timeline before the formula

M5A-E1

Question

A particle has $u = 2 \text{ m/s}$ and constant acceleration $a = 3 \text{ m/s}^2$. Find its displacement during the 4th second.



Two equivalent routes

ROUTE A - SUBTRACT

Compute $S(4) - S(3)$. This is the most transferable method.

$$S(4) = 32, S(3) = 19.5 \rightarrow s_4 = 12.5 \text{ m}$$

ROUTE B - SHORTCUT

$$s_4 = u + (a/2)(2 \cdot 4 - 1) = 2 + 1.5 \cdot 7 = 12.5 \text{ m.}$$

FIRST MOVE

Do not put $t = 4$ into $S(t)$ and stop. That gives the total from 0 to 4 s.

CONCEPT HELPER: TOTAL -> INTERVAL

A reusable three-step route

M5A-H1

When you see interval language

1

MARK THE CLOCK

Write the actual interval. "5th second" -> 4 to 5 s.

2

WRITE THE TOTAL

Use $S(t)$ for cumulative displacement from the start.

3

SUBTRACT

interval = later total - earlier total.

Recognition phrases

during the n th second

$(n-1) \rightarrow n$

during the 11th second

10 -> 11

between 3 s and 5 s

$S(5) - S(3)$

in the next t seconds

$S(2t) - S(t)$

in the first t seconds

$S(t) - S(0) = S(t)$

MEMORY PHRASE

TOTAL -> TOTAL -> DIFFERENCE.

MISCONCEPTION CLINIC: $t = n$ IS NOT THE n th SECOND

Repair the interval mistake

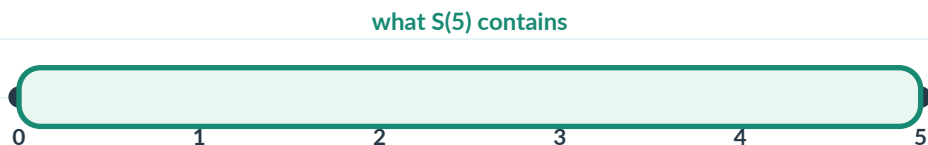
M5A-C4

Wrong idea

WRONG IDEA

"The question says 5th second, so I put $t = 5$ into $S(t)$."

That calculates the displacement accumulated from the start up to 5 s. It includes the first, second, third and fourth seconds too.



Counterexample and repair

what the 5th second contains



CORRECT MODEL

The 5th second is one one-second slice: $4 \rightarrow 5$ s. Therefore use $S(5) - S(4)$.

RETRY

What interval does "during the 11th second" mean?

APPLICATION: TWO n th-SECOND DISTANCES REVEAL a

Subtract the shortcut equations

M5A-C5

The pattern

Suppose the displacement during the n th second is known, and the displacement two seconds later is also known.

$$s_n = u + (a/2)(2n - 1)$$

$$s_{(n+2)} = u + (a/2)(2n + 3)$$

SUBTRACT THEM

u and n cancel cleanly: $s_{(n+2)} - s_n = 2a$. This is much faster than solving separately for u and n .

Numerical illustration

$$115.0 - 102.5 = 12.5 = 2a$$

$$a = 6.25 \text{ m/s}^2$$

WHY THIS IS DIFFICULT

The arithmetic is easy. The challenge is recognising that both data are interval displacements and choosing subtraction that cancels the hidden quantities.

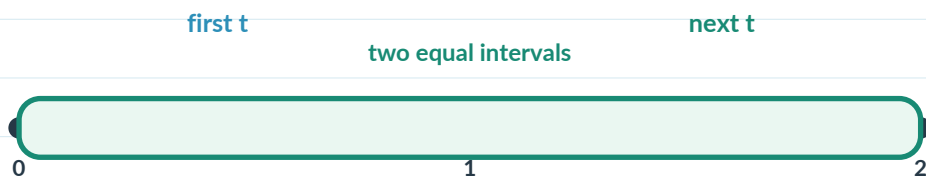
CLOSE VARIANT: THE NEXT t SECONDS

Same interval idea, a different width

M5A-C6

From rest with constant acceleration

A toy starts from rest and travels 10 m in the first t seconds. How far does it travel in the next t seconds?



Use the cumulative law

$$S(t) = \frac{1}{2} at^2 = 10$$

$$S(2t) = \frac{1}{2} a(2t)^2 = 40$$

$$\text{next } t = 40 - 10 = 30 \text{ m}$$

DO NOT READ "NEXT" AS TOTAL

The second piece is $S(2t) - S(t)$, not $S(2t)$.

CONNECTS FORWARD

This is also the doorway to the 1:3:5:7 successive-interval pattern, which will be developed in the next subtopic.

WHEN THE n th-SECOND SHORTCUT IS NOT SAFE

The subtraction idea is broader than the formula

M5A-C7

1. Acceleration is not constant

USE $S(\text{end})-S(\text{start})$, NOT THE SHORTCUT

The compact s_n formula comes from $S(t)=ut+\frac{1}{2}at^2$, so it assumes constant acceleration.

2. The interval is not one second

ADAPT THE CLOCK

For 2.5 $\rightarrow$ 4.0 s, use $S(4.0)-S(2.5)$. Do not force the n th-second formula.

3. Distance vs displacement after reversal

SIGN WARNING

$S(\text{end})-S(\text{start})$ gives signed displacement. If the object reverses during the interval and the question asks distance, split the motion at the turning point.

FAST CHECK

Constant a ? one-second interval? no reversal if distance is asked? If yes, the compact n th-second formula is safe.

M5A-G1

Question

A particle starts with $u = 4 \text{ m/s}$ and moves with constant acceleration 2 m/s^2 . Find the displacement during the 6th second.

6th second = 5 $\rightarrow$ 6 s



Hints - cover this column

H1 - NOTICE

"6th second" means 5 $\rightarrow$ 6 s.

H2 - MODEL

Constant acceleration, so $S(t) = ut + \frac{1}{2}at^2$ is valid.

H3 - START

Write $s_6 = S(6) - S(5)$.

BEFORE HINTS

Write the interval and decide whether you will use $S(6) - S(5)$ or the shortcut.

M5A-G2

Question

A uniformly accelerated body travels 54 m during its n th second and 66 m during its $(n+2)$ th second. Find the acceleration.

YOUR JOB

Do not solve for n or u unless you must. Write the two n th-second expressions and look for quantities that disappear when you subtract.

$s_n =$ _____

$s_{(n+2)} =$ _____

subtract: _____

Hints - reveal progressively

H1 - NOTICE

The two intervals are two seconds apart.

H2 - MODEL

$$s_{(n+2)} - s_n = 2a.$$

H3 - START

$$66 - 54 = 2a.$$

NOW YOU TRY + SELF-CHECK

Independent interval reasoning

M5A-CHECK

Independent transfer

A body starts from rest with constant acceleration. It covers 18 m in the first 3 s. Without using the nth-second shortcut directly, find the displacement during the next 3 s.

NO TIMELINE PROVIDED

Build the two intervals 0 → 3 and 3 → 6. Use cumulative displacement and subtract.

Check your understanding

- 1 What interval is the 11th second?
- 2 Why is $S(5)$ not the displacement during the 5th second?
- 3 How do you derive s_n from $S(t)$?
- 4 When does $s_{(n+2)} - s_n$ simplify to $2a$?
- 5 When can interval displacement differ from distance?

Subtopic self-check

Total-vs-interval distinction	PASS
Cumulative subtraction helper	PASS
nth-second derivation	PASS
Worked example	PASS
Causal misconception repair	PASS
Two-interval cancellation	PASS
Next-t interval variant	PASS
Model gates	PASS
Guided 1 / Guided 2 fade	PASS
Independent transfer	PASS
Typography/render QA	PENDING

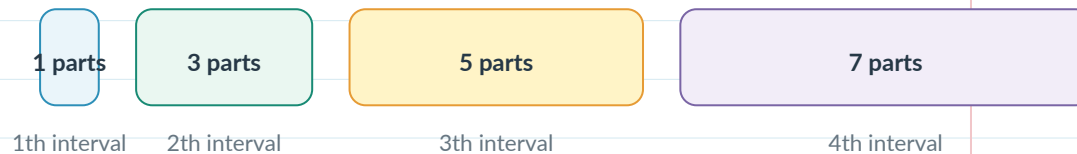
NEXT SUBTOPIC

M-ST05B: why successive equal-time distances from rest follow 1:3:5:7.

MOTION IN A STRAIGHT LINE

M-ST05B - Why Successive Distances Follow 1 : 3 : 5 : 7

From rest + constant acceleration + equal time intervals -> odd-number distance pattern.



MEMORY ANCHOR

SQUARE TOTALS -> ODD DIFFERENCES.
1,4,9,16 cumulative units produce 1,3,5,7 successive interval units.

SOURCE FOCUS

The supplied Motion Concept Book explicitly derives the 1:3:5:7 pattern from $S(t) \propto t^2$ and maps it to Q17, Q32 and Q59.

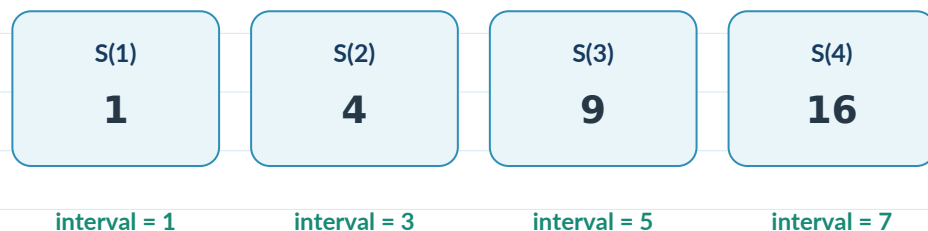
A TOY CAR FROM REST

See the pattern before the algebra

M5B-C1

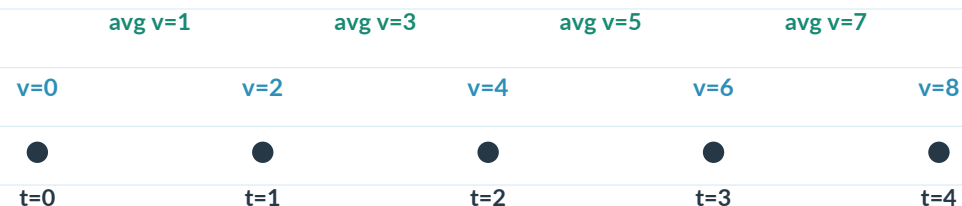
Imagine equal 1-second snapshots

A toy car starts from rest and accelerates uniformly. Let $a = 2 \text{ m/s}^2$. Then $S(t) = t^2$ metres.



Why the gaps get bigger

The car enters each new second with a larger velocity than before. Equal-length time intervals therefore contain larger average velocities.



WHAT THE CAR DOES

The first second adds 1 m, the second adds 3 m, the third adds 5 m and the fourth adds 7 m.

PHYSICAL STORY

The acceleration adds the same amount of velocity each second, so the average velocity of each successive second rises regularly.

SQUARE TOTALS -> ODD DIFFERENCES

The shortest derivation

M5B-C2

From rest

$$S(t) = \frac{1}{2} at^2$$

IGNORE THE COMMON FACTOR

For comparison, every cumulative displacement has the same factor $a/2$.
The time-squared part is what creates the pattern.

$$1^2, 2^2, 3^2, 4^2 \rightarrow 1, 4, 9, 16$$

Take first differences

$$1-0=1$$

$$4-1=3$$

$$9-4=5$$

$$16-9=7$$

THE LINK

A quadratic total produces an arithmetic sequence of first differences.
Physics gives the quadratic law; algebra exposes the odd numbers.

THE SAME RESULT FROM THE n th-SECOND FORMULA

[Connect to the previous subtopic](#)

M5B-C3

Start with the interval formula

$$s_n = u + (a/2)(2n - 1)$$

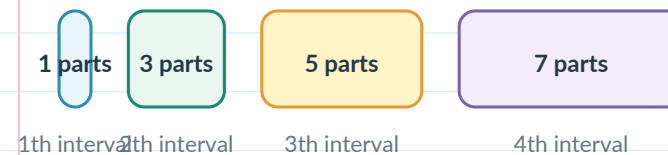
NOW USE "STARTS FROM REST"

From rest, $u = 0$. So $s_n = (a/2)(2n-1)$.

$$s_n \propto 2n - 1$$

Substitute $n = 1, 2, 3, 4$

$$2n-1 \rightarrow 1, 3, 5, 7$$



WHY TWO DERIVATIONS HELP

One route comes from square totals; the other comes from the n th-second formula. They must agree because they describe the same motion.

CONCEPT HELPER: FIRST INTERVAL SETS THE SCALE

Turn the ratio into a quick tool

M5B-H1

If the first-second distance is d

$$1\text{st} : 2\text{nd} : 3\text{rd} : 4\text{th} = 1 : 3 : 5 : 7$$

$$d, 3d, 5d, 7d, \dots$$

FAST USE

If a body from rest covers 4 m in the first second under constant acceleration, then the third-second distance is $5 \times 4 = 20$ m.

A more general equal-time version

The intervals do not have to be exactly one second. If the body starts from rest and acceleration is constant, successive equal time blocks Δt also follow 1:3:5:7.

$$S(k\Delta t) \propto k^2$$

RECOGNITION CUE

Look for starts from rest + constant acceleration + successive equal time intervals.

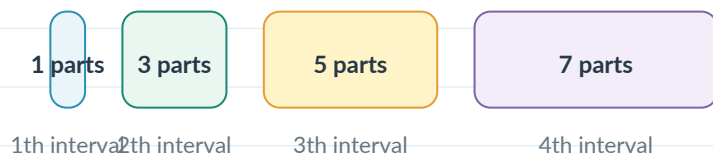
WORKED EXAMPLE: FIND A LATER INTERVAL FAST

Use the ratio, then check with $S(t)$

M5B-E1

Question

A body starts from rest with constant acceleration and covers 6 m during the first second. How far does it cover during the fourth second?



FIRST MOVE

The motion meets all three conditions for the odd-number pattern.
Fourth-second factor = 7.

Solution

$$s_4 = 7 \times s_1$$

$$s_4 = 7 \times 6 = 42 \text{ m}$$

PHYSICAL CHECK

A steadily accelerating body should cover much more distance in the fourth second than in the first. 42 m versus 6 m is consistent with increasing velocity.

M5B-C4

Wrong idea

WRONG IDEA

"Under constant acceleration, successive-second distances are always 1:3:5:7."

Constant acceleration alone is not enough. A nonzero initial velocity adds the same u contribution to every one-second interval.

$$s_n = u + (a/2)(2n-1)$$

Counterexample and repair

COUNTEREXAMPLE

Let $u=10$ m/s and $a=2$ m/s². Successive one-second displacements are 11, 13, 15, 17 m - an arithmetic sequence, but not 1:3:5:7.

CORRECT MODEL

The pure odd-number ratio needs $u=0$, constant acceleration, and equal time intervals measured successively from the start.

RETRY

Would 1:3:5:7 apply to a car already moving at 8 m/s when the clock starts?

APPLICATION: RECOVER THE TOTAL FROM ONE INTERVAL

Odd pattern + square total

M5B-C5

Question pattern

A body starts from rest with constant acceleration. Its displacement during the fourth second is 14 m. Find the total displacement in the first four

STEP 1 - FIND THE SCALE

Fourth-second factor is 7, so the first-second distance d satisfies $7d = 14$.
Hence $d = 2$ m.

STEP 2 - USE SQUARE TOTAL

After four equal intervals, cumulative distance factor is $4^2 = 16$. Total = $16d = 32$ m.

Why this is useful

interval factors: 1,3,5,7

total factors: 1,4,9,16

TWO REPRESENTATIONS, ONE MOTION

The interval sequence and cumulative square sequence are two views of the same quadratic displacement law.

WHEN THE ODD-NUMBER SHORTCUT FAILS

Use the full interval model instead

M5B-C6

1. Initial velocity is not zero

USE s_n , NOT 1:3:5

A nonzero u shifts every interval displacement.

2. Acceleration changes

NO FIXED ODD PATTERN

The quadratic total law no longer holds across the whole interval.

3. Time blocks are unequal

COMPARE CUMULATIVE TOTALS

1:3:5:7 belongs to successive equal-duration blocks. For unequal widths, evaluate $S(\text{end}) - S(\text{start})$ directly.

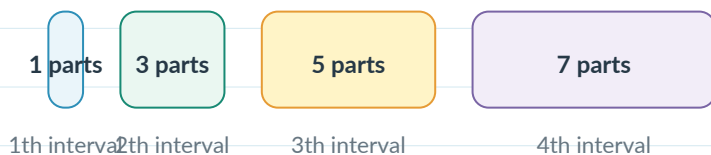
FAST GATE

Starts from rest? Constant a ? Equal successive time blocks? If all three are yes, the odd-number pattern is available.

M5B-G1

Question

A particle starts from rest under constant acceleration. It covers 3 m in the first second. Find the displacement during the fourth second.



BEFORE HINTS

Check the three conditions for the pattern before using it.

Hints - cover this column

H1 - NOTICE

Starts from rest + constant acceleration + equal 1 s blocks.

H2 - MODEL

1st : 4th = 1 : 7.

H3 - START

$s_4 = 7 \times 3.$

M5B-G2

Question

A body starts from rest with constant acceleration. It travels 20 m during the third second. Find the displacement during the fifth second.

YOUR JOB

Write the odd-number factors for the third and fifth seconds. Do not calculate acceleration unless you need it.

3rd-second factor = _____

5th-second factor = _____

ratio / answer = _____

Hints - reveal progressively

H1 - NOTICE

Third-second factor = 5.

H2 - MODEL

Fifth-second factor = 9.

H3 - START

$s_5 / s_3 = 9/5$.

M5B-CHECK

Independent transfer

A trolley starts from rest with constant acceleration. During the first 2 s it covers 8 m. Find how far it travels during the next 2 s. Solve once using the equal-interval odd-number pattern and once using cumulative displacement.

NO DIAGRAM PROVIDED

The equal intervals are 2 s wide, not 1 s. Decide whether the same 1:3 pattern still applies and justify it.

Check your understanding

- 1 Why do cumulative totals follow 1,4,9,16 from rest?
- 2 Why do successive differences become 1,3,5,7?
- 3 What three conditions are required for the pure odd ratio?
- 4 What happens if u is not zero?
- 5 Why can equal 2 s intervals still follow the same odd pattern?

Subtopic self-check

Real-life pattern	PASS
Square-to-odd derivation	PASS
n th-second link	PASS
Pattern helper	PASS
Worked example	PASS
Causal misconception repair	PASS
Recover-total variant	PASS
Model gates	PASS
Guided 1 / Guided 2 fade	PASS
Independent transfer	PASS
Typography/render QA	PENDING

NEXT SUBTOPIC

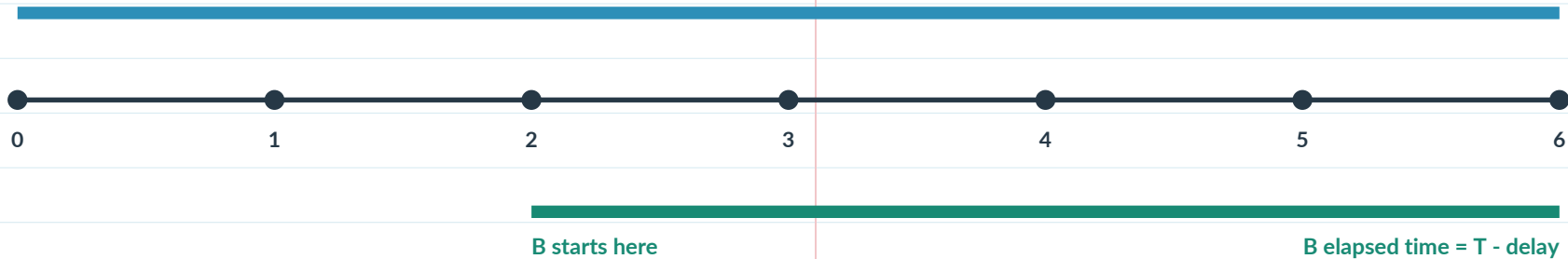
M-ST06A: delayed start and one common clock.

MOTION IN A STRAIGHT LINE

M-ST06A - Delayed Start and One Common Clock

Same global instant does not mean same elapsed time.

A active: elapsed time = T



MEMORY ANCHOR

GLOBAL TIME $\rightarrow$ LOCAL ELAPSED TIME.
If A has moved for T and B starts τ later, B has moved for $T - \tau$.

SOURCE FOCUS

The supplied Motion Concept Book makes the timeline the core model and anchors delayed-start reasoning to Q19.

TWO RUNNERS, ONE STARTS LATE

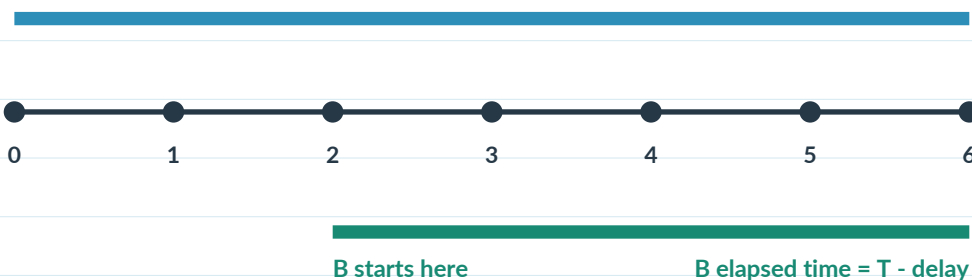
See the clock mismatch

M6A-C1

Real-life story

Runner A starts at the whistle. Runner B waits 2 s and then starts. At global time $T=6$ s, both are observed at the same instant.

A active: elapsed time = T



The common-clock translation

$$t_A = T$$

$$t_B = T - \tau$$

DO THIS BEFORE ANY KINEMATICS

Write each body's elapsed time under the same global clock. The equations are usually easy after this translation.

THE HIDDEN FACT

A has been moving for 6 s. B has been moving for only 4 s.

WHY USING THE SAME t FOR BOTH BODIES FAILS

A causal misconception repair

M6A-C2

Wrong idea

WRONG IDEA

"They are compared at the same instant, so both use the same t ."

The symbol t in a motion equation means elapsed time since that body's stage began, not automatically the global clock reading.

WHY IT FEELS PLAUSIBLE

Both positions are evaluated at one global instant, so it is easy to confuse clock time with motion duration.

Repair with a timeline

A active: elapsed time = T



B starts here

B elapsed time = $T - \text{delay}$



CORRECT MODEL

Use one global clock to locate the event, then convert that clock reading into each object's own elapsed time.

APPLICATION: SAME a , DIFFERENT START TIMES

Ratios become squared-time ratios

M6A-C3

Both start from rest with the same a

$$S_A = \frac{1}{2} aT^2$$

$$S_B = \frac{1}{2} a(T-\tau)^2$$

RATIO SHORTCUT

The common factor $a/2$ cancels. Position ratios reduce to squared elapsed-time ratios.

Example

A starts now. B starts 2 s later. At global time 6 s, compare their displacements if both start from rest with the same constant acceleration.

$$S_A : S_B = 6^2 : 4^2 = 36 : 16 = 9 : 4$$

FIRST MOVE

Do not calculate a unless the question needs it. Translate the times first; then look for cancellation.

M6A-H1

Three steps

1

GLOBAL

Choose one clock for the whole story.

2

LOCAL

Convert the global instant into each body's elapsed time.

3

EQUATION

Only now write each motion equation.

When not to use one clock blindly

MULTISTAGE WARNING

If a body itself changes stage, its local time may reset again for the new stage. Mark each stage boundary.

MEETING CONDITION

At a meeting event, positions are equal at the same global instant - but each position equation can contain a different elapsed time.

MEMORY PHRASE

same instant $\neq$ same elapsed time

M6A-G

Guided

Car A starts from rest at $t=0$ with constant acceleration a . Car B starts from the same point 3 s later, also from rest with the same acceleration. At global time T , write the ratio S_A/S_B .

H1 -> H2 -> H3

H1: mark A from 0 to T . H2: mark B from 3 to T . H3: use T and $T-3$ inside the same $\frac{1}{2}at^2$ model.

Independent transfer

Two identical carts start from the same point under the same constant acceleration. The second starts 4 s after the first. At some global instant, the first has travelled four times as far as the second. Find the elapsed time of the first cart.

NO DIAGRAM PROVIDED

Draw the global clock yourself. Let the first elapsed time be T and the second $T-4$.

MODEL CHECK

This ratio shortcut assumes both start from rest with the same constant acceleration.

M6A-CHECK

You should now be able to...

- 1 distinguish global time from local elapsed time
- 2 translate a delay into $T-\tau$
- 3 build two position equations at one meeting instant
- 4 spot when common factors cancel
- 5 avoid assigning the same t to delayed bodies
- 6 split further if a body changes stage

ExamSIDE status

CURRENT PRIMARY SET = 0

The JEE Main delayed-start examples in the current snapshot also require gravity/vertical-motion concepts. They are deferred to the gravity/vertical-motion subtopics rather than imported early.

AUDIT PRINCIPLE

A zero-primary subtopic is acceptable. We do not pad it with questions that require untaught physics.

MOTION IN A STRAIGHT LINE

M-ST06B - Two Braking Cars: Kinematics, Then Geometry

Solve each stopping motion independently; combine the distances only afterward.



MEMORY ANCHOR

STOP EACH CAR $\rightarrow$ THEN COMPARE WITH D .

SOURCE FOCUS

The supplied Motion book anchors this model-composition idea to Q25.

WHY THIS IS TWO PROBLEMS, NOT ONE

Kinematics first, geometry second

M6B-C1

Stage 1: each car stops



INDEPENDENT MOTIONS

For each car, use its own initial speed and deceleration to find its stopping distance.

Stage 2: compare with the gap

$$\text{remaining gap} = D - (s_1 + s_2)$$

GEOMETRY AFTER KINEMATICS

Only after the two stopping distances are known do you ask whether the cars collide, just touch, or stop with a gap.

M6B-C2

For uniform braking

$$s_{\text{stop}} = u^2 / (2|a|)$$

WHY ABSOLUTE a?

Stopping distance is a positive path length. The acceleration points opposite the motion, so use the deceleration magnitude in this compact form.

Two-car template

$$s1 = u1^2 / (2|a1|)$$

$$s2 = u2^2 / (2|a2|)$$

$$\text{gap} = D - s1 - s2$$

FIRST MOVE

Do not create one sign-heavy equation for both cars. Compute two positive stopping distances, then do the geometry.

WORKED EXAMPLE: WILL THEY COLLIDE?

Separate the physics from the geometry

M6B-E1

Question

Two cars approach each other from 300 m apart. Each moves at 20 m/s and each brakes uniformly at 2 m/s^2 . Find the gap when both stop.



Solution

$$s_1 = s_2 = \frac{20^2}{(2 \cdot 2)} = 100 \text{ m}$$

$$\text{gap} = 300 - 100 - 100 = 100 \text{ m}$$

PHYSICAL CHECK

The combined stopping distance is 200 m, less than the 300 m initial separation, so a positive gap must remain.

PREDICT

Each car needs the same stopping distance because their speeds and braking magnitudes are the same.

M6B-C3

Wrong approach

WRONG IDEA

“One car moves right and one left, so one stopping distance should be negative.”

That mixes displacement sign with the geometric length each car consumes from the separation.

WHY IT FEELS PLAUSIBLE

A single coordinate axis makes the two cars' signed displacements opposite. But the final collision/gap test needs how much of the initial separation each car uses up.

Repair

POSITIVE LENGTHS FIRST

Compute each stopping path length as a positive number. Then subtract their sum from the initial separation.

IF YOU USE SIGNED COORDINATES

That can also work, but keep the coordinate equations and the final geometric interpretation consistent. Do not switch conventions mid-solution.

M6B-C4

Three outputs from the same model

1

STOP

Find s_1 and s_2 .

2

COMPARE

Compute $D - (s_1 + s_2)$.

3

INTERPRET

Positive=gap; zero=just stop together; negative=collision before stop.

Variant: different brakes

If one car has a larger speed but also stronger brakes, do not guess which travels farther. Compare $u^2/|a|$ for each car.

RECOGNITION CUE

Two moving objects + braking + known initial gap -> solve stopping paths independently, then combine.

SAFE-SEPARATION FORM

Minimum D that avoids collision is $s_1 + s_2$.

M6B-G

Guided

Cars A and B approach each other from 250 m apart. A: $u=18$ m/s, $|a|=3$ m/s². B: $u=12$ m/s, $|a|=2$ m/s². Find the final gap if both brake immediately.

H1 -> H2 -> H3

H1: solve A alone. H2: solve B alone. H3: subtract the two stopping distances from 250 m.

Independent transfer

Two vehicles approach each other. Their stopping distances under current road conditions are 72 m and 98 m. What is the minimum initial separation that guarantees they stop without collision?

NO KINEMATICS NEEDED

This is the geometry layer only. Recognise that the minimum safe D equals the sum of stopping paths.

M6B-CHECK

Checklist

- 1 find each stopping distance independently
- 2 keep path lengths positive when using the compact stopping formula
- 3 compare $s_1 + s_2$ with D
- 4 interpret positive/zero/negative remaining gap
- 5 avoid mixing signed coordinate equations with unsigned geometry

ExamSIDE status

CURRENT PRIMARY SET = 1

JEE Main 2024, 9 Apr Evening is a direct two-braking-cars + separation problem.

NO PADDING

Related catch-up/meeting questions belong to other motion families and are not counted here merely to increase the question count.

MOTION IN A STRAIGHT LINE

M-ST06C - Velocity Change + Average Velocity: A Short Route

Sometimes definitions are cleaner than a full SUVAT route.

1. TIME FROM VELOCITY CHANGE

$$t = \Delta v / a = 20/4$$

2. DISTANCE FROM AVERAGE VELOCITY

$$s = v_{\text{avg}} t = 15t$$

MEMORY ANCHOR

$\Delta v + a \rightarrow$ time; average velocity + time $\rightarrow$ displacement.

SOURCE FOCUS

The supplied Motion Concept Book explicitly gives this two-step route and anchors it to Q33.

M6C-C1

Suppose you know...

KNOWN

constant acceleration a , total velocity change Δv , and average velocity v_{avg} over the same interval.

Then the route is direct

$$t = \Delta v / a$$

$$s = v_{\text{avg}} t$$

WHY THIS IS CLEAN

No need to introduce u and v separately if the question already gives the change and the average.

M6C-C2

Three quantities, one interval

ALIGN THE INTERVALS

Δv , a and v_{avg} must refer to the same time interval. If they come from different stages, split the motion first.

1. TIME FROM VELOCITY CHANGE

$$t = \Delta v / a = 50/50$$

2. DISTANCE FROM AVERAGE VELOCITY

$$s = v_{\text{avg}} t = 125t$$

Units tell the story

$$(m/s) / (m/s^2) = s$$

$$(m/s) \cdot s = m$$

PHYSICAL READING

Acceleration tells how fast velocity changes; dividing the total change by that rate gives the duration. Average velocity then converts duration into displacement.

WORKED EXAMPLE

Use change, rate, average

M6C-E1

Question

During one interval, a particle's velocity increases by 24 m/s under constant acceleration 6 m/s². Its average velocity over that same interval is 18 m/s. Find the displacement.

FIRST MOVE

Find the duration from the velocity change and acceleration. Do not invent initial/final velocities.

Solution

$$t = 24/6 = 4 \text{ s}$$

$$s = 18 \cdot 4 = 72 \text{ m}$$

CHECK

72 m in 4 s corresponds to 18 m/s average velocity, exactly as stated.

M6C-C3

Wrong instinct

WRONG IDEA

“Every constant-acceleration problem must begin by filling u , v , a , t , s .”

The UVATS strip is useful, but it is a tool - not a requirement. Sometimes the data are already compressed into Δv and average velocity.

WHY IT FEELS SAFE

A familiar template can feel safer, but introducing extra unknowns increases algebra and creates opportunities for sign errors.

Repair

ASK FOR THE SHORTEST EXECUTABLE ROUTE

What is directly known? What is the target? Can a definition connect them without reconstructing unnecessary variables?

NOT AN ANTI-FORMULA RULE

Use standard kinematic equations when they are the clean route. The goal is model economy, not avoiding formulas.

APPLICATION: ONE-SECOND DATA HIDES THE SAME ROUTE

A bridge to a harder interval question

M6C-C4

Given over $t \rightarrow t+1$ s

Suppose displacement is 125 m and velocity increases by 50 m/s during that one-second interval.

READ THE DATA

Because the interval length is 1 s, acceleration = $\Delta v / \Delta t = 50 \text{ m/s}^2$. Average velocity over the interval = displacement/time = 125 m/s.

Recover the endpoint velocities

$$v - u = 50$$

$$(u + v)/2 = 125$$

WHY THIS MATTERS

The definitions first give the interval's acceleration and average velocity. Only then do you reconstruct u and v if a later question requires them.

M6C-HG

Three-line helper

1

CHANGE

Identify Δv over the interval.

2

RATE

Use $t = \Delta v / a$ if a is constant.

3

AVERAGE

Use $s = v_{\text{avg}} t$.

MODEL GATE

All three pieces must refer to the same interval.

Guided

A body gains 30 m/s of velocity under constant acceleration 5 m/s². Its average velocity in that interval is 20 m/s. Find the displacement.

H1 -> H2 -> H3

H1: find time from change/rate. H2: $t = 30/5$. H3: multiply average velocity by that time.

INDEPENDENT VARIANT

What if the average velocity given belongs to a different stage? Then the shortcut is invalid until the stages are separated.

M6C-CHECK

Checklist

- 1 recognise $\Delta v/a$ as elapsed time
- 2 recognise $s=v_{\text{avg}} t$ as a definition
- 3 ensure both apply to the same interval
- 4 avoid reconstructing u and v unless needed
- 5 use units as a physical check
- 6 switch to a fuller model when data span different stages

ExamSIDE status

CURRENT PRIMARY SET = 1

JEE Main 2024, 30 Jan Morning starts from displacement and velocity increase over a one-second interval, then asks for a later interval. It strongly exercises this compact reasoning route.

SECONDARY LINK

It also connects to n th-second reasoning, but its primary owner here is the short-route/data-decoding concept because that is the first non-obvious step.

MOTION IN A STRAIGHT LINE

M-ST07A - What g Really Means

Gravity is constant acceleration near Earth's surface - not a new formula chapter.



MEMORY ANCHOR

g points downward. The sign is chosen by your axis: upward positive $\rightarrow a = -g$; downward positive $\rightarrow a = +g$.



SOURCE FOCUS

The supplied Motion Concept Book states that Q35-62 broadly use the same constant-acceleration equations with a replaced by $\pm g$.

DROP A PIECE OF CHALK

See g as a rate of velocity change

M7A-C1

Near Earth's surface

Ignoring air resistance, a freely falling object's downward velocity changes by about 9.8 m/s every second.

$$g \approx 9.8 \text{ m/s}^2$$

PHYSICAL MEANING

Each second, gravity changes the velocity by about 9.8 m/s in the downward direction.

The sign is a coordinate choice



UPWARD POSITIVE

Gravity points opposite +y, so $a = -g$.

SAME PHYSICS, DIFFERENT SIGNS

Coordinate systems change symbols, not motion

M7A-C2

Upward positive

$$a = -g$$

$$v = u - gt$$

INTERPRETATION

A falling object can have negative velocity and negative acceleration because both point downward under this axis choice.

Downward positive

$$a = +g$$

$$v = u + gt$$

SAME MOTION

Nothing physical changed. Only the sign convention changed.

GRAVITY USES THE SAME EQUATION FAMILY

Substitute $a = \pm g$

M7A-C3

Upward-positive form

$$v = u - gt$$

$$y = ut - \frac{1}{2}gt^2$$

$$v^2 = u^2 - 2gy$$

NOT NEW FORMULAS

These are the same constant-acceleration relations already learned, with $a = -g$.

Concept helper

VERTICAL SIGN STRIP

1. Choose + direction. 2. Mark gravity arrow downward. 3. Assign $a = +g$ or $-g$. 4. Translate u, v, y with the same axis.

MODEL GATE

This school-level model assumes near-Earth motion and neglects air resistance unless told otherwise.

M7A-C4

Wrong idea

WRONG IDEA

"If $a = -g$, gravity is negative or weaker."

The minus sign only says gravity points opposite the chosen positive direction.

WHY IT HAPPENS

Students often read algebraic sign as physical size instead of direction.

Correct model

MAGNITUDE VS DIRECTION

Magnitude: $g \approx 9.8 \text{ m/s}^2$. Direction: downward. Sign: determined by your axis.

RETRY

If downward is positive, what sign should acceleration due to gravity have?

M7A-G

Guided

A stone is thrown upward. Choose upward as positive. Write the signs of u , a and the velocity v while the stone is still rising.

H1 -> H2 -> H3

H1: draw $+y$ upward. H2: gravity arrow is downward. H3: $u > 0$, $a = -g$, and $v > 0$ until the top.

Self-check

- 1 state what g means physically
- 2 choose a vertical axis before writing signs
- 3 translate a into $\pm g$ consistently
- 4 explain why a negative sign is directional
- 5 recognise gravity as the same constant- a model

NEXT

M-ST07B: free fall from rest - why velocity grows linearly while distance grows quadratically.

MOTION IN A STRAIGHT LINE

M-ST07B - Free Fall from Rest

Velocity grows with t ; distance grows with t^2 .

$t=0s$



$v=0$

$t=1s$



$v=10$

$t=2s$



$v=20$

$t=3s$



$v=30$

MEMORY ANCHOR

$v \propto t$, $s \propto t^2$. Doubling fall time doubles speed but quadruples distance.

SOURCE FOCUS

The supplied Motion book anchors free-fall scaling to Q41-44, Q53, Q56, Q59 and Q60.

M7B-C1

Choose downward positive

$$u = 0$$

$$v = gt$$

$$s = \frac{1}{2} gt^2$$

WHY DOWNWARD POSITIVE HELPS

For a pure fall, all the main quantities become positive and the physical story is easy to read.

Two different growth laws

VELOCITY

Each second adds the same g to velocity -> linear growth with time.

DISTANCE

Later seconds are entered with larger velocity -> cumulative distance grows quadratically.

M7B-C2

From rest

$$h = \frac{1}{2} g t^2$$

$$t = \sqrt{(2h/g)}$$

SCALING

If height becomes 4 times as large, fall time becomes 2 times as large - not 4 times.

First half vs second half of the fall

Because the object speeds up, the second half of the distance takes less time than the first half.

PHYSICAL REASON

The body reaches the midpoint already moving. It does not start the second half from rest.

WORKED EXAMPLE: WHEN v NUMERICALLY EQUALS g

Velocity-distance route

M7B-E1

Question

A ball is dropped from rest from 18 m. Using $g=10 \text{ m/s}^2$, at what height above ground is $|v|$ numerically equal to $|g|$?

TRANSLATE

Numerically equal means $v=10 \text{ m/s}$. Use the time-free relation $v^2=2gs$ for the distance fallen.

Solution

$$10^2 = 2(10)s$$

$$s = 5 \text{ m}$$

$$\text{height above ground} = 18 - 5 = 13 \text{ m}$$

CHECK

The speed grows from 0 downward; reaching 10 m/s after only 5 m of fall is plausible for $g=10$.

MULTIPLE DROPS: SAME g , DIFFERENT ELAPSED TIMES

Connect free fall to the common clock

M7B-C3

Shower-drop story

Drops leave a nozzle at regular intervals. When the first drop reaches the floor, later drops have been falling for shorter times.



Same equation, different t

$$s_i = \frac{1}{2} g t_i^2$$

SECONDARY LINK TO M-ST06A

The common-clock idea supplies each drop's local elapsed time; free-fall physics supplies its position.

FIRST MOVE

Find the total fall time of the first drop. Divide it into equal release intervals. Then assign each later drop its own elapsed time.

M7B-C4

Wrong idea

WRONG IDEA

"If a body falls twice as long, it falls twice as far."

That would be constant-speed motion. Free fall from rest is accelerated motion.

$$s \propto t^2$$

Counterexample

1 s vs 2 s

With $g \approx 10$, $S(1) = 5$ m while $S(2) = 20$ m. Doubling time makes the distance four times as large.

CORRECT MODEL

Velocity grows linearly; cumulative displacement grows quadratically.

M7B-G

Guided

A stone is dropped from rest and falls for 3 s. Take $g=10 \text{ m/s}^2$. Find its speed and distance fallen.

H1 -> H2 -> H3

H1: downward positive. H2: $u=0$. H3: use $v=gt$ and $s=1/2gt^2$.

Self-check

- 1 use $u=0$ for a true drop from rest
- 2 explain vt and st^2
- 3 use $t/\sqrt{h}$ for height comparisons
- 4 assign separate elapsed times to regularly released drops
- 5 distinguish free-fall physics from the common-clock helper

NEXT

M-ST07C: last-distance and between-point problems are interval problems.

MOTION IN A STRAIGHT LINE

M-ST07C - Last-Distance and Between-Point Fall Problems

The last x metres are an interval, not a new fall from rest.



MEMORY ANCHOR

LAST / BETWEEN $\rightarrow$ SUBTRACT CUMULATIVE POSITIONS.

SOURCE FOCUS

The supplied Motion book explicitly maps the last-distance interval idea to Q44.

"LAST 6 m" IS NOT $S(0.2)$

The body is already moving

M7C-C1

Suppose the last 6 m takes 0.2 s

Let total fall time be T . The relevant interval is from $T-0.2$ to T , not from release to 0.2 s.

$$S(T) - S(T-0.2) = 6$$

Free-fall substitution

$$S(t) = \frac{1}{2}gt^2$$

$$\frac{1}{2}g[T^2 - (T-\Delta t)^2] = \Delta s$$

DIFFERENCE OF SQUARES

The interval structure simplifies cleanly because the cumulative law is quadratic.

WHY

At the start of the last 0.2 s, the object already has downward velocity.

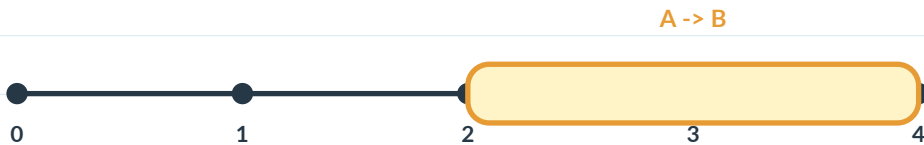
BETWEEN TWO POINTS

Same idea, different words

M7C-C2

Point A to point B

A falling body crosses point A, then 2 s later crosses point B, 80 m below A. The 80 m is an interval displacement.



Write the interval equation

$$S(T+2) - S(T) = 80$$

THEN FIND A

Once T is found, the distance of A from the release point is $S(T)$.

FIRST MOVE

Let the time to reach A be T. Then B is reached at T+2.

DROPLET SPACING IS ALSO AN INTERVAL DIFFERENCE

Different local times, same global instant

M7C-C3

Two drops observed together

An older drop has been falling longer than the next drop. Their vertical separation equals the difference between two cumulative free-fall distances.

$$\text{spacing} = 1/2g(t_{\text{old}}^2 - t_{\text{new}}^2)$$

CONNECTS TO M-ST06A

The common clock tells you each drop's elapsed time. This subtopic tells you to subtract their cumulative fall distances.

Recognition cue

WORDS TO WATCH

last x metres • between A and B • spacing between drops • during final Δt .

ONE CORE IDEA

Later cumulative position minus earlier cumulative position.

M7C-C4

Wrong idea

WRONG IDEA

“For the last 0.2 s, use $u=0$ and $t=0.2$.”

The body did not stop at the start of the last interval. It entered that interval with substantial downward velocity.

WHY IT HAPPENS

Students confuse a new calculation interval with a physical restart of the motion.

Repair

TWO SAFE ROUTES

Either subtract cumulative positions, or compute the velocity at the interval start and use it as the new initial velocity. Do not silently set $u=0$.

M7C-G

Guided

A stone is dropped from rest. It travels 70 m during a 2 s interval. Let the interval start T seconds after release. Write the equation needed to find T .

H1 -> H2 -> H3

H1: interval is $T \rightarrow T+2$. H2: use cumulative $S(t)$. H3: $S(T+2)-S(T)=70$.

Self-check

- 1 translate last/between wording into an interval
- 2 use $S(\text{end})-S(\text{start})$
- 3 avoid resetting $u=0$ mid-fall
- 4 connect droplet spacing to two elapsed times
- 5 switch to a start-velocity route only if done consistently

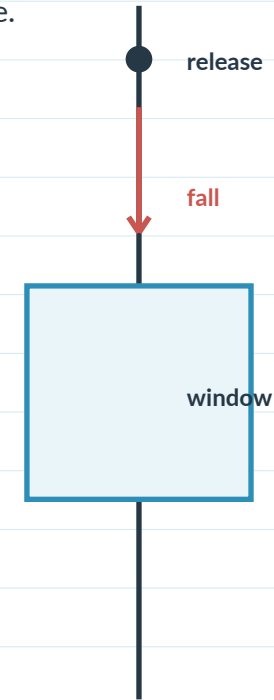
NEXT

M-ST07D: window crossing - same height, different entry speed.

MOTION IN A STRAIGHT LINE

M-ST07D - Window Crossing: Entry Speed Matters

The same window can take less time to cross when the object is dropped from higher above.



MEMORY ANCHOR

HIGHER RELEASE -> GREATER ENTRY SPEED -> LESS WINDOW TIME.

SOURCE FOCUS

The supplied Motion Concept Book explicitly develops this idea and anchors it to Q35.

SAME WINDOW, DIFFERENT RELEASE HEIGHTS

Separate approach motion from crossing motion

M7D-C1

Stage 1: reach the top of the window

A higher release point gives more fall distance before the window, so the object enters the window faster.

$$u_w^2 = 2g h_{\text{above}}$$

Stage 2: cross the window

$$H = u_w \tau + \frac{1}{2} g \tau^2$$

SAME H, DIFFERENT u_w

If window height H is fixed, a larger entry speed u_w requires a smaller positive crossing time τ .

ENTRY SPEED

The crossing problem does not start from rest at the top edge unless the object is released exactly there.

WHY HIGHER CAN MEAN FASTER THROUGH THE WINDOW

The hidden variable is entry speed

M7D-C2

Wrong intuition

WRONG IDEA

“Dropped from higher means the whole trip is longer, so crossing the window must take longer too.”

That confuses total journey time with the local time spent inside the window.

WHY IT FEELS PLAUSIBLE

A greater release height does increase total fall time, but it also makes the object faster by the time it reaches the window.

Correct local comparison

COMPARE THE SAME H

Inside the same window, both cases cover the same distance H under the same g . The case with larger entry speed finishes that local interval sooner.

WORKED EXAMPLE

Compare two release heights

M7D-E1

Question

A 5 m tall window is crossed by a falling ball. Case A: the ball is released from rest at the top edge. Case B: it is released 15 m above the top edge. Take $g=10$. Which case has the shorter crossing time?

CASE A

At the top edge $u_w=0$. Solve $5=5\tau^2 \rightarrow \tau=1$ s.

Case B

$$u_w = \sqrt{(2 \cdot 10 \cdot 15)} = \sqrt{300}$$

NO NEED FOR EXACT τ

Because $u_w > 0$ and the window height/ g are unchanged, the positive crossing time must be less than 1 s.

CONCEPT HELPER: APPROACH -> ENTRY -> CROSSING

Do not reset the motion

M7D-H1

Three steps

1

APPROACH

Use release height to find speed at top edge.

2

ENTRY

Carry that speed into the window stage.

3

CROSS

Use $H = u_w \tau + \frac{1}{2} g \tau^2$ for the local interval.

When not to use this model

AIR RESISTANCE / NONCONSTANT g

The school-level relation assumes constant downward g and negligible air resistance across the motion.

DO NOT RESET $u=0$

The object enters the window with the speed acquired above it.

M7D-G

Guided

A ball is dropped 20 m above the top of a 4 m window. Take $g=10$. Write the two equations needed to determine its window-crossing time.

H1 -> H2 -> H3

H1: find entry speed after 20 m. H2: $v^2=2gh$. H3: use $4=u_w\tau+5\tau^2$.

Self-check

- 1 separate approach motion from crossing motion
- 2 carry entry speed into the second stage
- 3 explain why higher release reduces local window time
- 4 avoid resetting $u=0$ at the window
- 5 apply the same constant- g model across both stages

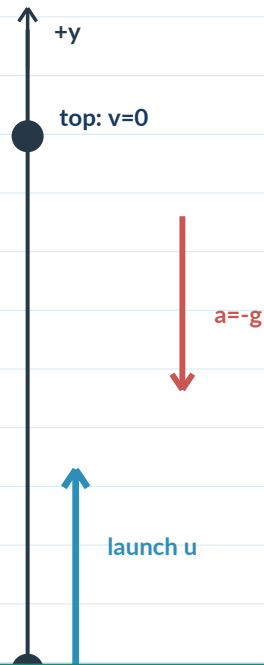
NEXT

Unit 8: vertical throw - highest point, symmetry, and same-height passages.

MOTION IN A STRAIGHT LINE

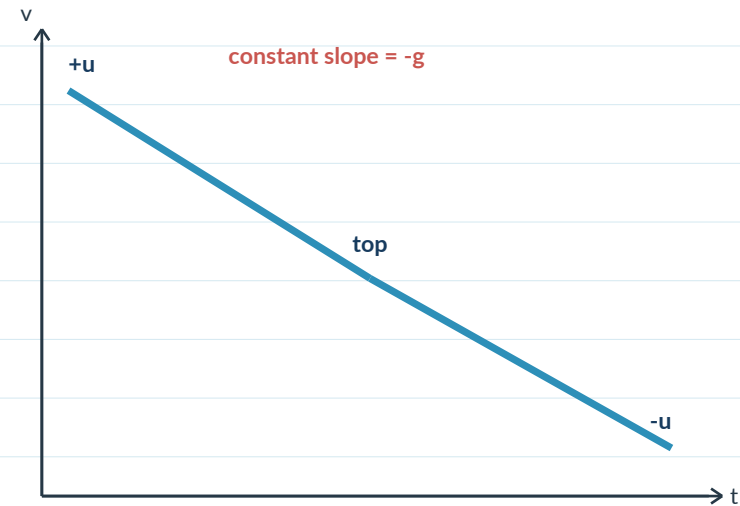
M-ST08A - Highest Point: $v = 0$, $a = -g$

The object pauses in velocity, not in acceleration.



MEMORY ANCHOR

TOP $\rightarrow v=0$, but gravity still acts.



SOURCE FOCUS

The supplied Motion book explicitly marks Q38 and later vertical-motion questions for this misconception.

THROW A BALL UPWARD

Velocity shrinks to zero; acceleration does not

M8A-C1

Use upward positive

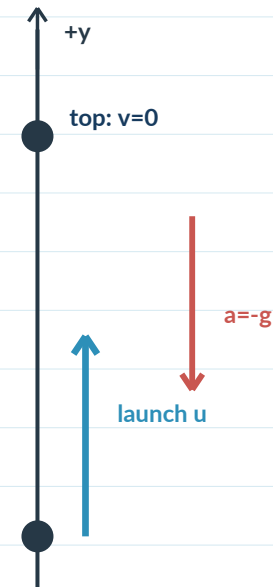
$$a = -g$$

$$v = u - gt$$

WHAT HAPPENS EACH SECOND

Gravity subtracts about g from the upward velocity every second. Eventually v reaches zero for an instant.

At the top



TURNING POINT

$v=0$ is the condition for the turning instant. The nonzero $-g$ is exactly what makes v become negative immediately afterward.

MISCONCEPTION CLINIC: $v=0$ DOES NOT IMPLY $a=0$

Velocity and acceleration answer different questions

M8A-C2

Wrong idea

WRONG IDEA

"At the highest point the ball stops, so acceleration must also be zero."

Velocity describes the object's current motion. Acceleration describes how that velocity is changing.

COUNTEREXAMPLE

At the top, $v=0$ but $a=-g$. A moment later the velocity is downward, so something had to change it.

Correct model



GRAPH CHECK

The v - t line crosses zero with a nonzero negative slope. That slope is acceleration.

WORKED EXAMPLE: VELOCITY AT TWO TIMES

Keep the sign logic simple

M8A-E1

Question

A ball is thrown vertically upward at 150 m/s. Take $g=10 \text{ m/s}^2$. Compare its velocities after 3 s and 5 s.

$$v(3)=150-10\cdot3=120 \text{ m/s}$$

$$v(5)=150-10\cdot5=100 \text{ m/s}$$

PHYSICAL CHECK

Both are still upward because the time to the top is 15 s.

First-step helper

VERTICAL VELOCITY STRIP

Choose + upward $\rightarrow a=-g \rightarrow$ use $v=u-gt \rightarrow$ check whether v is positive, zero, or negative.

WHEN NOT TO USE $v=0$

Only use $v=0$ when the question refers to the highest point / turning instant, not merely any later time.

M8A-G

Guided

A stone is thrown vertically upward with $u=30$ m/s. Take $g=10$. Find the time when its velocity first becomes zero and state its acceleration at that instant.

H1 -> H2 -> H3

H1: top means $v=0$. H2: $a=-g$. H3: $0=30-10t$.

Self-check

- 1 distinguish velocity from acceleration
- 2 use $v=0$ only at the turning instant
- 3 keep $a=-g$ through the top
- 4 read the constant negative slope of $v-t$
- 5 predict sign of v before and after the top

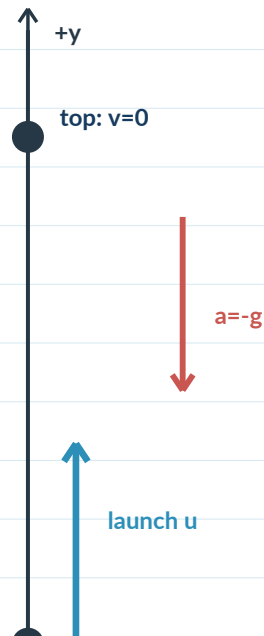
NEXT

M-ST08B: time to top, maximum height, and same-level flight time.

MOTION IN A STRAIGHT LINE

M-ST08B - Time to Top, Maximum Height, Flight Time

Three results, one turning-point condition: $v=0$ at the top.



$$t_{\text{up}} = u/g \quad H = u^2/(2g) \quad T = 2u/g$$

MEMORY ANCHOR

TOP CONDITION $\rightarrow v=0$. Use it once for time, once for height.

SOURCE FOCUS

Source anchors: Q36, Q45, Q48-50, Q55, Q58.

BUILD THE THREE RESULTS

Do not memorise them separately

M8B-C1

Time to top

$$0 = u - gt_{\text{up}}$$

$$t_{\text{up}} = u/g$$

Maximum height

$$0 = u^2 - 2gH$$

$$H = u^2/(2g)$$

Same-level flight time

SYMMETRY

If launch and landing heights are equal, time down = time up.

$$T = 2u/g$$

MODEL GATE

Do not use $T=2u/g$ when the landing height differs from the launch height.

SCALING WITHOUT RECALCULATING

Read the powers

M8B-C2

Same g , change u

$$H \propto u^2$$

$$T \propto u$$

DOUBLE u

Maximum height becomes 4×, while same-level flight time becomes 2×.

Same u , change g

$$H \propto 1/g$$

$$t_{\text{up}} \propto 1/g$$

COMPARE PARAMETERS

Before computing, identify what changed: u , g , or launch/landing height.

WORKED EXAMPLE: JUGGLER RELEASE RATE

Time to top sets the rhythm

M8B-E1

Question

A juggler throws the next ball exactly when the previous ball reaches its highest point. If the juggler throws n balls each second, find the maximum height

TRANSLATE THE RATE

Time between throws = $1/n$. That equals t_{up} .

Solution route

$$u/g = 1/n \rightarrow u = g/n$$

$$H = u^2/(2g) = g/(2n^2)$$

WHY THIS WORKS

The problem hides the upward flight time inside a throwing frequency.

M8B-C3

Wrong idea

WRONG IDEA

"Total flight time is always $2u/g$."

That formula assumes the object returns to the launch height.

COUNTEREXAMPLE

Throw from the top of a tower and land on the ground: the downward journey extends below the launch level, so the total time is longer.

Correct model

CHECK THE ENDPOINT

If final height = initial height, use symmetry. Otherwise solve the position equation for the actual landing level.

M8B-G

Guided

A ball is thrown upward at 40 m/s from ground. Take $g=10$. Find time to top, maximum height, and total same-level flight time.

H1 -> H2 -> H3

H1: top gives $v=0$. H2: $t_{\text{up}}=u/g$. H3: use $H=u^2/(2g)$, then double t_{up} .

Self-check

- 1 derive t_{up} from $v=0$
- 2 derive H with the no-time relation
- 3 use $T=2t_{\text{up}}$ only for same-level return
- 4 read Hu^2 and Tu
- 5 decode frequency or timing language into t_{up}

NEXT

M-ST08C: the same height is crossed twice - two times, one quadratic.

MOTION IN A STRAIGHT LINE

M-ST08C - Same Height Twice

One height, two passage times: upward and downward.



MEMORY ANCHOR

ONE HEIGHT $\rightarrow$ TWO ROOTS. The two times are the two solutions of the position equation.

SOURCE FOCUS

The supplied Motion book anchors this quadratic/symmetry structure to Q37.

WHY THE SAME HEIGHT APPEARS TWICE

Upward passage and downward passage

M8C-C1

Position equation

$$y = ut - \frac{1}{2}gt^2$$

FIX y

For one chosen height below the maximum, this becomes a quadratic equation in t .

$$\frac{1}{2}gt^2 - ut + y = 0$$

Physical picture



TWO PASSAGES

The earlier root is on the way up; the later root is on the way down.

SUM OF THE TWO TIMES

Quadratic algebra meets flight symmetry

M8C-C2

For $At^2+Bt+C=0$

$$t_1 + t_2 = -B/A$$

For vertical throw

$$t_1 + t_2 = 2u/g$$

MEANING

The sum equals the total same-level flight time. Algebra and symmetry tell the same story.

Speed at the two passages

SAME MAGNITUDE, OPPOSITE SIGN

At the same height, the no-time relation gives the same v^2 . Upward passage has $+|v|$; downward passage has $-|v|$.

MISCONCEPTION CLINIC: SAME HEIGHT DOES NOT MEAN SAME VELOCITY

Direction matters

M8C-C3

Wrong idea

WRONG IDEA

"At the same height the velocity is the same."

The speed magnitude is the same under the ideal model, but the directions are opposite.

UP VS DOWN

Going up: $v > 0$. Coming down: $v < 0$, for upward-positive coordinates.

Correct model

$$v_{\text{up}} = +\sqrt{u^2 - 2gy}$$

$$v_{\text{down}} = -\sqrt{u^2 - 2gy}$$

CHECK THE QUESTION WORD

If it asks speed, use magnitude. If it asks velocity, include direction/sign.

M8C-EG

Worked idea

A ball reaches maximum height h . At height $h/3$ it is seen once going up and once coming down. The two times are roots of one quadratic.

FIRST MOVE

Use $H = u^2/(2g)$ to express u^2 in terms of gh , then solve the height equation or use root relations.

Guided transfer

A ball is thrown upward at u . It crosses height y at t_1 and t_2 . Show without solving the quadratic completely that $t_1 + t_2 = 2u/g$.

H1 -> H2 -> H3

H1: rearrange to a quadratic. H2: identify A and B. H3: use sum of roots $-B/A$.

NEXT

M-ST08D: a fraction of speed maps to a different fraction of maximum height.

MOTION IN A STRAIGHT LINE

M-ST08D - Speed Fraction vs Height Fraction

Height depends on the reduction in v^2 , not the reduction in v .

$$v^2 = u^2 - 2gy$$

MEMORY ANCHOR

SQUARE THE SPEED FRACTION FIRST.

SOURCE FOCUS

The supplied Motion book works the $u/2$ case and anchors it to Q57.

HALF THE SPEED DOES NOT MEAN HALF THE HEIGHT

Use v^2 , not v

M8D-C1

Suppose $v = u/2$

$$(u/2)^2 = u^2 - 2gy$$

$$y = 3/4 \cdot u^2/(2g)$$

$$y = 3H/4$$

REMAINING HEIGHT

$$H - y = H/4.$$

General fraction helper

if $v = ku$, then $y/H = 1 - k^2$

EXAMPLE

If speed falls to $0.8u$, height reached is $1 - 0.64 = 0.36$ of H .

M8D-C2

Wrong idea

WRONG IDEA

"Speed has fallen by half, so half the maximum height remains."

The no-time relation contains v^2 . The relevant change is in the square of speed.

COUNTEREXAMPLE

At $v=u/2$, only $H/4$ remains above the object, not $H/2$.

Correct model

FRACTION ROUTE

1. Form $k=v/u$. 2. Square it. 3. Height fraction reached = $1-k^2$. 4. Remaining height fraction = k^2 .

M8D-EG

Worked

A ball is thrown upward. At some point its speed is 60% of launch speed. What fraction of maximum height has it reached?

$$y/H = 1 - 0.6^2 = 0.64$$

ANSWER

64% of maximum height.

Guided

When the speed has fallen to one-third of u , what fraction of H remains above the body?

H1 -> H2 -> H3

H1: $k=1/3$. H2: remaining fraction = k^2 . H3: answer = $1/9$.

NEXT

M-ST08E: Earth vs Moon - keep the same equation and change only g .

MOTION IN A STRAIGHT LINE

M-ST08E - Earth vs Moon: Change Only g

Same governing equation; isolate the parameter that changed.

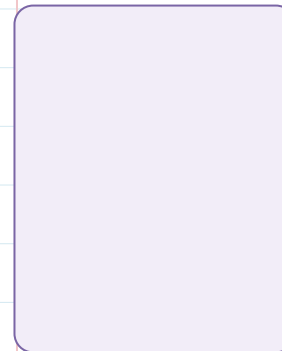
H x1



Earth

g

H x6



Moon

$g/6$

MEMORY ANCHOR

SAME $u \rightarrow H \ 1/g$.

SOURCE FOCUS

The supplied Motion book anchors environment comparison to Q36 and Q49, with a warning that the Moon ratio must be supplied or already known.

M8E-C1

One equation

$$H = u^2/(2g)$$

COMPARE

If u is the same, H is inversely proportional to g .

Earth vs Moon example

$$g_{\text{Moon}} \approx g_{\text{Earth}}/6$$

$$H_{\text{Moon}} \approx 6H_{\text{Earth}}$$

SOURCE CAUTION

Use the 1/6 numerical ratio only when it is supplied or already known; the source notes that Q49 does not visibly state it.

SAME HEIGHT, DIFFERENT REQUIRED SPEED

Reverse the question

M8E-C2

Solve for u

$$u = \sqrt{2gH}$$

SMALLER g -> SMALLER u

To reach the same height on the Moon, less launch speed is needed.

Ratio helper

$$u_M/u_E = \sqrt{g_M/g_E}$$

IF $g_M = g_E/6$

Required speed ratio is $1/\sqrt{6}$.

M8E-C3

Wrong idea

WRONG IDEA

"Earth and Moon need different motion formulas."

The governing constant-acceleration relation is the same. Only the value of g changes in this model.

WHY IT HAPPENS

Context changes can distract the learner from identifying which physical parameter actually changed.

Correct model

SAME LAW -> CHANGE ONE PARAMETER

Write the same equation twice, label Earth/Moon quantities, then form a ratio before inserting numbers.

M8E-G

Guided

A ball launched upward with the same speed reaches 20 m on Earth. If lunar g is one-sixth of Earth's, how high would the school-level model predict on the Moon?

H1 $\rightarrow$ H2 $\rightarrow$ H3

H1: same u . H2: $H1/g$. H3: one-sixth $g \rightarrow$ six times H .

Self-check

- 1 keep the governing equation unchanged
- 2 identify whether u , H , or g is held fixed
- 3 form ratios before calculating
- 4 use $H1/g$ for same u
- 5 use $u\sqrt{g}$ for same H
- 6 state the assumed g ratio instead of inventing it

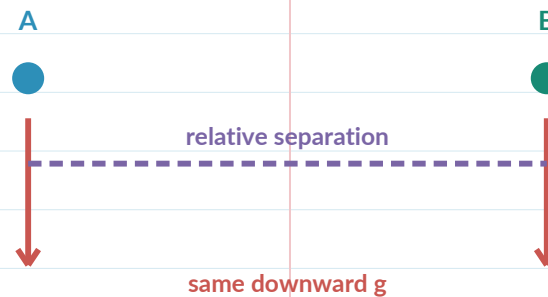
NEXT

Unit 9: two-body and relative motion - common gravity can cancel.

MOTION IN A STRAIGHT LINE

M-ST09A - Relative Acceleration: Common Gravity Cancels

Gravity still acts on both bodies; it disappears only from the separation equation.



MEMORY ANCHOR

SAME g ON BOTH $\rightarrow$ RELATIVE $a = 0$. Their relative velocity stays constant while both are in free flight.

SOURCE FOCUS

Source anchors: Q39, Q51 and related two-body items.

WHY GRAVITY CAN DISAPPEAR FROM A RELATIVE EQUATION

Subtract equal accelerations

M9A-C1

Define relative position

$$\mathbf{x}_{A/B} = \mathbf{x}_A - \mathbf{x}_B$$

$$\mathbf{a}_{A/B} = \mathbf{a}_A - \mathbf{a}_B$$

$$(-g) - (-g) = 0$$

What follows

CONSTANT RELATIVE VELOCITY

Once both are freely falling under the same g , their relative velocity no longer changes.

FIRST MOVE

If two bodies are in the same gravitational field, try a relative-motion view before writing two long quadratic equations.

IMPORTANT

Gravity did not vanish physically. It affects both bodies equally, so it cancels only when we describe their separation.

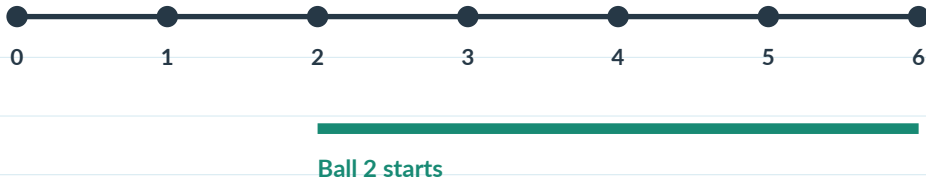
DELAYED LAUNCH: COMMON CLOCK + RELATIVE MOTION

Two ideas combine

M9A-C2

At the second launch

Ball 1 active



After both are in flight

$$a_{\text{rel}} = 0$$

STEP 2

Use constant relative velocity to close the existing separation. This avoids solving two full position quadratics from scratch.

STEP 1

Find the first ball's position and velocity at the instant the second ball is launched.

CONNECTS BACK

M-ST06A supplies the delayed-start clock; M-ST09A supplies the relative-motion simplification.

MISCONCEPTION CLINIC: $a_{\text{rel}}=0$ DOES NOT MEAN BOTH BODIES HAVE $a=0$

Relative is not absolute

M9A-C3

Wrong idea

WRONG IDEA

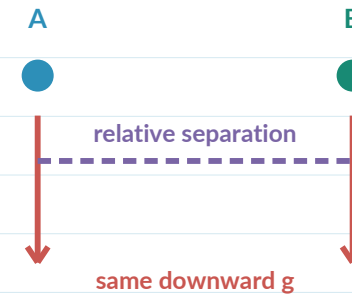
"Relative acceleration is zero, so gravity is not acting."

Each body still accelerates downward at g . Only the difference of their accelerations is zero.

CHECK

In the ground frame both velocity values keep changing. Their difference stays constant.

Correct model



LANGUAGE

Absolute acceleration: $-g$ for each. Relative acceleration: 0 between them.

M9A-G

Guided

Ball 1 is projected upward at 50 m/s. Two seconds later Ball 2 is projected upward from the same point at 50 m/s. Take $g=10$. Find when Ball 2 meets Ball 1.

H1 -> H2 -> H3

H1: at $t=2$, find Ball 1's position and speed. H2: after $t=2$, $a_{\text{rel}}=0$. H3: relative speed is constant; close the separation.

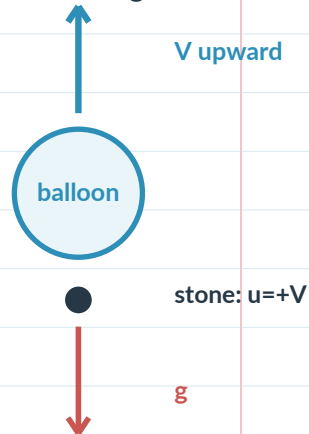
Self-check

- 1 form $x_{A/B}$ and $a_{A/B}$
- 2 recognise equal g cancellation
- 3 distinguish absolute and relative acceleration
- 4 combine delayed start with relative motion
- 5 use constant relative speed only after both share the same acceleration

MOTION IN A STRAIGHT LINE

M-ST09B - Dropped from a Moving Platform

“Dropped” means no extra throw relative to the carrier - not zero ground velocity.



MEMORY ANCHOR

RELEASED FROM MOVING CARRIER -> INHERIT CARRIER VELOCITY.

SOURCE FOCUS

Source anchors: Q52 and Q62.

WHAT “DROPPED” REALLY MEANS

Reference frame first

M9B-C1

Relative to the balloon

$v_{\text{stone/balloon}} = 0$ at release

NO EXTRA THROW

The hand does not give the stone a new velocity relative to the balloon.

Relative to the ground

$u_{\text{stone}} = V_{\text{balloon}}$

IF BALLOON RISES

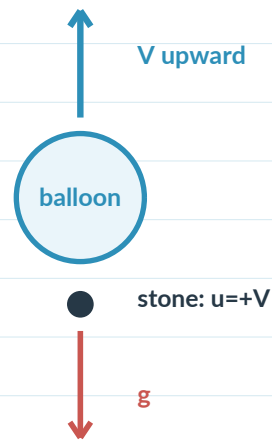
With upward positive and balloon speed $+V$, the stone begins free flight with $u=+V$ and $a=-g$.

THE STONE FIRST KEEPS GOING UP

Inertia before reversal

M9B-C2

Immediately after release



Equations in ground frame

$$y = y_0 + Vt - \frac{1}{2}gt^2$$

$$v = V - gt$$

DO NOT RESET $u=0$

The free-flight stage begins with the carrier's instantaneous velocity.

STORY

The stone is already moving upward. Gravity slows that upward motion, brings v to zero, then makes it fall.

M9B-C3

Wrong idea

WRONG IDEA

"Dropped means $u=0$."

That is true only when the releasing hand/platform is at rest in the chosen frame.

COUNTEREXAMPLE

A stone released from a rising balloon has upward ground velocity at the instant of release.

Correct helper

FRAME $\rightarrow$ CARRIER $\rightarrow$ OBJECT

1. Choose the observer/frame. 2. Read carrier velocity in that frame. 3. Add any release velocity relative to the carrier. For a simple drop, the relative term is zero.

M9B-G

Guided

A balloon rises at 10 m/s and releases a stone at height 75 m. Take $g=10$. Write the stone's ground-frame initial velocity and the equation for its height after t seconds.

H1 -> H2 -> H3

H1: ground frame. H2: $u=+10$. H3: $y=75+10t-5t^2$.

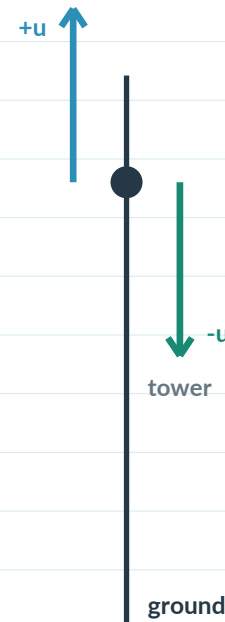
Self-check

- 1 distinguish relative and ground velocity
- 2 inherit carrier velocity at release
- 3 explain why the stone can initially keep rising
- 4 write $y(t)$ with the inherited u
- 5 avoid assuming every 'drop' begins from rest

MOTION IN A STRAIGHT LINE

M-ST09C - Upward vs Downward Launch: Same Impact Speed

Initial direction changes the journey, but not the final speed magnitude at the same lower level.



MEMORY ANCHOR

SAME $|u|$ + SAME HEIGHT DROP $\rightarrow$ SAME final speed magnitude.

SOURCE FOCUS

Source anchors: Q46 and Q54.

WHY THE FINAL SPEEDS MATCH

The no-time equation removes the sign of u

M9C-C1

Both start at the same tower top

$$v_f^2 = u^2 + 2gH$$

KEY DETAIL

Whether the initial velocity is $+u$ or $-u$, its square is u^2 .

What does change?

PATH HISTORY

The upward-thrown ball rises first, returns, then falls. The downward-thrown ball begins falling immediately.

WHAT DOES NOT CHANGE

At the common ground level, ideal final speed magnitudes are equal.

MISCONCEPTION CLINIC: LONGER PATH DOES NOT MEAN GREATER FINAL SPEED

Height change controls v^2

M9C-C2

Wrong idea

WRONG IDEA

"The upward-thrown ball travels farther, so it must hit faster."

In this ideal constant-g model, the speed-at-height relation depends on initial speed magnitude and net vertical displacement, not total path length.

WHY IT FEELS PLAUSIBLE

Distance travelled often matters in resistive or energy-loss situations. Here, air resistance is neglected.

Correct model

$$v_f = \sqrt{u^2 + 2gH}$$

MODEL GATE

If air resistance is important, the path can matter and this equality need not hold.

M9C-G

Guided

From the same tower top, one ball is thrown upward at 12 m/s and another downward at 12 m/s. Ignore air resistance. Compare their speeds at the ground.

H1 -> H2 -> H3

H1: same $|u|$. H2: same net height H . H3: write $v_f^2 = u^2 + 2gH$ for each.

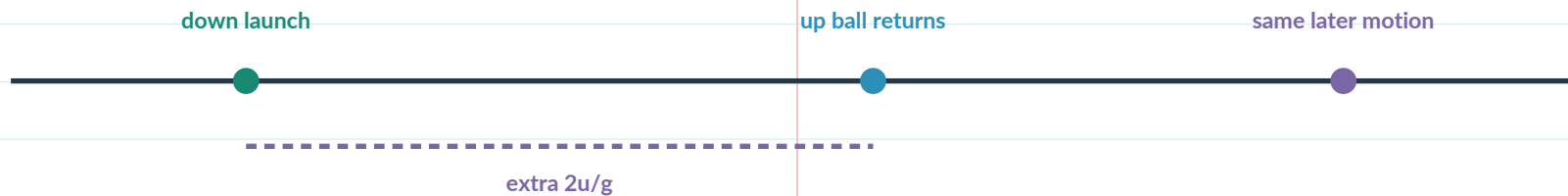
Self-check

- 1 separate speed magnitude from velocity sign
- 2 use the no-time relation for same endpoint height
- 3 explain why u^2 removes launch direction
- 4 distinguish path length from net displacement
- 5 state the no-air-resistance assumption

MOTION IN A STRAIGHT LINE

M-ST09D - Time Gap: Upward vs Downward Launch from a Tower

The upward launch contains one extra symmetric excursion.



MEMORY ANCHOR

EXTRA TIME = $2u/g$. The upward ball rises and returns to launch level before matching the downward ball's starting state.

SOURCE FOCUS

Source anchor: Q47.

WHY THE TIME DIFFERENCE IS SIMPLE

Shift the same downward journey

M9D-C1

Downward-launched ball

STARTS THE DOWNWARD JOURNEY NOW

Initial downward speed magnitude u from the tower top.

Once it returns

SAME STATE

At the tower top on the way down, it has speed magnitude u downward - exactly the state of the other ball at its launch.

Upward-launched ball

EXTRA LOOP

It first spends u/g rising and u/g returning to the tower top.

$$\Delta T = 2u/g$$

M9D-C2

Let times be t_1 and t_2

From the same tower, equal speed magnitude u is used once upward and once downward. Let their ground-hit times be t_1 and t_2 .

$$t_1 - t_2 = 2u/g$$

SAME HEIGHT EQUATIONS

Combining the two tower-height equations also gives the drop-from-rest time t_0 .

Result

$$t_0 = \sqrt{t_1 t_2}$$

WHY IT IS USEFUL

You can find the release-from-rest time without first solving separately for tower height and launch speed.

MISCONCEPTION CLINIC: UPWARD TIME IS NOT JUST DOWNWARD TIME + u/g there are two extra legs

M9D-C3

Wrong idea

WRONG IDEA

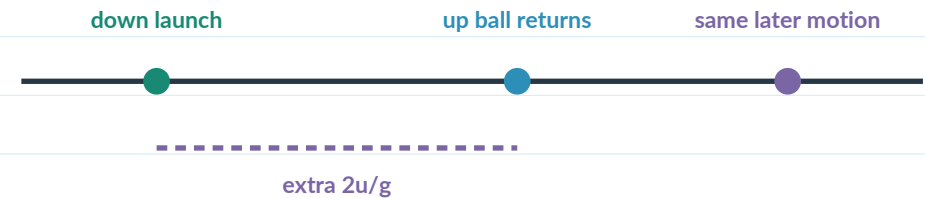
"The upward launch only adds the time to reach the top, u/g ."

It must also spend another u/g returning from the top to the original launch level.

TWO SYMMETRIC LEGS

rise to top + return to launch level = $u/g + u/g$.

Correct model



STATE MATCHING

After $2u/g$, the upward-launched ball has the same position and downward speed as the downward-launched ball had initially.

M9D-G

Guided

A ball thrown upward from a tower takes 6 s to reach the ground. Thrown downward from the same point with the same speed, it takes 1.5 s. Find the time if simply dropped.

H1 -> H2 -> H3

H1: use the paired-time relation. H2: $t_0 = \sqrt{t_{\text{up}} t_{\text{down}}}$. H3: $\sqrt{6 \cdot 1.5}$.

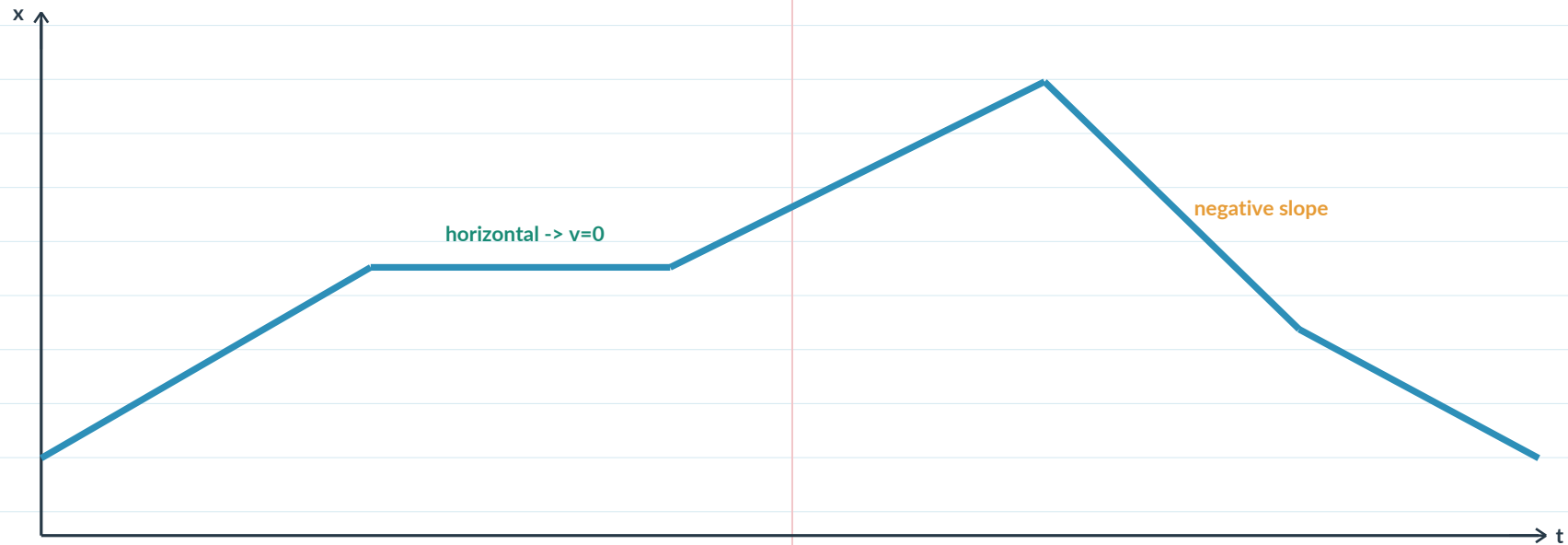
Self-check

- 1 explain the extra symmetric $2u/g$ excursion
- 2 recognise the same-state time shift
- 3 use $t_1 - t_2 = 2u/g$
- 4 use $t_0 = \sqrt{t_1 t_2}$ when the setup matches
- 5 avoid using the result for unequal speed magnitudes

MOTION IN A STRAIGHT LINE

M-ST10A - Position-Time Graph: Slope Is Velocity

Graphs are another language for the same motion - read the axes before using a rule.



MEMORY ANCHOR

x-t graph height = position; x-t graph slope = velocity.

SOURCE GROUNDING

A steeper x-t line means greater velocity magnitude. Horizontal means $v=0$. For a curve, chord slope gives average velocity; tangent slope gives instantaneous velocity. Source anchors: Q63, Q66, Q67.

RECOG

Recognition cues

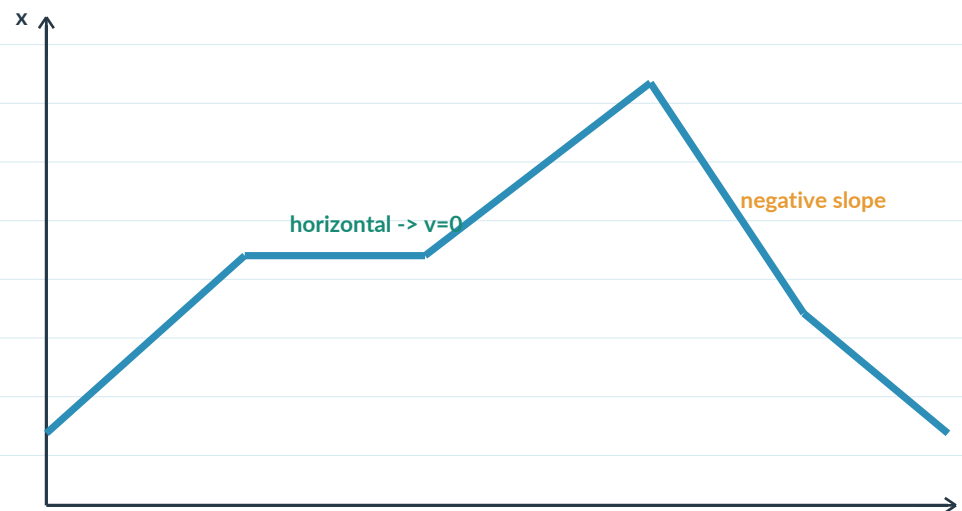
1 position/displacement on vertical axis + time on horizontal

2 average velocity over an interval

3 instantaneous velocity from a curved x-t graph

4 horizontal or negative-slope x-t segments

Build the representation



FIRST MOVE

First move: label axes. Then ask: slope over an interval, or tangent slope at one instant?

WHY

Why

EXPLAINER

Slope is change in position divided by change in time, exactly the definition of velocity. A tangent narrows the interval toward one instant.

Repair / what this is not

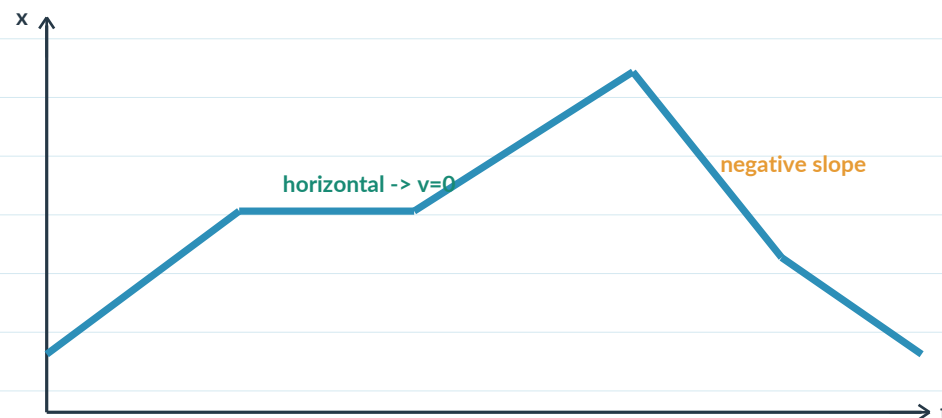
COUNTEREXAMPLE / REPAIR

Graph height is position; graph slope is velocity. A high horizontal line means far from origin but stationary.

Common wrong idea

MISCONCEPTION

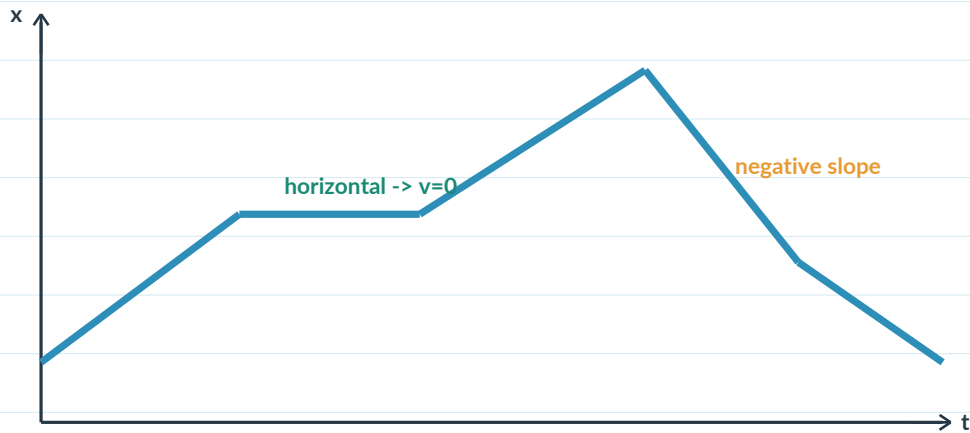
“Higher on an x-t graph means faster.”



GUIDED

Guided 1 - full visual

A toy's x-t graph rises linearly, becomes horizontal for 2 s, then falls linearly. What does each segment say about velocity?



H1 -> H2 -> H3

H1: read axis labels. H2: slope sign gives velocity sign. H3: horizontal slope is zero.

Guided 2 - faded visual

A curved x-t graph is shown. Explain how you would estimate average velocity from 2-5 s and instantaneous velocity at 4 s.

Axes / model: _____

First relation: _____

Physical check: _____

H1 -> H2 -> H3

H1: average uses endpoints. H2: instantaneous uses a tangent. H3: compare slope signs/magnitudes.

CHECK

Independent transfer

A cyclist's x - t graph has the same position at $t=0$ and $t=12$ s but changes in between. What is the average velocity over the full interval? What extra information would you need for distance?

NO HINT INITIALLY

Draw or annotate the representation yourself before writing an equation.

Self-check

- 1 I separate graph height from slope.
- 2 I can get average velocity from a chord.
- 3 I can interpret tangent slope.
- 4 I recognise rest and reversal from slope.

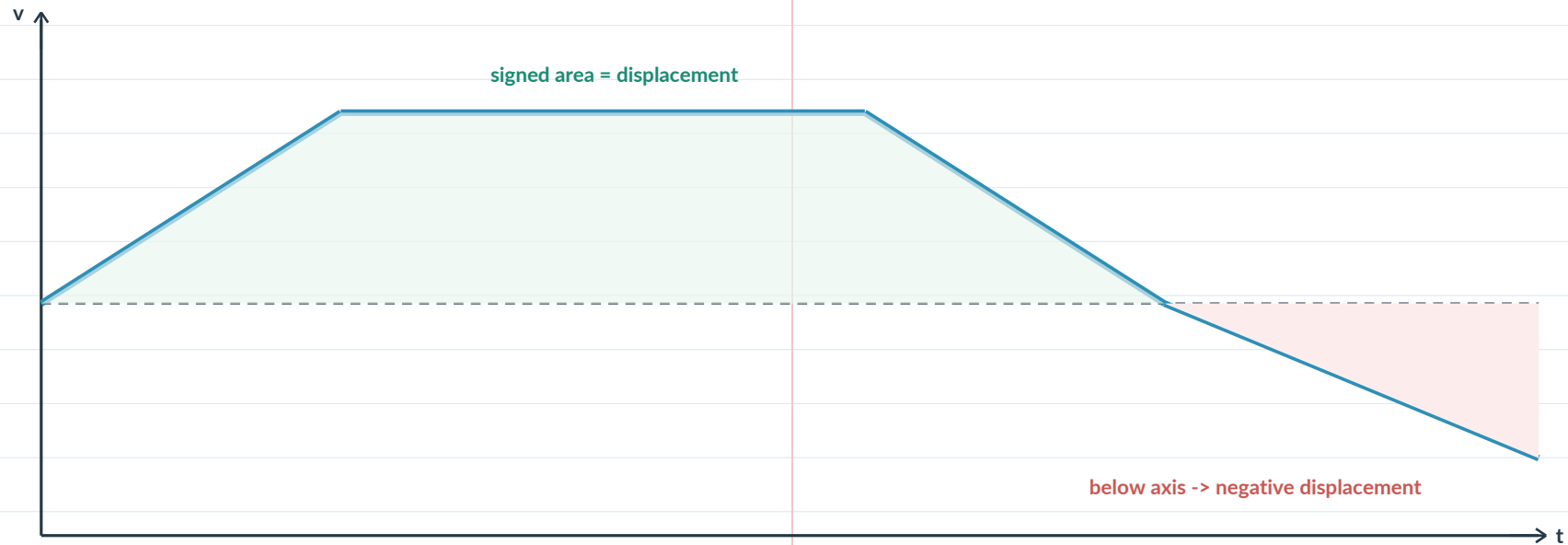
EXAMSIDE LINK STATUS

Current primary set: 1. Secondary linked items: 1.

MOTION IN A STRAIGHT LINE

M-ST10B - Velocity-Time Graph: Slope, Area, Distance

Graphs are another language for the same motion - read the axes before using a rule.



MEMORY ANCHOR

v-t slope = acceleration; signed area = displacement; absolute area = distance.

SOURCE GROUNDING

Area has units $(\text{m/s})(\text{s}) = \text{m}$. Under constant acceleration, rectangle + triangle reproduces $s = ut + \frac{1}{2}at^2$. Source anchors: Q63, Q64, Q68.

RECOG

Recognition cues

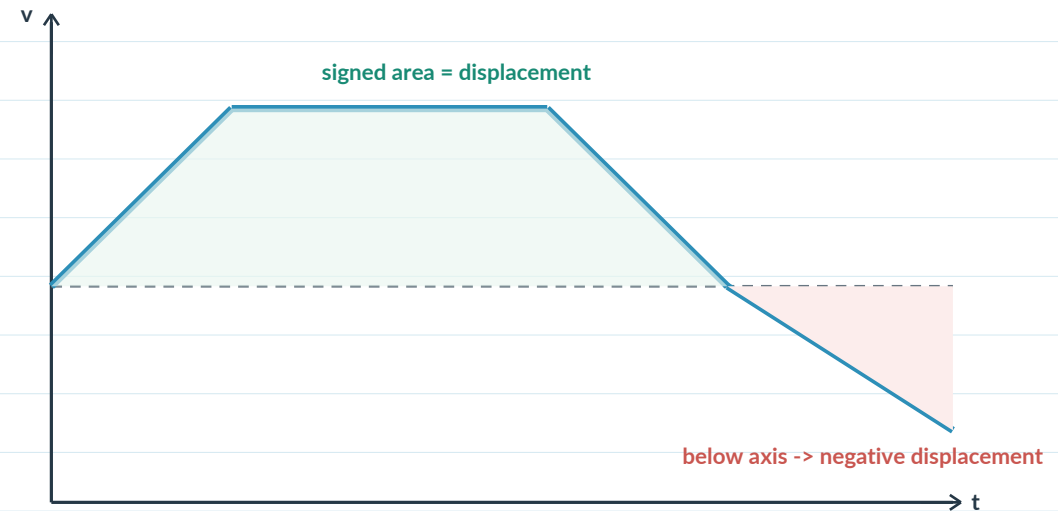
1 velocity on vertical axis + time on horizontal

2 distance/displacement from a graph

3 acceleration from graph slope

4 regions above and below the time axis

Build the representation



FIRST MOVE

First move: mark the time interval, split the area into simple shapes, and decide signed area or absolute area.

WHY

Why

EXPLAINER

Slope is change in velocity per time, exactly acceleration. Area multiplies velocity by time, giving displacement units.

Repair / what this is not

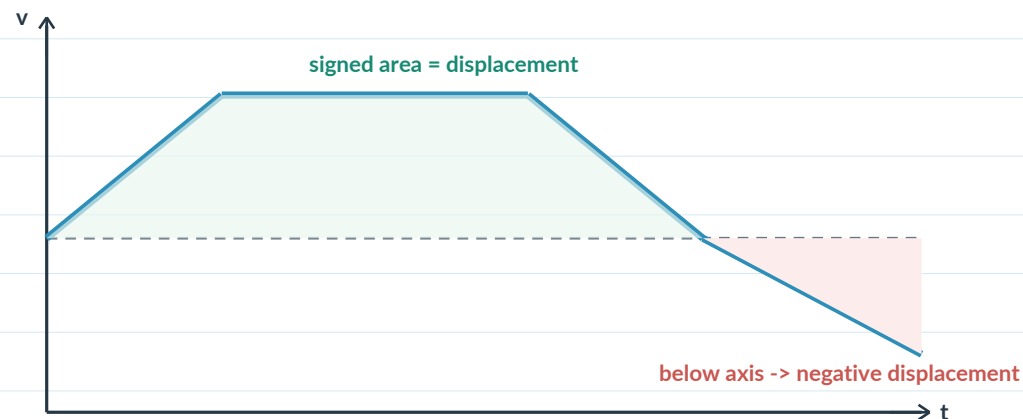
COUNTEREXAMPLE / REPAIR

Signed v-t area is displacement. Distance is the sum of absolute area magnitudes when direction reverses.

Common wrong idea

MISCONCEPTION

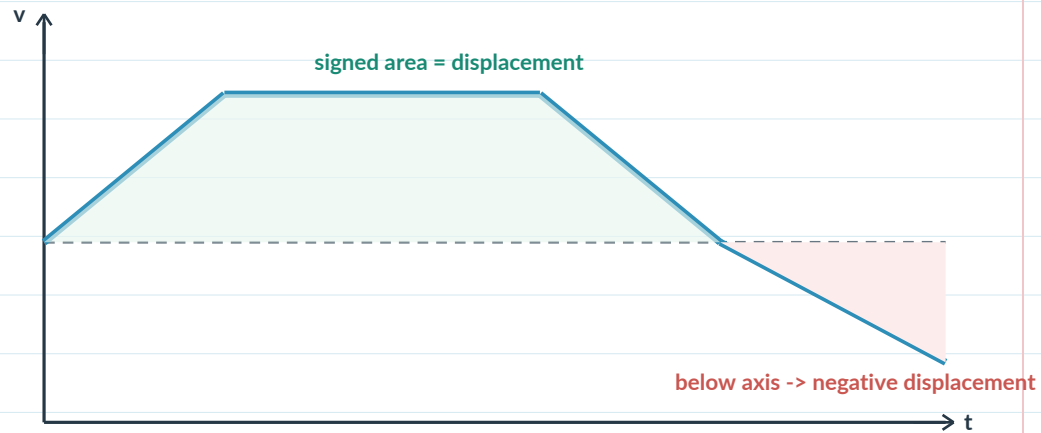
“Area under v-t always gives distance.”



GUIDED

Guided 1 - full visual

A v-t graph has +6 m/s for 4 s and -2 m/s for 3 s. Find displacement and distance.



Guided 2 - faded visual

A straight v-t line rises from 2 to 10 m/s in 4 s. Find acceleration and displacement from the graph.

Axes / model: _____

First relation: _____

Physical check: _____

H1 -> H2 -> H3

H1: split above/below axis. H2: signed area for displacement. H3: absolute areas for distance.

H1 -> H2 -> H3

H1: $\text{slope} = (10-2)/4$. H2: area is trapezium. H3: units m/s^2 and m.

CHECK

Independent transfer

A v-t graph has equal positive and negative signed areas. What are net displacement and average velocity? Can distance still be nonzero?

NO HINT INITIALLY

Draw or annotate the representation yourself before writing an equation.

Self-check

- 1 I separate displacement from distance.
- 2 I use slope for acceleration.
- 3 I use signed/absolute areas correctly.
- 4 I can rebuild $s=ut+\frac{1}{2}at^2$ from area.

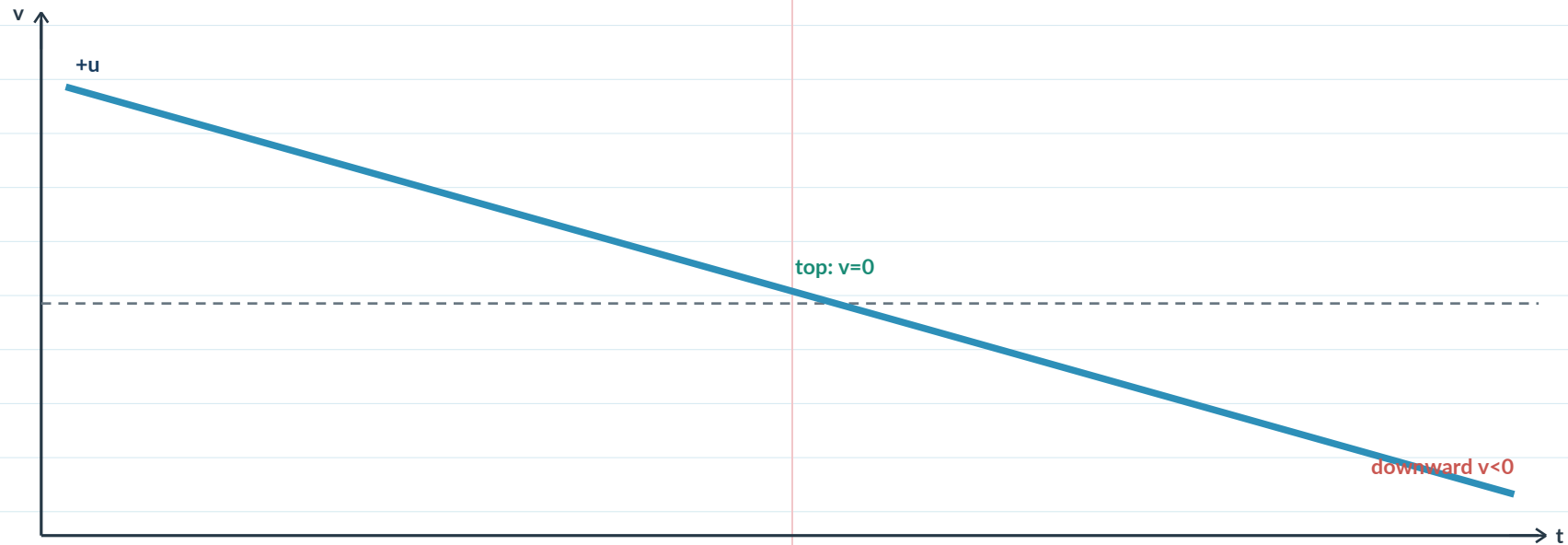
EXAMSIDE LINK STATUS

Current primary set: 6. Secondary linked items: 5.

MOTION IN A STRAIGHT LINE

M-ST10C - Vertical Projection on a v-t Graph

Graphs are another language for the same motion - read the axes before using a rule.



MEMORY ANCHOR

Vertical throw $\rightarrow$ one straight v-t line with slope $-g$.

SOURCE GROUNDING

With upward positive, $v=u-gt$ is a straight line of slope $-g$. Crossing $v=0$ is the top; below the axis means downward velocity, not negative speed. Source anchors: Q64.

WHEN DO I SEE THIS?

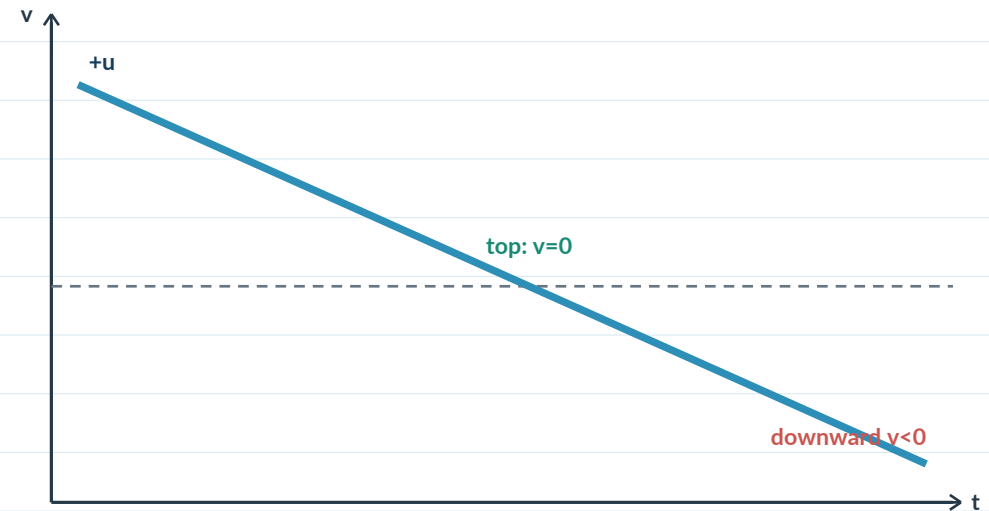
Recognition before formula

RECOG

Recognition cues

- 1 vertical throw or fall represented on v-t axes
- 2 highest point appears where v crosses zero
- 3 upward-positive sign convention
- 4 impact or reversal shown as a change in graph

Build the representation



FIRST MOVE

First move: choose + upward. The line slope must remain $-g$ while free flight continues.

WHY

Why

EXPLAINER

The equation $v=u-gt$ is linear in t , so the v - t graph must be one straight line of slope $-g$.

Repair / what this is not

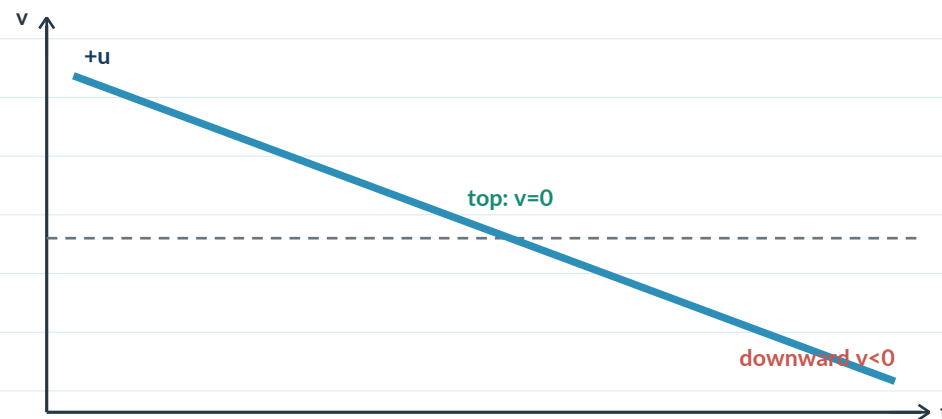
COUNTEREXAMPLE / REPAIR

Negative velocity means motion opposite the chosen positive direction. Speed is a nonnegative magnitude.

Common wrong idea

MISCONCEPTION

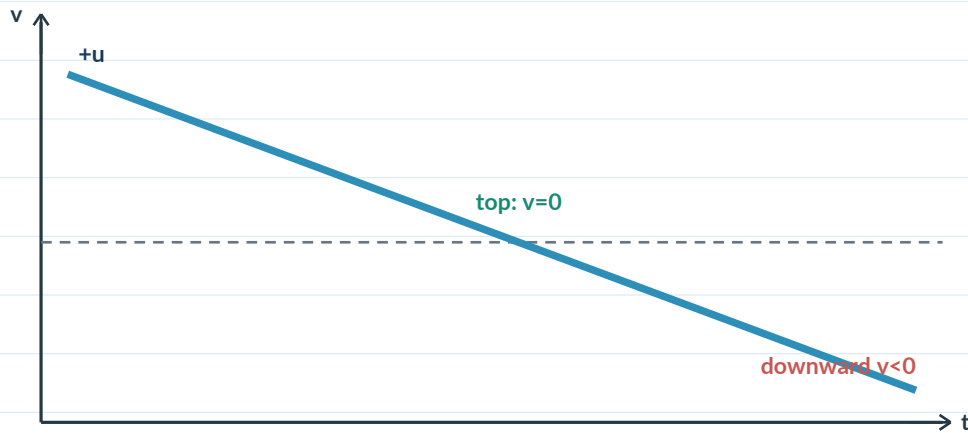
"Below the t -axis means negative speed."



GUIDED

Guided 1 - full visual

A ball is thrown upward at 30 m/s with $g=10$. Sketch its v-t graph through 6 s and label the top.



H1 -> H2 -> H3

H1: slope is $-g$. H2: $v=0$ at $t=u/g$. H3: continue line below axis after the top.

Guided 2 - faded visual

A downward-moving body hits the ground and sticks. Describe the v-t graph immediately before and after impact.

Axes / model: _____

First relation: _____

Physical check: _____

H1 -> H2 -> H3

H1: choose sign. H2: free-flight segment has fixed slope. H3: collision can create a jump.

CHECK

Independent transfer

A vertical throw v - t line crosses zero at 2.5 s. Without knowing u , what is the total same-level flight time in the ideal model?

NO HINT INITIALLY

Draw or annotate the representation yourself before writing an equation.

Self-check

- 1 I keep slope $-g$ through the top.
- 2 I know $v=0$ does not mean $a=0$.
- 3 I interpret negative velocity physically.
- 4 I can model an impact jump.

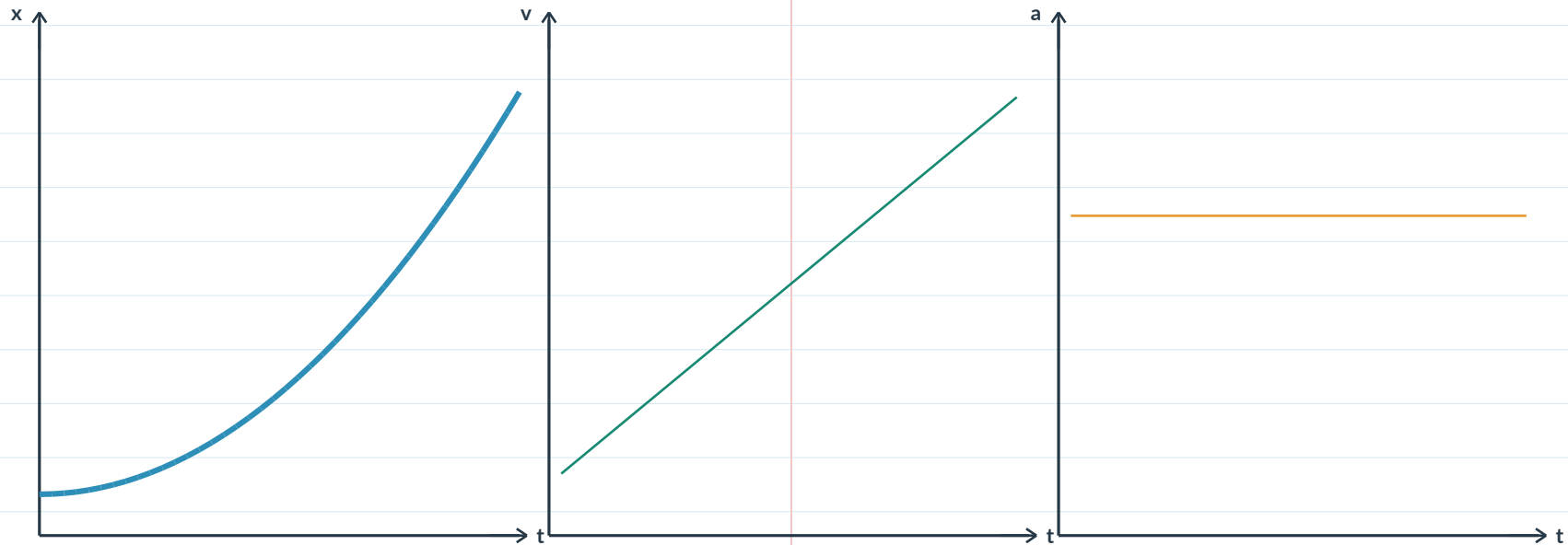
EXAMSIDE LINK STATUS

Current primary set: 2. Secondary linked items: 1.

MOTION IN A STRAIGHT LINE

M-ST10D - Constant Acceleration: x-t Parabola, v-t Line, a-t Constant

Graphs are another language for the same motion - read the axes before using a rule.



MEMORY ANCHOR

constant $a \rightarrow a$ -t horizontal, v -t straight, x -t quadratic.

SOURCE GROUNDING

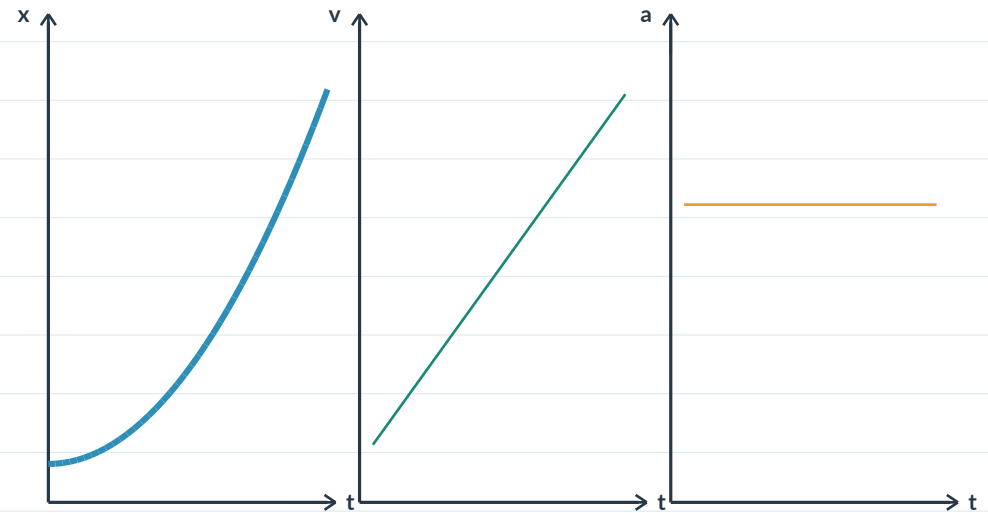
The t^2 term in $x = x_0 + ut + \frac{1}{2}at^2$ makes the position-time graph parabolic. For vertical projection the parabola bends downward. Source anchors: Q66.

RECOG

Recognition cues

- 1 constant/uniform acceleration
- 2 compare x-t, v-t and a-t representations
- 3 from rest: x grows like t^2
- 4 qualitative graph matching

Build the representation



FIRST MOVE

First move: start from the equation family: a constant $\rightarrow$ v linear $\rightarrow$ x quadratic.

WHY

Why

EXPLAINER

Constant a integrates conceptually to a steadily changing velocity; cumulative position then acquires the t^2 term.

Repair / what this is not

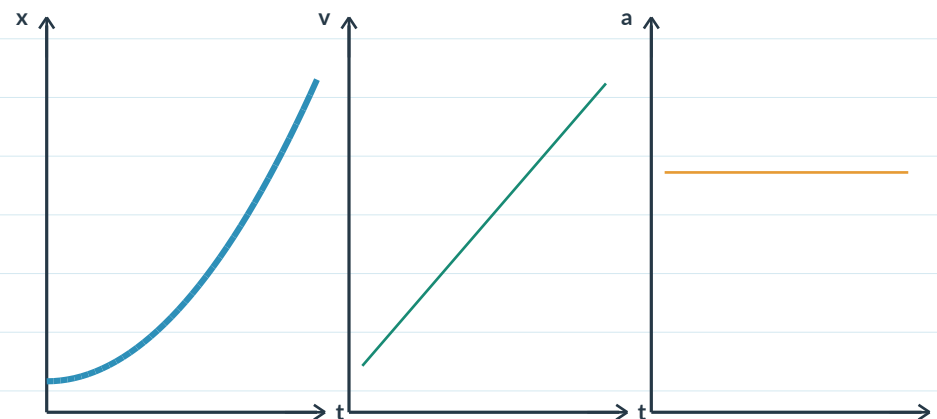
COUNTEREXAMPLE / REPAIR

Only v - t is straight under constant a . x - t is quadratic unless $a=0$.

Common wrong idea

MISCONCEPTION

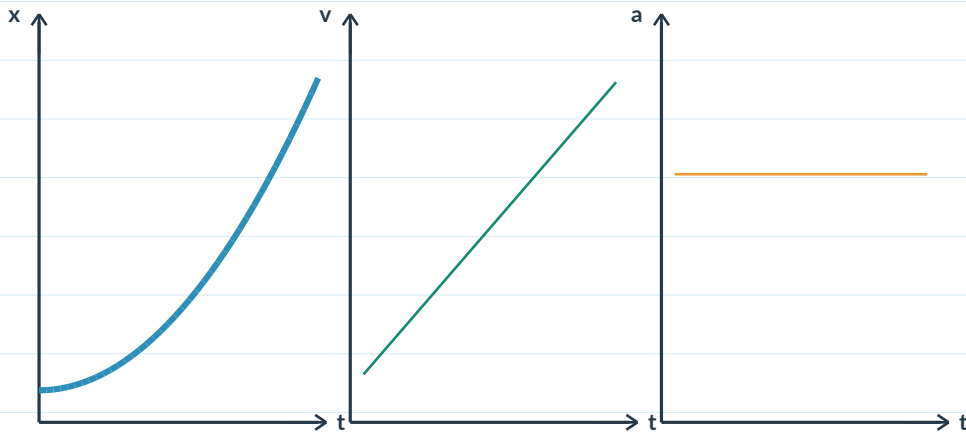
"Constant acceleration gives a straight x - t graph."



GUIDED

Guided 1 - full visual

A particle starts from rest with constant positive a . Sketch a - t , v - t and x - t on one page.



H1 -> H2 -> H3

H1: a is constant. H2: v changes linearly. H3: x accumulates quadratically.

Guided 2 - faded visual

A body's x - t graph is a downward-opening parabola. What can you infer about the sign of acceleration?

Axes / model: _____

First relation: _____

Physical check: _____

H1 -> H2 -> H3

H1: curvature indicates acceleration sign. H2: tangent slope is velocity. H3: turning point can have $v=0$ but curvature remains.

CHECK

Independent transfer

Which representation makes constant acceleration easiest to see directly: x - t , v - t or a - t ? Explain how the other two must then look.

NO HINT INITIALLY

Draw or annotate the representation yourself before writing an equation.

Self-check

- 1 I link a - t , v - t and x - t .
- 2 I recognise a parabola under constant a .
- 3 I can infer curvature sign.
- 4 I can switch representations.

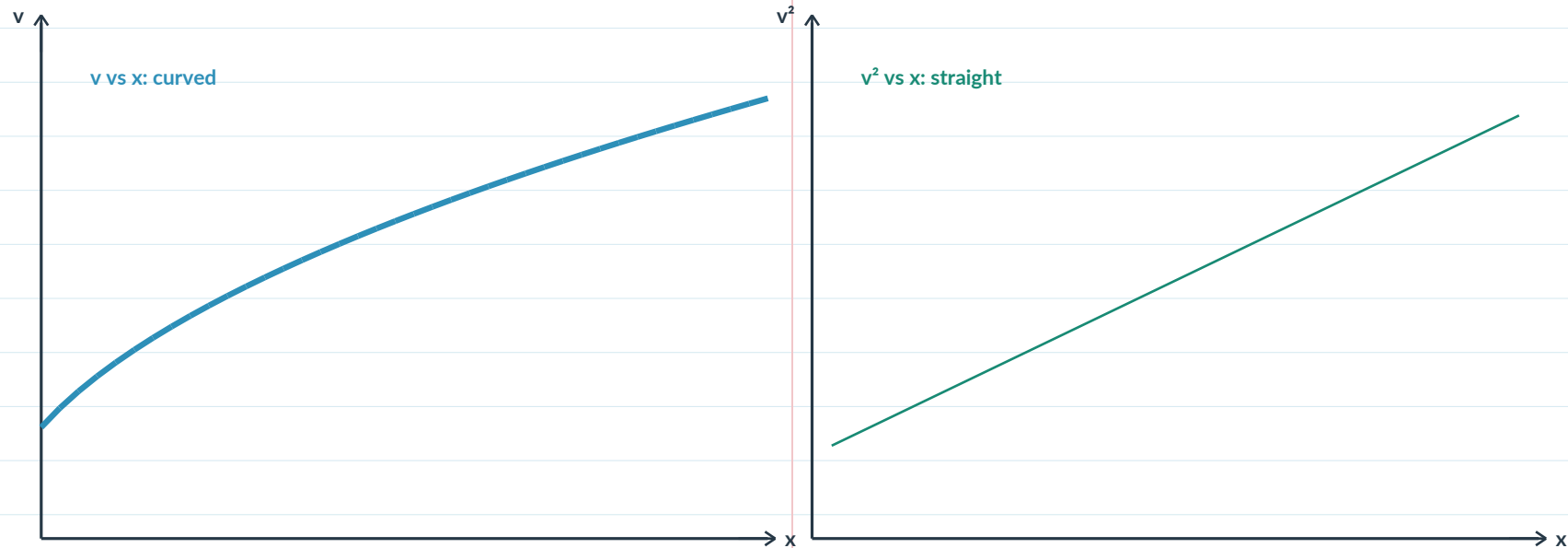
EXAMSIDE LINK STATUS

Current primary set: 2. Secondary linked items: 3.

MOTION IN A STRAIGHT LINE

M-ST10E - Velocity vs Displacement: Read the Axes Before Naming the Curve

Graphs are another language for the same motion - read the axes before using a rule.



MEMORY ANCHOR

v^2 vs x is linear for constant acceleration; v vs x generally is not.

SOURCE GROUNDING

For free fall from rest $v^2=2gs$. Thus v vs s is square-root shaped, while v^2 vs s is a straight line with slope $2g$. Always inspect the axes. Source anchors: Q65.

RECOG

Recognition cues

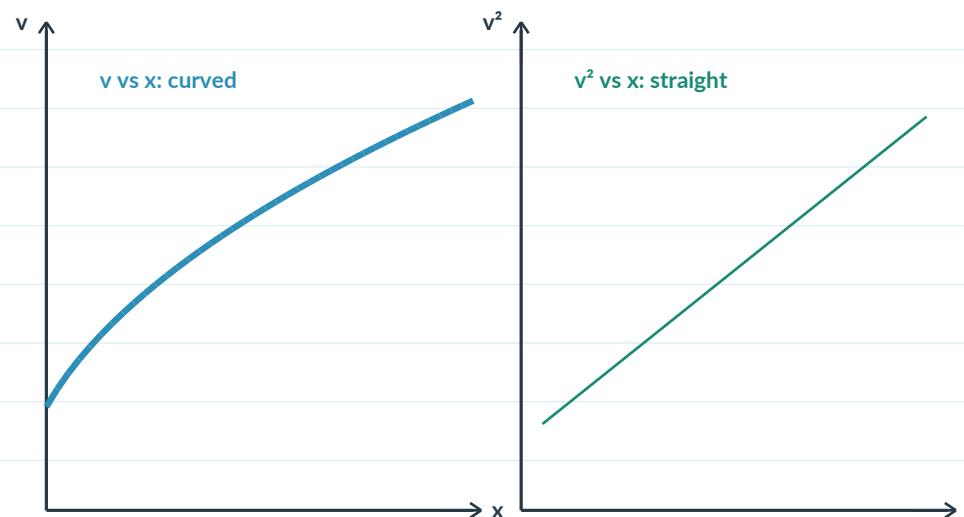
1 horizontal axis is displacement/position, not time

2 v - x or v^2 - x relation

3 constant acceleration hidden in $v^2 = u^2 + 2ax$

4 graph conversion marked extension/challenge

Build the representation



FIRST MOVE

First move: read exactly what is plotted. v - x and v^2 - x are different graphs.

WHY

Why

EXPLAINER

The no-time relation is linear in v^2 , not v . That is why squaring the plotted velocity can turn a curve into a line.

Repair / what this is not

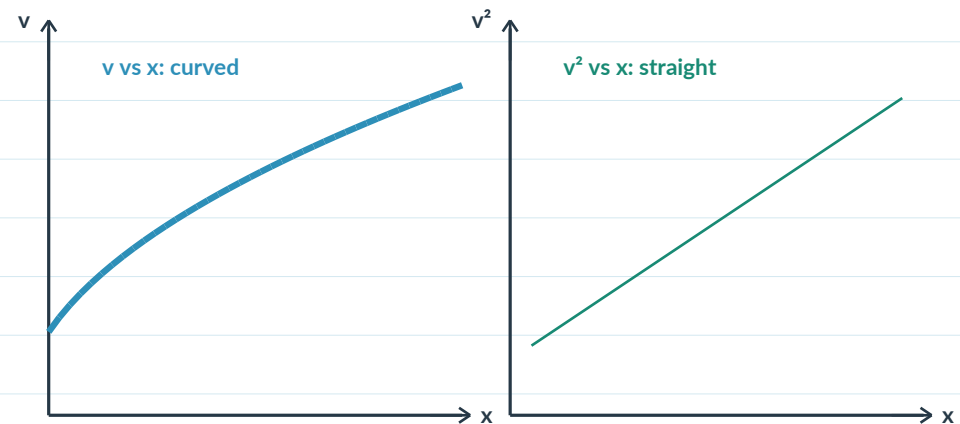
COUNTEREXAMPLE / REPAIR

Always inspect axes. v^2 - x is linear under constant a ; v - x generally has a square-root shape. The $a=v \, dv/dx$ step is labelled foundation extension.

Common wrong idea

MISCONCEPTION

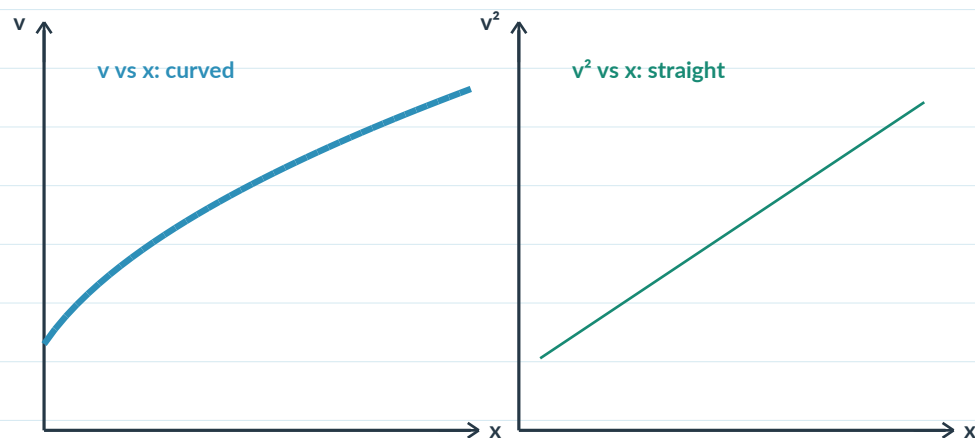
"A velocity-displacement graph must be a straight line under constant acceleration."



GUIDED

Guided 1 - full visual

For $v^2=4+6x$, what is the slope of v^2 - x and what acceleration does it represent?



Guided 2 - faded visual

A v - x graph is straight with negative slope. Explain why the corresponding a - x graph need not have the same slope or shape.

Axes / model: _____

First relation: _____

Physical check: _____

H1 -> H2 -> H3

H1: identify whether vertical axis is v or v^2 . H2: compare to $v^2=u^2+2ax$. H3: slope= $2a$.

H1 -> H2 -> H3

H1: $a=v \, dv/dx$ is extension. H2: multiply v by graph slope dv/dx . H3: interpret sign and intercept.

CHECK

Independent transfer

A straight v^2 - x graph has slope -8 SI units. What is acceleration? What happens to speed as x increases?

NO HINT INITIALLY

Draw or annotate the representation yourself before writing an equation.

Self-check

- 1 I read axes before naming shape.
- 2 I know v^2 - x is linear for constant a .
- 3 I extract a from slope/2.
- 4 I label $a = v \, dv/dx$ as extension.

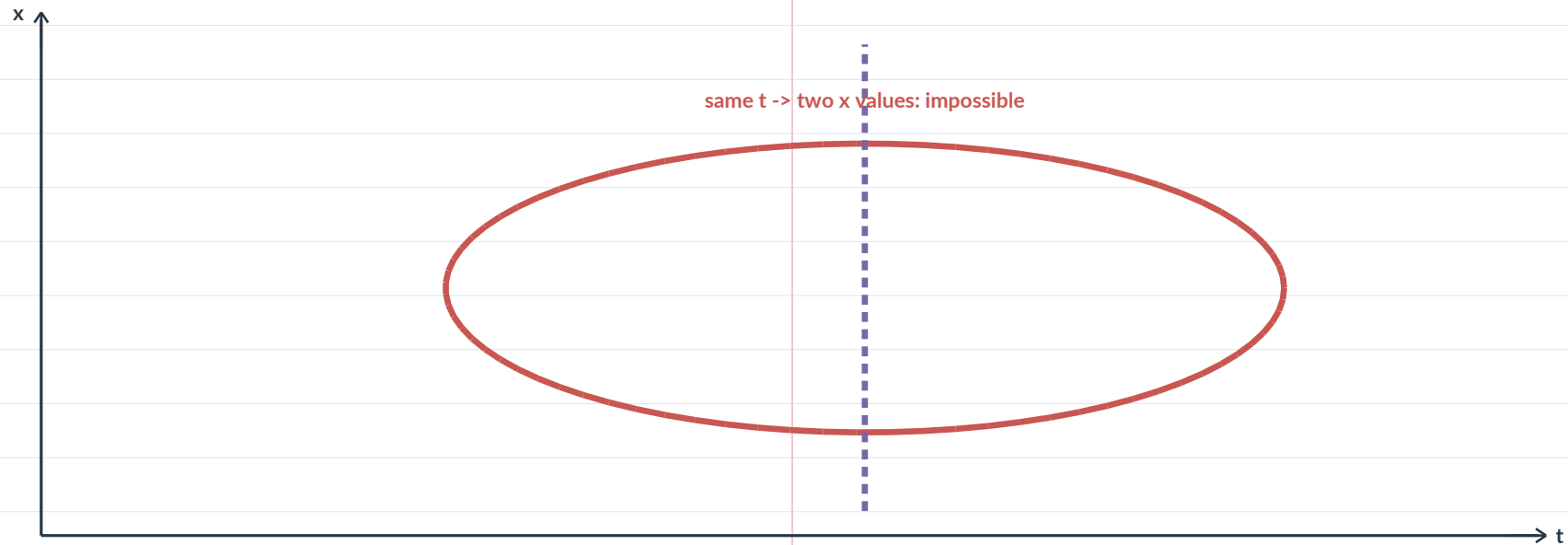
EXAMSIDE LINK STATUS

Current primary set: 4. Secondary linked items: 2.

MOTION IN A STRAIGHT LINE

M-ST10F - Impossible and Inconsistent Motion Graphs

Graphs are another language for the same motion - read the axes before using a rule.



MEMORY ANCHOR

one particle + one instant $\rightarrow$ one position and one velocity.

SOURCE GROUNDING

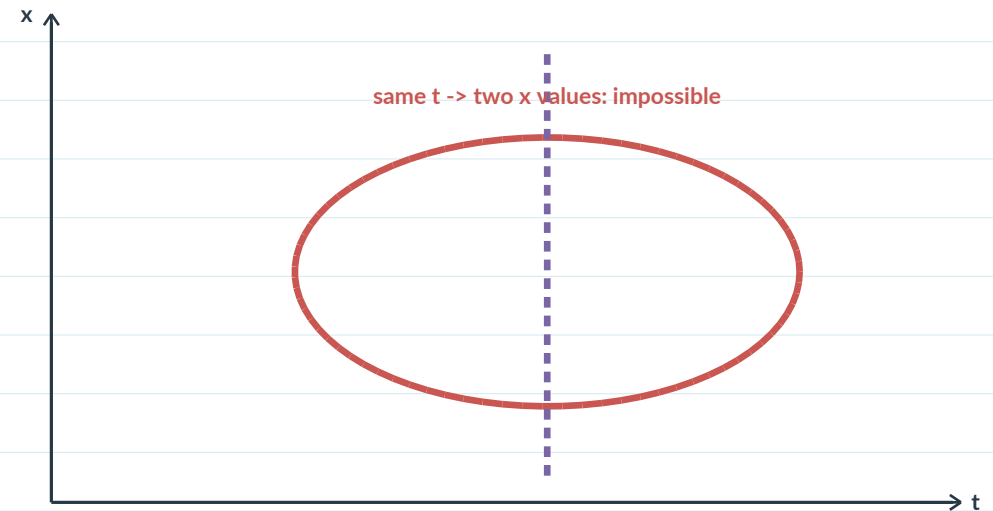
x - t and v - t histories must be single-valued in time. A closed loop fails the vertical-line test. A vertical v - t segment means a finite velocity change in zero time - unbounded acceleration in the ideal limit. Source anchors: Q67, Q68.

RECOG

Recognition cues

- 1 'possible graph?' or 'same motion?'
- 2 closed loop on x - t or v - t
- 3 two values at the same time
- 4 multiple graph representations must tell one story

Build the representation



FIRST MOVE

First move: apply the vertical-line test in time-based graphs, then check cross-representation consistency.

WHY

Why

EXPLAINER

A motion graph is a function of time: one time input cannot map to two positions or two velocities for one particle in one frame.

Repair / what this is not

COUNTEREXAMPLE / REPAIR

A discontinuity may model an impulse, but two different values at the same time are not a valid single-valued history.

Common wrong idea

MISCONCEPTION

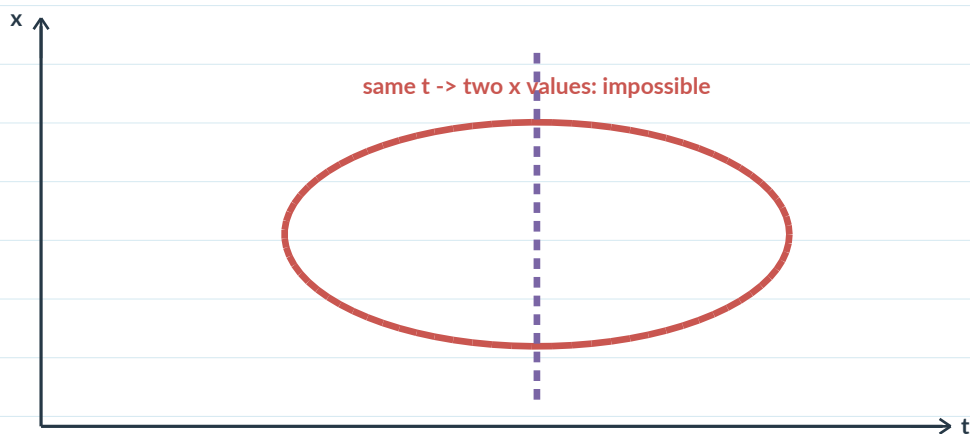
“Any smooth-looking curve can represent motion.”



GUIDED

Guided 1 - full visual

A proposed x-t graph is a closed loop. Explain exactly why it cannot be one particle's motion history.

H1 $\rightarrow$ H2 $\rightarrow$ H3

H1: choose one time. H2: draw a vertical line. H3: more than one intersection means impossible.

Guided 2 - faded visual

Four graphs claim to describe the same motion. Write a checklist for converting each into position/velocity/acceleration statements before comparing.

Axes / model: _____

First relation: _____

Physical check: _____

H1 $\rightarrow$ H2 $\rightarrow$ H3

H1: translate each graph separately. H2: compare initial state. H3: compare turning point and direction.

CHECK

Independent transfer

A v-t graph contains a loop. A second candidate contains one vertical jump but is otherwise single-valued. Which feature is fundamentally impossible for a one-particle history?

NO HINT INITIALLY

Draw or annotate the representation yourself before writing an equation.

Self-check

- 1 I use the vertical-line test.
- 2 I distinguish jump from multi-valued loop.
- 3 I compare initial states across graphs.
- 4 I demand one consistent motion story.

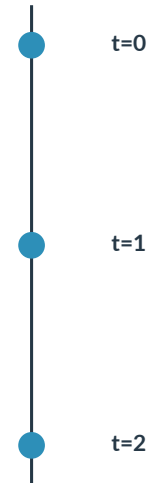
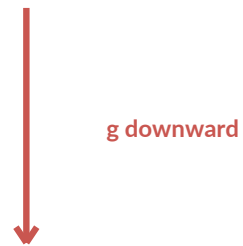
EXAMSIDE LINK STATUS

Current primary set: 2. Secondary linked items: 0.

MOTION IN A STRAIGHT LINE

M-ST11A - Average Velocity Numerically Related to g

FOUNDATION EXTENSION | Short derivation lab grounded to the supplied source bank.



MEMORY ANCHOR

Translate the verbal number condition into an equation before solving.

SOURCE GROUNDING

For free fall from rest, $v=gt$ and average velocity $= (0+v)/2 = gt/2$. If average velocity is numerically $g/2$, then $t=1$ s and $v=g$. Source anchor: Q40.

APPLY

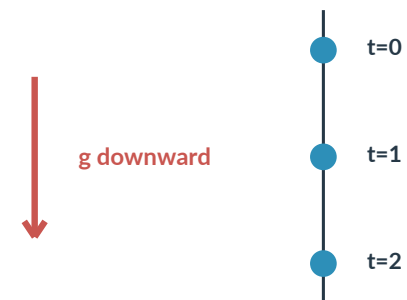
When do I see this?

- 1 free fall from rest
- 2 average velocity is compared numerically with g or a multiple of g
- 3 wording asks for time / final velocity from that numerical condition

First move

FIRST EXECUTABLE MOVE

Write $v=gt$ and $v_{avg}=gt/2$ first. Only then translate the numerical statement into an equation.



HELPER

Helper

$$v = gt$$

$$v_{\text{avg}} = gt/2$$

NUMERICAL CONDITION

If v_{avg} has numerical value $g/2$, then $gt/2 = g/2 \rightarrow t = 1 \text{ s}$.

Physical / mathematical check

CHECK

If $t = 1 \text{ s}$ in free fall from rest, $v = g$ and the average of 0 and g is $g/2$. The story is internally consistent.

WHY THIS MATTERS

The source lab is testing reconstruction: translate a condition, combine known relations, or expose a common factor instead of memorising a special-case formula.

BOUND

Misconception clinic

WRONG IDEA

“Numerically related to g means the quantities have the same units.”

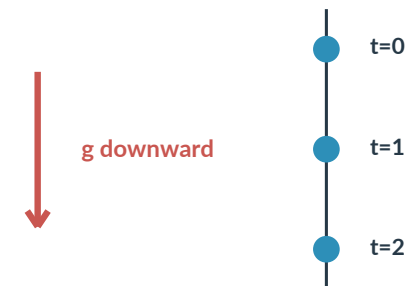
REPAIR

The comparison is about numerical values in a stated unit convention. Keep dimensions conceptually separate.

Variant / anti-trigger

VARIANT / WHEN NOT TO USE

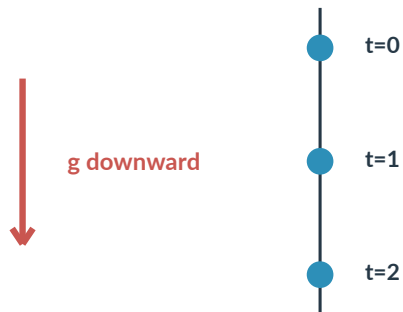
Variant: if v_{avg} is numerically $3g/2$, then $gt/2 = 3g/2 \rightarrow t = 3 \text{ s}$. Anti-trigger: this shortcut assumes free fall from rest with constant g .



GUIDED

Guided 1 - full support

A body falls freely from rest. Its average velocity over the elapsed time is numerically equal to g . Find the elapsed time.



Guided 2 - faded support

A freely falling body from rest has average velocity numerically equal to $2g$. What is its final velocity in terms of g ?

Recognition cue: _____

Model / equations: _____

First algebra step: _____

H1 -> H2 -> H3

H1: free fall from rest. H2: $v_{\text{avg}} = gt/2$. H3: set $gt/2 = g$.

H1 -> H2 -> H3

H1: use the same average-velocity expression. H2: solve time first. H3: final $v = gt$.

CHECK

Independent transfer

A free-fall problem states that the average velocity has numerical value 14.7 in SI units while $g=9.8$. Find the elapsed time and final speed.

NO HINT INITIALLY

Write the physical relations first. The goal of Unit 11 is to construct the needed result, not recall a named shortcut.

Self-check

- 1 I translate a numerical verbal condition into an equation.
- 2 I can derive v_{avg} for free fall from rest.
- 3 I keep units conceptually distinct.
- 4 I can generalise to another multiple of g .

EXTERNAL TRANSFER STATUS

Current ExamSIDE primary = 0; secondary cross-links = 0.

MOTION IN A STRAIGHT LINE

M-ST11B - When Velocity and Displacement Are Numerically Equal

FOUNDATION EXTENSION | Short derivation lab grounded to the supplied source bank.

20

m/s

≠

20

m

same number, different units

MEMORY ANCHOR

same numerical value does not mean same physical quantity.

SOURCE GROUNDING

For free fall from rest, $v=gt$ and $s=1/2 gt^2$. If their numerical values match in SI units, $t=2$ s and $v=2g$. The dimensions remain different. Source anchor: Q42.

APPLY

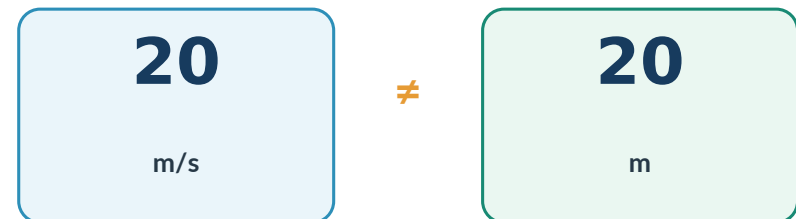
When do I see this?

- 1 two unlike quantities are said to be 'numerically equal'
- 2 units are specified or implied
- 3 the quantities can be written as functions of the same time

First move

FIRST EXECUTABLE MOVE

Write both physical expressions with units kept visible. Equate only their numerical values in the stated unit system.



same number, different units

HELPER

Helper

$$v = gt$$

$$s = \frac{1}{2} gt^2$$

NUMBERS, NOT DIMENSIONS

gt and $\frac{1}{2}gt^2$ can have the same numerical value in SI units while carrying different units.

Physical / mathematical check

CHECK

At $t=2$ s with $g=10$, $v=20$ m/s and $s=20$ m. The numbers match; the units do not.

WHY THIS MATTERS

The source lab is testing reconstruction: translate a condition, combine known relations, or expose a common factor instead of memorising a special-case formula.

BOUND

Misconception clinic

WRONG IDEA

"If velocity and displacement are numerically equal, they are physically equal."

REPAIR

Equality of numbers such as 20 and 20 does not erase the difference between metres and metres per second.

Variant / anti-trigger

VARIANT / WHEN NOT TO USE

Variant: if numerical v equals 2 times numerical s , write $gt=2(1/2gt^2)$ and solve. Anti-trigger: do not equate quantities when the problem never states a numerical relation.

20

m/s

≠

20

m

same number, different units

GUIDED

Guided 1 - full support

In SI units, a freely falling body from rest has the same numerical values of speed and displacement. Find the time.

20

m/s

≠

20

m

same number, different units

Guided 2 - faded support

In SI units, the numerical speed is twice the numerical displacement. Find the nonzero time.

Recognition cue: _____

Model / equations: _____

First algebra step: _____

H1 -> H2 -> H3

H1: 'same numerical values'. H2: write gt and $1/2gt^2$. H3: cancel gt only after excluding $t=0$.

H1 -> H2 -> H3

H1: translate 'twice'. H2: $gt = 2(1/2gt^2)$. H3: solve the nonzero root.

CHECK

Independent transfer

A problem says, in SI units, that the numerical displacement is three times the numerical speed during free fall from rest. Form the equation and solve for nonzero time.

NO HINT INITIALLY

Write the physical relations first. The goal of Unit 11 is to construct the needed result, not recall a named shortcut.

Self-check

- 1 I understand numerical equality vs dimensional equality.
- 2 I keep the $t=0$ root in mind before choosing the physical root.
- 3 I can change 'equal' to a ratio condition.
- 4 I can check with actual SI numbers.

EXTERNAL TRANSFER STATUS

Current ExamSIDE primary = 0; secondary cross-links = 2.

MOTION IN A STRAIGHT LINE

M-ST11C - Air Resistance Proportional to Mass: Why Mass Cancels

FOUNDATION EXTENSION | Short derivation lab grounded to the supplied source bank.



both forces scale with mass $\rightarrow$ same acceleration

MEMORY ANCHOR

If every force term carries the same factor m , divide the whole equation by m .

SOURCE GROUNDING

With downward positive and upward resistance km , $ma = mg - km = m(g - k)$, so $a = g - k$. Different masses can have the same acceleration under this stated model. Source anchor: Q41.

APPLY

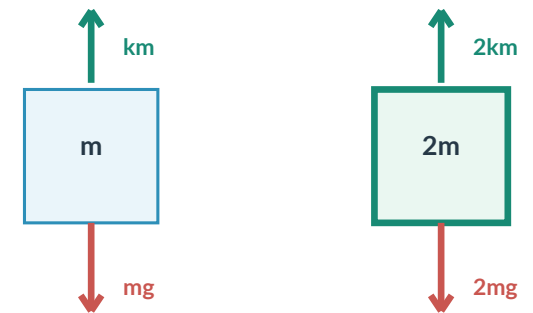
When do I see this?

- 1 force model explicitly says resistance is proportional to mass
- 2 gravity and resistance act in opposite directions
- 3 question asks whether acceleration depends on mass

First move

FIRST EXECUTABLE MOVE

Draw the force arrows and write Newton's second law before cancelling anything.



both forces scale with mass -> same acceleration

HELPER

Helper

$$ma = mg - km$$

$$a = g - k$$

WHY MASS CANCELS

Both gravity and the stated resistance are proportional to m , so the common factor divides out.

Physical / mathematical check

CHECK

If m doubles, both mg and km double. Net force doubles too, while acceleration remains unchanged.

WHY THIS MATTERS

The source lab is testing reconstruction: translate a condition, combine known relations, or expose a common factor instead of memorising a special-case formula.

BOUND

Misconception clinic

WRONG IDEA

"Mass always cancels in falling-with-resistance problems."

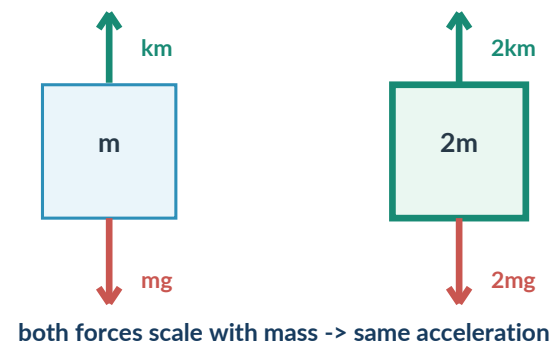
REPAIR

Mass cancels only under the stated force model $F_{\text{res}}=km$. Other drag laws may depend on speed, area, or be independent of mass.

Variant / anti-trigger

VARIANT / WHEN NOT TO USE

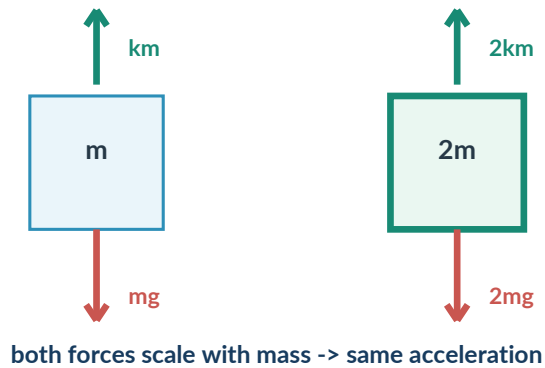
Variant: with resistance $0.2mg$, acceleration is $0.8g$. Anti-trigger: a constant force R gives $a=g-R/m$, so mass does not cancel.



GUIDED

Guided 1 - full support

A body falls with downward weight mg and upward resistance $0.25mg$. Find its acceleration.



Guided 2 - faded support

Two masses m and $3m$ fall under gravity with upward resistance km . Compare their accelerations.

Recognition cue: _____

Model / equations: _____

First algebra step: _____

H1 -> H2 -> H3

H1: resistance is a fraction of mg . H2: net downward force = $mg - 0.25mg$. H3: divide by m .

H1 -> H2 -> H3

H1: both forces contain m . H2: write $ma = m(g - k)$. H3: compare after cancellation.

CHECK

Independent transfer

A model gives downward weight mg and upward resistance $c\,m$, where c has acceleration units. Explain why two different masses have the same acceleration, then state one drag model where they would not.

NO HINT INITIALLY

Write the physical relations first. The goal of Unit 11 is to construct the needed result, not recall a named shortcut.

Self-check

- 1 I draw forces before cancelling.
- 2 I know exactly why m cancels.
- 3 I state the force-model assumption.
- 4 I can name a case where mass would not cancel.

EXTERNAL TRANSFER STATUS

Current ExamSIDE primary = 0; secondary cross-links = 1.

MOTION IN A STRAIGHT LINE

M-ST11D - Ratio of nth-Second Distance to Total Distance

FOUNDATION EXTENSION | Short derivation lab grounded to the supplied source bank.

successive equal-time distances: 1, 3, 5, ...



nth interval / total through n = $(2n-1)/n^2$

MEMORY ANCHOR

construct the ratio from two known relations; do not memorize another formula.

SOURCE GROUNDING

From rest, total through n seconds $S_n = 1/2 a n^2$ and nth-second distance $s_n = 1/2 a (2n-1)$, so $s_n/S_n = (2n-1)/n^2$. Source anchor: Q34.

APPLY

When do I see this?

- 1 motion starts from rest with constant acceleration
- 2 question compares one interval distance with cumulative total or another interval
- 3 nth-second / consecutive-second language

First move

FIRST EXECUTABLE MOVE

Write the two quantities being compared separately: cumulative S_n and interval s_n . Form the ratio only afterward.

successive equal-time distances: 1, 3, 5, ...



nth interval / total through n = $(2n-1)/n^2$

HELPER

Helper

TOTAL THROUGH n

$$S_n = \frac{1}{2} a n^2$$

ONLY THE n th INTERVAL

$$s_n = \frac{1}{2} a (2n-1)$$

$$s_n/S_n = (2n-1)/n^2$$

Physical / mathematical check

CHECK

For $n=4$, interval factors are 1,3,5,7 and total factor is 16, so $s_4/S_4=7/16$, matching the formula.

WHY THIS MATTERS

The source lab is testing reconstruction: translate a condition, combine known relations, or expose a common factor instead of memorising a special-case formula.

BOUND

Misconception clinic

WRONG IDEA

"The nth-second ratio is another formula I must memorize."

REPAIR

Rebuild the ratio from S_n and s_n . If $u \neq 0$, the pure from-rest odd-sequence form changes.

Variant / anti-trigger

VARIANT / WHEN NOT TO USE

Variant: consecutive interval ratio is $s_{(n-1)}/s_n = (2n-3)/(2n-1)$. Anti-trigger: cumulative S_n and interval s_n are different objects.

successive equal-time distances: 1, 3, 5, ...



nth interval / total through $n = (2n-1)/n^2$

GUIDED

Guided 1 - full support

From rest under constant acceleration, what fraction of the first 5-second total displacement occurs during the 5th second?

successive equal-time distances: 1, 3, 5, ...



$\text{nth interval / total through } n = (2n-1)/n^2$

Guided 2 - faded support

For motion from rest, find s_7/s_6 without first finding a .

Recognition cue: _____

Model / equations: _____

First algebra step: _____

H1 -> H2 -> H3

H1: compare interval with total. H2: s_5 and S_5 . H3: fraction=9/25.

H1 -> H2 -> H3

H1: both are interval distances. H2: s_{2n-1} . H3: ratio=(13)/(11).

CHECK

Independent transfer

A body starts from rest with constant acceleration. During which second does the interval displacement first exceed one-fifth of the total displacement accumulated up to that second?

NO HINT INITIALLY

Write the physical relations first. The goal of Unit 11 is to construct the needed result, not recall a named shortcut.

Self-check

- 1 I separate cumulative and interval distance.
- 2 I rebuild s_n/S_n .
- 3 I derive consecutive-interval ratios.
- 4 I avoid using the from-rest form when u is nonzero.

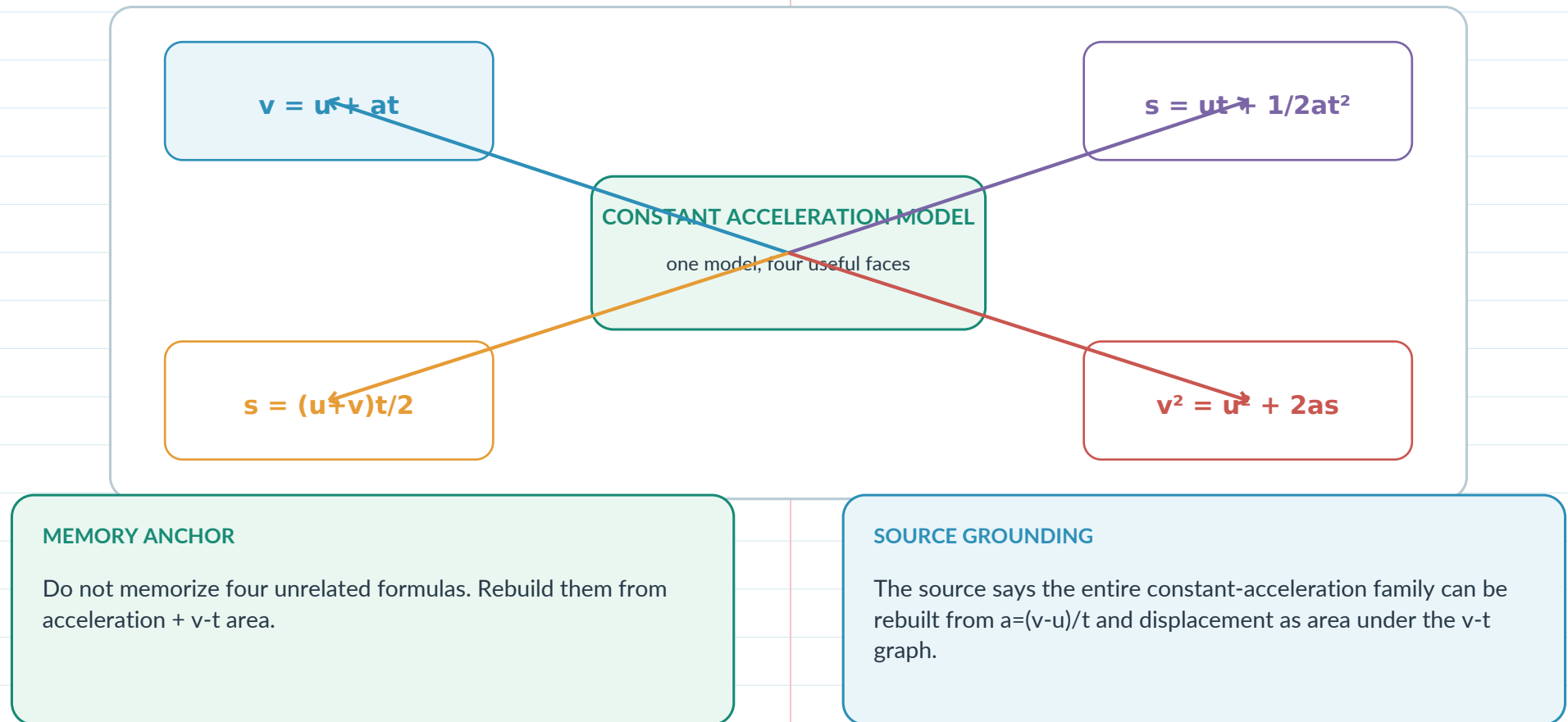
EXTERNAL TRANSFER STATUS

Current ExamSIDE primary = 1; secondary cross-links = 2.

MOTION IN A STRAIGHT LINE

M-ST12A - Rebuild the Equation Family from Two Ideas

CONSOLIDATION | Make formula choice a modelling decision, not a memory contest.



M12A-C1

Idea 1: acceleration

$$a = (v-u)/t$$

$$v = u + at$$

PHYSICAL STORY

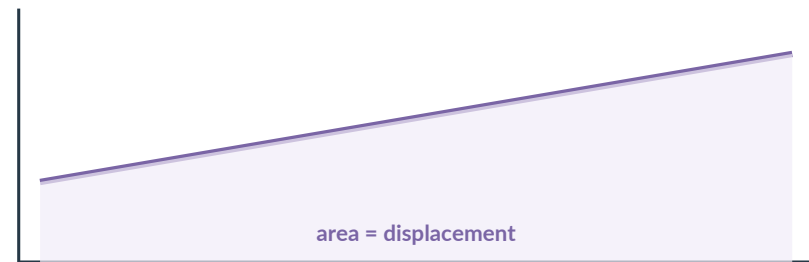
Final velocity = starting velocity + accumulated change in velocity.

Idea 2: area under v-t

IDEA 1: acceleration definition

$$a = (v-u)/t \rightarrow v = u + at$$

IDEA 2: area under v-t



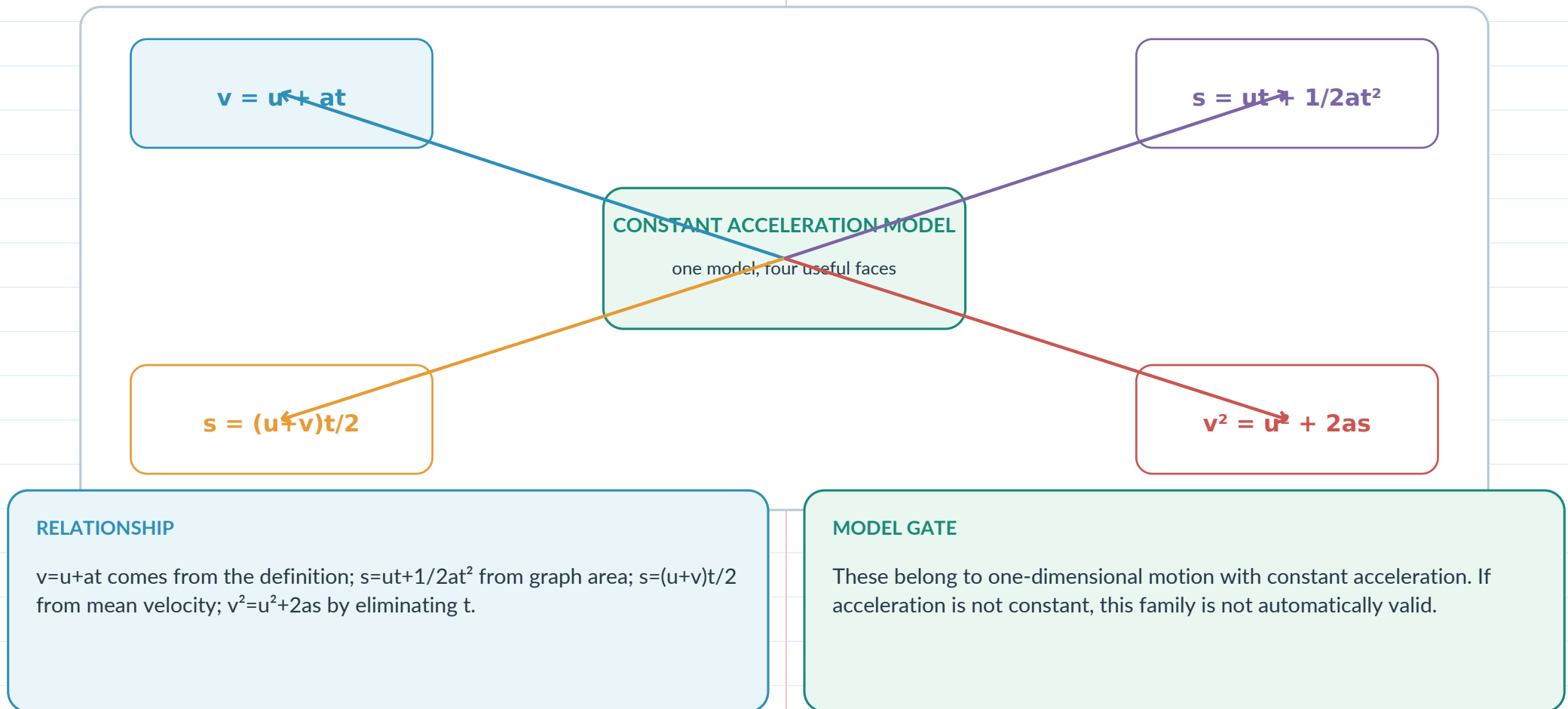
WHY AREA

Velocity x time has units of displacement. The graph makes the 1/2 visible as a triangle, not a magic coefficient.

BUILD THE FOUR FACES

One model, several useful eliminations

M12A-C2



M12A-C3

Wrong idea

WRONG IDEA

"I need to remember the first, second, third and fourth equations separately."

That encourages label-searching instead of model recognition.

REPAIR

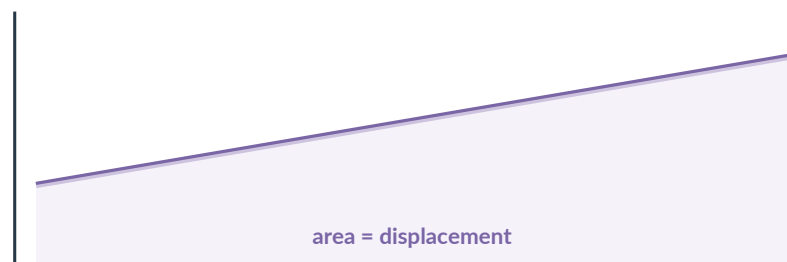
Remember the model and the two starting ideas. Rebuild the face you need.

Visual reconstruction

IDEA 1: acceleration definition

$$a = (v-u)/t \rightarrow v = u + at$$

IDEA 2: area under v-t



CHECK

Can you explain where the $1/2$ and t^2 come from without naming a formula?

M12A-G

Guided 1 - full support

A car starts at u and accelerates uniformly at a for time t . Rebuild its displacement relation from a v - t graph.

IDEA 1: acceleration definition

$$a = (v-u)/t \rightarrow v = u + at$$

IDEA 2: area under v - t

area = displacement

Guided 2 - faded

Starting from $v=u+at$ and $s=(u+v)t/2$, eliminate t to recover the time-free relation.

Substitute $t =$ _____

Simplify product: _____

Final relation: _____

H1 -> H2 -> H3

H1: draw initial velocity rectangle. H2: add acceleration triangle. H3: area = $ut + 1/2(at)t$.

H1 -> H2 -> H3

H1: $t=(v-u)/a$. H2: use $(u+v)(v-u)=v^2-u^2$. H3: rearrange.

M12A-T

Independent transfer

Without looking back, reconstruct all four constant-acceleration relations from the two starting ideas. Next to each relation, write what variable it avoids.

NO HINT

If one step is forgotten, go back to meaning: velocity change per time and area under v - t .

Self-check

- 1 I can derive $v=u+at$.
- 2 I can derive displacement from v - t area.
- 3 I can derive average-velocity form.
- 4 I can eliminate time to get v^2 relation.
- 5 I can state the constant- a model gate.

MOTION IN A STRAIGHT LINE

M-ST12B - Which Equation Should I Use?

CONSOLIDATION | Make formula choice a modelling decision, not a memory contest.

IF THIS IS ABSENT	USE	WHY
s absent	$v = u + at$	no displacement needed
v absent	$s = ut + \frac{1}{2}at^2$	no final velocity needed
t absent	$v^2 = u^2 + 2as$	direct speed-distance relation
a absent	$s = \frac{(u+v)t}{2}$	endpoint velocities + time

MEMORY ANCHOR

Model first. Then choose the relation that leaves one unnecessary variable absent.

interv

SOURCE GROUNDING

The source selection matrix maps absent s, v, t, a, and nth-second wording to the most economical relation, with an explicit warning that modelling comes first.

THE DECISION ROUTINE

WORDS -> MODEL -> UVATS -> EXCLUDE -> CHOOSE

M12B-C1

Before equations

- 1 MODEL** Is acceleration constant?
- 2 SIGNS** Choose + direction.
- 3 TRANSLATE** rest -> $u=0$; stop -> $v=0$.
- 4 TARGET** Circle the asked quantity.
- 5 EXCLUDE** Which variable can stay absent?
- 6 CHOOSE** Use the relation that avoids it.
- 7 CHECK** units, sign, magnitude, model.

Then use the selector

IF THIS IS ABSENT	USE	WHY
s absent	$v = u + at$	no displacement needed
v absent	$s = ut + \frac{1}{2}at^2$	no final velocity needed
t absent	$v^2 = u^2 + 2as$	direct speed-distance
a absent	$s = \frac{(u+v)t}{2}$	endpoint velocities +
nth second	$S(n) - S(n-1)$	interval, not cumulative

CORE RULE

The matrix is an aid after modelling, not a substitute for modelling.

FORMULA CHOICE MADE VISIBLE

Cross out the variable you do not need

M12B-C2

Example A: stopping time

Car: $u=20$ m/s, $a=-5$ m/s², stops. Target: t .

u	v	a	t	s
20	0	-5	?	-

CHOOSE

s is unnecessary $\rightarrow$ use $v=u+at$.

Example B: stopping distance

Same car, same braking. Target: s .

u	v	a	t	s
20	0	-5	-	?

CHOOSE

t is unnecessary $\rightarrow$ use $v^2=u^2+2as$.

ANTI-TRIGGERS: WHEN THE MATRIX IS NOT ENOUGH

Classification before selection

M12B-C3

Do not force UVATS onto...

NONCONSTANT a

If acceleration changes with time or position, the standard constant- a equations are not automatically valid.

TWO-BODY / DELAYED START

Build the common timeline or relative model first; then each stage may use constant- a relations.

Special structures

n th SECOND

Use $S(n)-S(n-1)$. Do not confuse 'during the n th second' with total at $t=n$.

GRAPH QUESTION

Read slope/area directly when the representation already gives the physics more cleanly than algebra.

GUIDED 1 + GUIDED 2

Make the equation obvious

M12B-G

Guided 1 - full support

A train at 12 m/s accelerates uniformly at 2 m/s² for 5 s. Find displacement.

u

12

v

-

a

2

t

5

s

?

Guided 2 - faded

A body speeds from 8 to 20 m/s over 84 m with constant acceleration. Find a.

Known: _____

Unnecessary variable: _____

Equation: _____

H1 -> H2 -> H3

H1: constant a. H2: v is not needed. H3: choose $s=ut+\frac{1}{2}at^2$.

H1 -> H2 -> H3

H1: u,v,s known. H2: time is unnecessary. H3: use $v^2=u^2+2as$.

M12B-T

I'M STUCK - WHICH EQUATION?

- 1 **MODEL** Is acceleration constant?
- 2 **SIGNS** Choose + direction.
- 3 **TRANSLATE** rest $\rightarrow u=0$; stop $\rightarrow v=0$.
- 4 **TARGET** Circle the asked quantity.
- 5 **EXCLUDE** Which variable can stay absent?
- 6 **CHOOSE** Use the relation that avoids it.
- 7 **CHECK** units, sign, magnitude, model.

Independent mixed transfer

For each mini-story, do not solve. Only name the model, target, unnecessary variable, and first equation: (a) braking distance, (b) delayed second car meets first, (c) 7th-second distance, (d) v-t graph displacement.

SUCCESS CRITERION

The correct answer is not merely an equation name. It is a justified route: model $\rightarrow$ representation $\rightarrow$ equation.

MOTION IN A STRAIGHT LINE

M-ST12C - Reconstruction Test: Can You Explain the Physics?

CONSOLIDATION | Make formula choice a modelling decision, not a memory contest.

t^2 Why does acceleration contribute t^2 to displacement?

v^2 Why does the time-free relation contain squared velocities?

u^2 Why is stopping distance proportional to u^2 ?

1,3,5 Why do equal-time distances from rest follow odd numbers?

$v=0$ How can $v=0$ while a is not zero?

g cancels Why can common gravity disappear from relative motion?

MEMORY ANCHOR

If you can explain why the equation has that shape, you can rebuild it when memory fails.

SOURCE GROUNDING

The source reconstruction test asks why t^2 appears, why velocities are squared, why stopping distance scales with u^2 , why odd distances appear, why $v=0$ can coexist with $a \neq 0$, why g can cancel relatively, and why v - t area has displacement dimensions.

SEVEN RECONSTRUCTION QUESTIONS

Explain before calculating

M12C-C1

t^2 Why does acceleration contribute t^2 to displacement?

v^2 Why does the time-free relation contain squared velocities?

u^2 Why is stopping distance proportional to u^2 ?

1,3,5 Why do equal-time distances from rest follow odd numbers?

$v=0$ How can $v=0$ while a is not zero?

g cancels Why can common gravity disappear from relative motion?

area Why does v - t area have displacement units?

RULE

Answer in ordinary language first. Then rebuild the equation or graph consequence.

WHY

This detects whether the learner owns the structure or only remembers formula labels.

EXPLAINER 1: WHY t^2 AND WHY v^2 ?

Two mathematical fingerprints

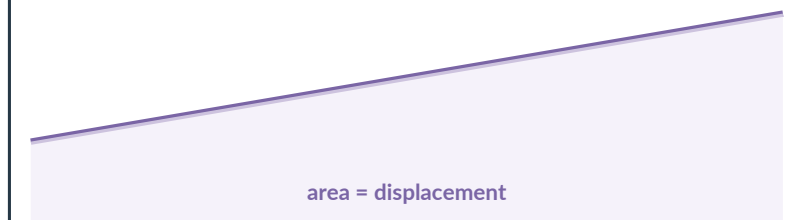
M12C-C2

Why t^2 in displacement?

IDEA 1: acceleration definition

$$a = (v-u)/t \rightarrow v = u + at$$

IDEA 2: area under v-t



ORDINARY LANGUAGE

Acceleration makes velocity grow with time. Displacement accumulates that growing velocity, so the extra contribution grows like time x time.

Why squared velocities?

$$t = (v-u)/a$$

$$s = (u+v)t/2$$

ELIMINATE t

Multiplying $(u+v)(v-u)$ produces $v^2 - u^2$. The squares are an algebraic fingerprint of removing time.

EXPLAINER 2: STOPPING, ODD DISTANCES, TURNING POINTS

Physical consequences

M12C-C3

Stopping distance

$$0 = u^2 + 2as$$

SCALING

For fixed braking magnitude, s_{stop} is proportional to u^2 . Double speed $\rightarrow$ four times distance.

Turning point

$$v=0, a \neq 0$$

At the top of vertical throw, velocity is zero for an instant but gravity still changes velocity, causing reversal.

ODD DISTANCES

From rest, cumulative t^2 totals have first differences 1,3,5,... . The odd pattern is the discrete signature of a quadratic law.

EXPLAINER 3: RELATIVE g + v - t AREA

Representation and subtraction

M12C-C4

Why common g can cancel

$$a_{\text{rel}} = (-g) - (-g) = 0$$

IMPORTANT

Gravity still acts on each body. It disappears only from the relative acceleration because the same vector is subtracted.

Why v - t area is displacement

$$(\text{m/s}) \times \text{s} = \text{m}$$

SIGNED AREA

Positive and negative velocity regions contribute opposite displacement signs. Distance needs absolute areas.

M12C-G

Guided 1

Why does doubling initial speed quadruple ideal stopping distance under the same braking acceleration?

H1 -> H2 -> H3

H1: stop -> $v=0$. H2: time is unnecessary. H3: solve $v^2=u^2+2as$ for s and inspect the power of u .

Guided 2 - faded

Explain why successive one-second distances from rest form an arithmetic sequence without quoting the n th-second formula.

Cumulative law: _____

First differences: _____

Conclusion: _____

H1 -> H2 -> H3

H1: totals scale as n^2 . H2: subtract consecutive squares. H3: differences are $2n-1$.

M12C-T

Teach-back challenge

Choose any three prompts from the reconstruction test. Explain each to a younger student using a physical story, a diagram or graph, and at most

PASS STANDARD

A correct explanation must include the model assumption, the physical mechanism, and one check or counterexample.

Mastery self-check

- 1 I can explain t^2 physically.
- 2 I can explain v^2 algebraically and physically.
- 3 I can justify stopping-distance scaling.
- 4 I can explain odd interval distances.
- 5 I can explain $v=0$ with a nonzero a .
- 6 I can explain relative g cancellation.
- 7 I can explain signed graph area.

APPENDIX A - SOURCE Q1-Q68 TO CONCEPT MAP

Part 1

SRC

Q	Concept link(s)	Core idea / source location
1	M-ST01A	distance vs displacement
2	M-ST01A	distance vs displacement
3	M-ST01B	arc distance vs chord displacement
4	M-ST01A	closed path: zero displacement
5	M-ST01B	arc distance vs chord displacement
6	M-ST02A	average speed vs average velocity
7	M-ST02A	average speed vs average velocity
8	M-ST02B	equal-time vs equal-distance average
9	M-ST02B	equal-time vs equal-distance average
10	M-ST02C	complete cycles + final partial cycle
11	M-ST03A, M-ST04A	acceleration / constant-a opening
12	M-ST03A, M-ST04A	acceleration / constant-a opening
13	M-ST04A	acceleration / constant-a opening
14	M-ST03A, M-ST04A	acceleration / constant-a opening
15	M-ST04D	stopping distance scales with u^2
16	M-ST05A	nth-second interval

APPENDIX A - SOURCE Q1-Q68 TO CONCEPT MAP

Part 2

SRC

Q	Concept link(s)	Core idea / source location
18	M-ST04D	equal-distance change in v^2
19	M-ST06A	delayed-start common clock
20	M-ST04B, M-ST05A	interval displacement from cumulative position
21	M-ST04D	equal-distance change in v^2
22	M-ST04A	evaluate $v(t)$ under constant a
23	M-ST04C	constant- a average velocity
24	M-ST04E	read u and a from $x(t)$ coefficients
25	M-ST06B	two stopping distances + geometry
26	M-ST04E	read u and a from $x(t)$ coefficients
27	M-ST05A, M-ST05B	consecutive interval difference reveals a
28	M-ST03A	$v=4t \rightarrow$ constant acceleration
29	M-ST03B, M-ST04A	same speed opposite direction / sign
30	M-ST03A	constant speed can still change velocity direction
31	M-ST03A	average acceleration from Δv
32	M-ST05B	1:3:5 pattern
33	M-ST06C	$\Delta v/a$ then $v_{avg} t$

APPENDIX A - SOURCE Q1-Q68 TO CONCEPT MAP

Part 3

SRC

Q	Concept link(s)	Core idea / source location
35	M-ST07D	window crossing entry speed
36	M-ST08B, M-ST08E	same max height under different g
37	M-ST08C	same height at two times
38	M-ST08A	top: $v=0$, $a=-g$
39	M-ST09A	common g cancels relatively
40	M-ST04C, M-ST11A	average velocity in free fall
41	M-ST11C	resistance proportional to mass
42	M-ST11B	numerical equality of v and s
43	M-ST07B	free fall from rest
44	M-ST07C	last-distance interval
45	M-ST08B	time to top
46	M-ST09C	same final speed from tower
47	M-ST09D	upward launch extra $2u/g$
48	M-ST08B	same-level flight time
49	M-ST08E	H proportional to $1/g$
50	M-ST08B	H scales with u^2

APPENDIX A - SOURCE Q1-Q68 TO CONCEPT MAP

Part 4

SRC

Q	Concept link(s)	Core idea / source location
52	M-ST09B	moving-platform release inherits velocity
53	M-ST07B	fall time $\sqrt{h}$
54	M-ST09C	same final speed from tower
55	M-ST08B	H scales with u^2
56	M-ST07B	free fall from rest
57	M-ST08D	speed fraction vs height fraction
58	M-ST08B	H scales with u^2
59	M-ST05B, M-ST07B	odd free-fall distances
60	M-ST07B	free fall from rest
61	M-ST07A, M-ST08B	sign convention for vertical motion
62	M-ST09B	moving-platform release inherits velocity
63	M-ST10A, M-ST10B	x-t slope / v-t slope & area
64	M-ST10C	vertical v-t straight line
65	M-ST10E	v^2 vs displacement axis logic
66	M-ST10D	x-t parabola
67	M-ST10F	impossible multi-valued motion graphs

APPENDIX B - SOURCE QUALITY / INTEGRATION NOTES

Preserve uncertainty

QC

1

Q29 appears to require a uniform-acceleration assumption that is not visibly explicit in the scanned stem.

2

Q49 needs the Moon/Earth gravity ratio if a numerical answer is expected; the scanned stem does not visibly state it.

3

Q52 is partly obscured; treatment is limited to the clearly identifiable moving-release structure.

4

Q65 is safest handled through $v^2=2gs$; the named graph shape depends on which quantity is on which axis.

5

The scan places printed pages 74 and 75 out of sequence; mapping follows question number, not file page order.

6

Integration preflight found a real curriculum gap: source Unit 4.4 / Q24 and Q26 were missing from the rebuilt subtopic sequence. M-ST04E was created to close it.